CompoundIQ Research Report

Compounding Pharmacy Prescription Ordering UX

How Leading Platforms Handle Medication Configuration

April 7, 2026

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Executive Summary

This report presents comprehensive research across seven critical domains that inform the design and development of CompoundIQ, a next-generation compounding pharmacy prescription ordering platform. The research spans existing portal UX analysis, structured sig building standards, medication data models, protocol template systems, progressive disclosure UX patterns, regulatory requirements, and real-world compounding prescription examples.

The Market Opportunity Is Clear. The compounding pharmacy ordering portal landscape is dominated by LifeFile, a platform whose usability issues are well-documented — pharmacy trainers explicitly warn providers that it is "finicky." No existing platform offers titration scheduling, protocol bundles, or progressive disclosure UI patterns. These three gaps represent CompoundIQ's primary competitive advantages.

Structured Sig Standards Exist But Are Underadopted. The NCPDP Structured and Codified Sig format can structurally represent 95% of prescription directions, but only 10–32% of e-prescriptions use structured fields. Compounded medication sigs — especially titration schedules — are almost universally free text. Surescripts' Sig IQ now augments 12.7% of transactions with ML-based parsing, signaling that the industry is moving toward automated structuring of free-text sigs. CompoundIQ should generate structured sig data internally while outputting clean free text for transmission.

The Data Model Must Be Ingredient-Centric. Compounded medications have no NDC codes and no RxCUI identifiers. The fundamental data entity is the ingredient, not the finished product. A compound is a configurable recipe: ingredients × concentrations × dosage form × route. The recommended entity-relationship model centers on a CompoundTemplate entity that links to IngredientConfig entities, each specifying an active ingredient with its concentration range, available salt forms, and dosage form constraints.

Protocol Templates Are the Killer Feature. Functional medicine clinics run multi-medication protocols (e.g., Testosterone + Anastrozole + HCG for male hormone optimization) that require 3–5 coordinated prescriptions per encounter. Currently, providers manually assemble these from individual templates or Chart Parts. A purpose-built protocol template system — with one-click instantiation of all medications, sex-specific variants, and integrated titration timelines — would save 10+ minutes per encounter and dramatically reduce prescribing errors.

Progressive Disclosure Is Essential for Complex Ordering. Research from Epic, Cerner, and academic studies shows that cascading progressive disclosure — where each field selection reveals the next relevant field — reduces order friction by 50–80% compared to form-dump approaches. CompoundIQ's prescribing interface should follow a 10-level cascade: Medication Category → Drug → Salt/Ester → Dosage Form → Route → Concentration → Volume/Quantity → Dose per Administration → Frequency → Titration Schedule.

Regulatory Compliance Is Non-Negotiable and Complex. Electronic compounding prescriptions must satisfy 21 CFR 1306 field requirements, 21 CFR 1311 EPCS requirements (two-factor authentication, pre-signing review screens, audit trails), Section 503A clinical necessity documentation, HIPAA transmission security, and a patchwork of 50 state board regulations. The platform must support both EPCS electronic transmission (via NCPDP SCRIPT v2017071) and HIPAA-compliant fax fallback for pharmacies not on the Surescripts network.

The 10-Level Cascade Tree Is the Core UI Innovation. Analysis of real-world compounding prescriptions for semaglutide, testosterone, LDN, and BPC-157 reveals a universal cascading dependency structure. Each medication selection drives constraints on every subsequent field. This cascade is both the data model and the UX model — it is the single most important architectural decision for CompoundIQ.

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Section 1: Existing Compounding Pharmacy Ordering Portals — UX & Workflow Analysis

Date: April 7, 2026

Scope: Prescriber-facing ordering portals for major U.S. compounding pharmacies — catalog structure, form design, titration handling, and UX patterns.

Executive Summary

The compounding pharmacy ordering portal landscape is bifurcated between two dominant paradigms:

1. **Fixed-catalog / pre-configured SKU portals** (503B office-use ordering): Prescribers or clinic staff browse a defined product catalog with pre-set concentrations, volumes, and formulations. The order process resembles an e-commerce checkout. Empower Pharmacy's 503B portal and Olympia Pharmacy's DrScript-powered portal are the clearest examples.
2. **LifeFile-based prescription portals** (503A patient-specific ordering): Prescribers write individual patient prescriptions using a pharmacy-hosted web app. Medications are searchable within the pharmacy's catalog, and the prescriber sets quantity, sig (dosage instructions), refills, and day supply. LifeFile is the dominant platform powering this workflow, used by Empower Pharmacy (503A), Belmar Pharmacy, Strive Pharmacy, University Compounding Pharmacy, and others.

Titration schedules (e.g., Semaglutide 10-week escalation) are generally **not handled by the portal itself** — they are managed via free-text SIG/directions fields, or by issuing sequential prescriptions at different dose levels, or through the pharmacy's "multi-order" batch function.

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1. DrScript / Olympia Pharmacy Network

Platform Identity

DrScript is a proprietary portal software built by **503 Success LLC** (domain: `503success.com`), deployed under the subdomain `olympiapharmacy.drscriptportal.com`. It powers the prescriber ordering interface for Olympia Pharmaceuticals, a major 503B outsourcing facility based in Orlando, FL. The same platform is licensed to other 503 pharmacies.

Portal Structure

- **Type:** Fixed-catalog / SKU-based (503B outsourcing facility model)
- **Access:** Prescribers must set up an account with Olympia; the portal is free for registered customers.
- The portal URL `olympiapharmacy.drscriptportal.com` currently shows "Ordering has been disabled" — reflecting the FDA enforcement action on compounded GLP-1s that began in early 2025. The portal infrastructure remains live. (Olympia DrScript Portal)

Medication Catalog

The Olympia medication directory at `olympiapharmacy.com/medication-directory` provides the publicly visible catalog structure: (Olympia Medication Directory)

Field	Description
Medication Name	Full compound name (e.g., "Biest (80/20) 0.25%")
Category	Therapeutic area tags (e.g., Hormones, Weight Management, IV Kits)
Route	IV, IM, Oral, Topical
Concentration	Per-mL specification (e.g., "Estradiol 0.5mg, Estriol 2mg")
Volume	Vial/container size (e.g., 30mL, "Other")

Catalog filters include Category, Route, and Volume dropdowns. Each distinct SKU (concentration × volume combination) is a separate catalog entry. For example, Anastrozole exists as four separate line items: 0.25mg, 0.5mg, 0.75mg, and 1.5mg.

Ordering Workflow

Based on 503 Success's description of its platform and Olympia's provider documentation:

1. **Log in** to portal (account required, set up via sales/marketing contact)
2. **Browse or search catalog** — filtered by Category, Route, or Volume; or search by medication name
3. **Select a SKU** — a specific pre-formulated concentration and container
4. **Enter patient/order details** — patient name, shipping destination (clinic or patient), quantity, billing profile (credit card on file)
5. **Review and submit** — preview order before submission
6. **Order management** — view order history, track shipments, check expiration dates (IPC Partnership page)

Key UX Characteristics

- **Catalog-based:** No customization of formulation — prescribers select from pre-defined SKUs.
- **Dropdown/selection-driven:** Medication type, concentration, and volume are chosen from list, not typed.
- **503B model means no individual patient Rx required** for office-use; orders go to clinic stock.
- **Favorites/templates:** Not publicly documented but typical of the 503 Success platform.
- **Titration handling:** Not handled in-portal. Providers order different vial strengths as needed for dose escalation; each is a separate SKU.

Provider Resources

Olympia offers custom prescription forms for many medications and provides online walkthrough tutorials via screen-sharing for new accounts. (Olympia Provider Resources)

2. WellsPx3 / Wells Pharmacy Network

Platform Identity

WellsPx3 is Wells Pharmacy Network's proprietary prescriber portal, accessible at `orders.wellsrx.com`. Wells also operates **WPNScripts** (`wplscripts.wellsrx.com`), a newer electronic prescribing portal. Wells Pharmacy is a 503A/503B compounding pharmacy based in Ocala, FL. The portal is **free for all registered Wells customers**. (WellsPx3 Home)

Wells is notable for being the **only DEA 1311-certified solution** for electronic ordering of both controlled and non-controlled compounded substances. (WellsPx3 About)

Dual-Portal Architecture

- **WellsPx3 Electronic Ordering** — For physicians only; active Rx submission.
- **WellsPx3 Order Management** — For both physicians and veterinary prescribers; post-order tracking. Patients of BodyLogicMD practices may also access Order Management.
- **WPNScripts** — Newer portal (`wplscripts.wellsrx.com`, version 2026-03-27) for electronic prescription submission, potentially replacing WellsPx3 for some functions.

Ordering Workflow (WellsPx3 EHR Integration via P2P)

The most documented workflow is through a **P2P (Practice to Practice) EHR integration**, where Wells appears as an embedded iframe within the EHR: (Wells Pharmacy: Order New Rx — YouTube)

1. In the patient chart, navigate to **Ordering → Medications → select "Wells Pharmacy"** from the drop-down
2. The Wells iframe opens. If the user is linked to multiple Wells locations, select the appropriate location.
3. Click **"New RX"** to initiate a prescription.
4. **Patient name and date of birth auto-populate** from the EHR chart.
5. Select the **date the patient was last seen**.
6. Select **visit method**: In-office or Telehealth.
7. **Type the medication or drug name** → an auto-populated drop-down list appears → select the correct option (e.g., "Tirzepatide 4.5 mL").
8. Once medication is selected:
   - Prescription details populate (dose, directions auto-generate based on entered values)
   - Optionally include **injection supplies** (checkbox)
   - Add **notes to pharmacy** (free text)
   - Choose shipping destination: **patient** or **clinic** (address auto-fills from EHR)
9. Click **"Save and Ready to Sign"** → prescriber reviews and clicks **"Sign and Send to WPN"** → prescription goes live.
10. Wells prescriptions appear in **blue** in the EHR medication history for visual identification.

Clone (reorder) workflow: Click the clone icon next to an existing prescription → all details auto-populate → edit as needed → sign and send.

Order Management Dashboard Fields

Post-submission, the Order Management dashboard shows: (Wells Px3 Order Management PDF)

Field	Description
Order Number	Unique identifier (also invoice number)
Patient	Patient name
Line Count	Number of Rx items in the order
Added	Date entered into system
State	Workflow stage (Entered → Verified → Adjudicated → Printed/Processed → Approved → Shipped)
Shipping	Shipping terms (Ground, 2nd Day, O/N, etc.)
Shipping Date	Date order left facility
Tracking No	UPS/FedEx tracking link
Expected by	Estimated arrival date

EPCS (Controlled Substances)

Wells Px3 is DEA 1311-certified for EPCS. The workflow adds: (Wells EPCS Training PDF)

- Two-factor authentication setup (software or hardware token via Experian identity proofing)
- Access Control Setup requiring two authorized users
- When writing a controlled substance Rx, "e-Send" is enabled only for EPCS-registered pharmacies
- Two-factor "Sign" screen appears for final authentication

Key UX Characteristics

- **Hybrid:** Catalog-driven medication selection (type-ahead dropdown from Wells formulary) + free-text fields for directions and pharmacy notes.
- **EHR-embedded:** The preferred workflow runs inside the provider's existing EHR (P2P integration) rather than requiring a separate portal login.
- **Clone/reorder function:** Reduces friction for repeat prescriptions.
- **Titration handling:** No dedicated titration tool. Each dose escalation = new Rx via New RX flow or clone with modified dose. Wells accepts multi-refill prescriptions with SIG notes for provider-managed dose changes.
- **Templates/favorites:** Not explicitly documented in public materials; the clone feature serves a similar purpose.
- **Color coding:** Wells medications highlighted in blue within the integrated EHR view.

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3. LifeFile / ePowerRx — The Dominant Platform

Platform Identity

LifeFile (lifefile.net) is a cloud-based pharmacy management system founded in 2006, widely recognized as the dominant e-prescribing platform in the compounding pharmacy sector. It is used by some of the largest compounding pharmacies in the nation. (LifeFile website) (RXinsider profile)

LifeFile provides **four portals**: Pharmacy Portal, Provider Portal, Patient Portal App, and Sales Portal.

Note: LifeFile and the older "ePowerRx" are both compounding e-prescribing platforms. LifeFile is the more widely deployed modern system; the name "ePowerRx" appears in historical references to compounding e-prescribing. For most active deployments, LifeFile is the operational platform.

Pharmacies Using LifeFile Provider Portal

- Empower Pharmacy (503A patient-specific)
- Belmar Pharmacy (all locations)
- Strive Pharmacy
- University Compounding Pharmacy (San Diego)
- South End Pharmacy
- Numerous other independent compounding pharmacies

Provider Portal — Detailed Step-by-Step Workflow

The following is reconstructed from Strive Pharmacy's documented LifeFile training session (Strive Sessions Ep. 2 — YouTube) and South End Pharmacy's LifeFile tutorial (Prescribe in a Flash — YouTube):

Login: Accessible via the pharmacy's website (e.g., strivepharmacy.com → person icon → username/password).

Homepage tabs:

1. Announcements / Critical Updates
2. Billing (phone/email contacts)
3. Shipping policies & turnaround times
4. **Digital Catalog** (all products and pricing, updated in real time, 24/7 access)
5. Help Center (resources and guides)

New Order Workflow:

Step 1 — Select/Create Patient

- Click the large **"Select Patient"** box
- Search existing patients by last name → info auto-populates
- OR click **"New Patient"** and enter required fields:
- First name, Last name
- Patient address and shipping address
- Patient email address

• Date of birth

• Click **"Save Patient"**

Step 2 — Patient "Bubble Screen"

• After selecting the patient, a set of action bubbles appears

• For new orders, click the first bubble: **"ERX"**

Step 3 — Order Processing Information

• **Pick-up or Ship?** Select delivery type

• Patient info section auto-populates from profile

• **Delivery address:** Click "Copy Patient Information" (ship to patient) OR "Copy Doctor Information" (ship to clinic)

Step 4 — Select Medication

• Browse the digital catalog or search by medication name

• Click the medication to select it

• Choose **strength** and **vial size** from dropdowns

• The rest of the prescription partially auto-populates

Step 5 — Complete Prescription Fields

• **Quantity** (in milliliters, not number of vials)

• **Refills** (prescriber's discretion)

• **Day supply** (prescriber's discretion)

• **Directions (SIG):** Preloaded popular options available as buttons; can type custom directions if none match

• **Include syringes?** Checkbox for applicable injectable medications

• Any special instructions

Step 6 — Add to Cart & Submit

• Click green **"Add to Order"** button

• Cart counter shows "Medications in current order: X"

• Can add additional medications for the same patient

• Click **"Review and Sign"** → review draft order → click **"Submit"**

Refill Workflow:

• Navigate to patient → locate previous order → click to open → "Review and Sign" → submit

• Alternatively: edit/update details if dosing has changed

Advanced Features:

Order Linking (ship multiple orders together):

• Place multiple orders normally, but use **"Save and Continue Later"** instead of Submit

• Return to saved orders, check boxes for orders to link, click **"Link Shipping"** → Submit together

• All linked orders ship in the same package to the same address

Multi-Order Management (same medication for multiple patients):

• From left control panel: **"Multi-Order Management"** → "New Multi-Order"

• Select delivery/shipping information (usually shipping to clinic)

• Set recipient type (prescriber) and payer

- Add patients from a list
- Click "Edit RX List" → "Add RX" → configure the medication, strength, directions (same for all patients in the batch)
- Click "Submit Order" → batch is sent to pharmacy

Custom Order:

- For non-catalog formulations, same ERX flow but medication is entered as a custom/free-text compound
- Must include clinical difference statement if a commercially available equivalent exists (per FDA 503A requirements)

Key UX Characteristics

- **Primarily catalog-based** with custom order option: Standard prescriptions use a searchable catalog of pre-defined formulations; custom orders allow free-text entry of formulation components.
- **Mostly dropdown-driven:** Medication, strength, and vial size use dropdowns; SIG directions have a hybrid of preloaded options + free text.
- **MyRX List:** A personal favorites list of frequently used prescriptions for one-click reuse.
- **Real-time pricing:** Digital catalog displays current pricing, updated regularly.
- **EPCS compliant:** Supports electronic prescriptions for Schedule II–V controlled substances.
- **Patient compliance management:** Built-in features to track patient adherence.
- **B2B API integration:** EMR systems can push orders directly to LifeFile via API (e.g., Cer.bo EHR integration). (FxMedSupport LifeFile Integration)

Titration Handling

No dedicated titration schedule builder exists within LifeFile as of current public documentation. Dose escalation is managed by:

1. Writing sequential prescriptions at each dose level (new ERX for each step)
2. Using detailed free-text SIG (e.g., "Week 1-4: inject 0.25mg weekly; Week 5-8: inject 0.5mg weekly...")
3. Multi-order function to batch-send the same dose to multiple patients simultaneously
4. "Save and Continue Later" + order linking to group sequential dose prescriptions

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4. Empower Pharmacy Ordering Portal

Platform Identity

Empower Pharmacy (Houston, TX) is one of the largest compounding pharmacies in the U.S., operating both a 503A compounding pharmacy (PCAB-accredited) and a 503B outsourcing facility (FDA-registered). It accordingly offers **two separate ordering portals** for providers. (Empower Portal Login)

Two-Portal Architecture

Portal 1: 503A Patient-Specific Orders → LifeFile

- Providers use a LifeFile account (auto-created for all Empower customers) to submit individual patient prescriptions.
- Step-by-step instructions are provided on the 503A Orders page. (Empower 503A Orders)
- Features: Real-time product availability, order tracking, EPCS-compliant technology, robust reporting, patient compliance management.
- Ordering methods for 503A: Portal (LifeFile), fax to (832) 678-4419, call in at (346) 229-5386, or via EMR integration.

Portal 2: 503B Office-Use Bulk Orders → Proprietary Office-Use Portal

- Separate login from LifeFile: "Office-Use Account Login."
- Designed for bulk ordering of pre-manufactured batches for clinic stock (no individual patient Rx required).
- Ordering methods for 503B: Portal, fax to (832) 678-4419, call at (346) 229-5385, or email officeuse@empowerpharmacy.com.

Custom Order Form Program (503B)

Empower offers a **Custom Order Form Program** that allows clinics to request a personalized, clinic-branded order form configured to their specific prescribing patterns: (Empower Custom Order Form Program)

1. Clinic fills out a request form
2. Dedicated Empower representative contacts clinic to gather product preferences
3. Empower team creates a custom form tailored to that clinic
4. Clinic receives a ready-to-use, clinic-branded order form

This effectively creates a filtered/curated sub-catalog for high-volume clinics, reducing the cognitive load of browsing the full formulary.

Formulary Structure

Empower describes itself as having "the nation's largest compounding formulary." Products are browsable on the website by category. The 503B portal presents a catalog of available bulk SKUs. The 503A LifeFile portal presents the 503A formulary within the LifeFile catalog interface. (Empower Providers page)

Notable formulary categories include: GLP-1 agonists (Semaglutide/Cyanocobalamin, Tirzepatide/Niacinamide), hormones (Testosterone, Progesterone), peptides (Sermorelin, NAD+), IV nutrition, sexual health, dermatology.

Semaglutide Handling

Empower's 503A catalog includes Semaglutide/Cyanocobalamin Injection in multiple concentrations (1 mg/mL and 5 mg/mL; 1 mL and 2.5 mL vials) and Semaglutide/Methylcobalamin ODT in three strength combinations (2mg/0.1mg, 5mg/0.1mg, 12mg/0.1mg). (Semaglutide/Cyanocobalamin product page)

For titration, Empower's product documentation describes a standard escalation protocol but notes prescribers set the actual titration through the SIG on individual prescriptions. The standard approach: 0.25mg weekly × 4 weeks → 0.5mg × 4 weeks → 1mg × 4 weeks → up to 2.5mg maximum. The portal

does not build a titration schedule; each dose level requires a separate prescription through LifeFile.

Key UX Characteristics

- **Dual portal model** cleanly separates 503A (patient-specific) from 503B (office-use/bulk).
- **503A uses LifeFile** — same UX as other LifeFile pharmacies described above.
- **503B portal is catalog-based** — browse pre-manufactured batch SKUs.
- **Custom order forms** are available as a clinic-specific overlay on the 503B catalog.
- **No templates or titration builders** evident in public documentation.
- Account qualification required for controlled substances.

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5. ReviveRX Ordering Interface

Platform Identity

ReviveRX is a 503A licensed mail-order compounding pharmacy based in Houston, TX (3831 Golf Dr., Houston, TX 77018), serving 45 states plus Puerto Rico. It positions itself as a "partnership-first" pharmacy for healthcare providers in hormone therapy, wellness, weight management, functional medicine, and concierge/telehealth practices. (ReviveRX website) (ReviveRX About)

Portal Documentation

ReviveRX's prescriber portal has very limited public documentation. The available information suggests:

- **Onboarding process:** Provider submits a short provider form → dedicated representative contacts them → account is set up → prescribing can begin. Ordering preferences and clinical considerations are discussed during account setup.
- **Prescription submission:** "Your practice can begin sending prescriptions, according to your instructions, as soon as onboarding is complete."
- **Portal quality:** One provider testimonial states: "their interface dashboard for ordering is my favorite." No further specification available.
- **Direct-to-patient shipping:** All orders ship directly to patients within ~3 days.
- **Support model:** Dedicated representative + pharmacy support team for each provider; providers are not expected to self-serve without support.

Inferred Characteristics

ReviveRX's 503A-only model (no 503B mentioned) and emphasis on personalized partnerships suggests a **prescription-driven workflow** rather than a fixed-SKU catalog. The "ordering preferences" discussed during onboarding suggest some degree of protocol or template configuration. Given ReviveRX's Houston location and 503A focus, the portal may be LifeFile-based (LifeFile is heavily used by Houston-area compounding pharmacies), but this is not confirmed in public sources. (ReviveRX Partners)

Key UX Characteristics

- **Prescription-driven** (503A, patient-specific)

- **High-touch onboarding** with dedicated representative
- **Ordering preferences** configured at account setup (suggests some template/protocol capability)
- **Minimal self-service documentation** — support-forward model

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6. Belmar Pharmacy Ordering System

Platform Identity

Belmar Pharma Solutions is a multi-location 503A compounding pharmacy network with locations in Colorado (Golden), Arizona, Florida, Wisconsin, Kansas, and Utah (as MedQuest Pharmacy). Their prescriber portal is **powered by LifeFile**. (Belmar Portal Login)

Portal Structure

- **Platform:** LifeFile (titled "LifeFile | Belmar Provider Portal" on their portal page)
- **Free** for all registered prescribers
- Two specific login endpoints: **Belmar Select Outsourcing Portal** and **Belmar Pharmacy – Colorado Portal**
- One-on-one training available by contacting `lifefile@belmarpharmasolutions.com`

Portal Features

As documented on Belmar's portal page and prescription submission guide: (Belmar Portal Login page) (Belmar How to Write a Prescription)

- Sends new prescriptions and refills securely and accurately
- Real-time order status and tracking
- EPCS-compliant capabilities
- User-friendly prescribing tools for quick, easy ordering
- Easy refill entry
- Delivery tracking for patients
- **Personalized medication lists customizable to your practice** (indicates a "MyRX"-style favorites list)

Handy How-To's on the portal page:

1. How to Add a New Patient
2. How to Submit a New Order
3. How to Submit an ezRefill

Required Prescription Fields

For any compounded prescription at Belmar, providers must supply: (Belmar Submit a Prescription)

Category	Fields Required
Patient Information	First name, last name, DOB, gender, address, cell phone, email, known drug allergies, patient identifier (applicable states)
Prescription Information	Date, drug(s), dose, dosage form, directions (frequency + route), quantity, refills
Prescriber Information	First/last name, practice address, phone, fax, NPI#, state license#, DEA license, state controlled substance registration (where applicable)
Prescriber Signature	Pen-to-paper for controlled substances; EPCS preferred
Payment Type	Patient pay or clinic pay

Submission Methods

1. Phone (toll-free numbers by location)
2. Fax (toll-free numbers by location)
3. EMR e-prescribe (via NCPDP number by location)
4. **LifeFile Prescriber Portal** (contact required to set up access)

Pellet/Testosterone Injectable Office-Use Orders

For pellet and testosterone injectable bulk orders, Belmar uses a separate **Belmar Select Outsourcing Portal** (also LifeFile-based). New customers must apply separately for this account. (Belmar Patient-Specific 503A page)

Key UX Characteristics

- **LifeFile-based** — same UX as described in Section 3.
- **Personalized medication lists** per practice (LifeFile's MyRX feature).
- **EPCS-compliant** for controlled substances.
- **Multi-location complexity:** Six separate pharmacy locations with different phone/fax numbers; portal may allow location selection.
- **Titration handling:** Not built into the portal; handled via free-text SIG or sequential prescriptions.

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7. Additional Compounding Pharmacy Portals

7.1 Strive Pharmacy

Portal platform: LifeFile (confirmed), accessed via strivepharmacy.com → person icon login.

Strive Pharmacy is notable for providing the most comprehensive public documentation of the LifeFile provider portal through their **Strive Sessions webinar series**, including a full step-by-step prescribing walkthrough (Strive Sessions Ep. 2 — YouTube) and a blog post summarizing LifeFile features. (Strive LifeFile Webinar Blog)

Key features documented by Strive:

- Digital catalog tab (all products and pricing, updated daily)
- Multi-Order Management for batch same-medication orders across multiple patients
- Order linking (shipping multiple orders in one package)
- Custom Order entry for non-catalog formulations
- Clinical difference statement documentation (required per 503A rules when a commercial equivalent exists)
- MyRX list for frequently prescribed medications

Strive's webinar included an example of prescribing **semaglutide sublingual drops** via the multi-order management workflow.

7.2 Tailor Made Compounding (TMC)

Portal platform: Proprietary EMR/portal system. TMC launched a **new Provider Portal in November 2024**, replacing a legacy portal at `emr.tailormadecompounding.com`. (TMC Legacy Portal Page)

- Access via: `tailormadecompounding.com` → secure, HIPAA-compliant online ordering portal
- Features: Electronic prescriptions, patient record tracking, order management
- Account setup requires: Clinic name, prescriber information (multiple prescribers can be added), license uploads, communication preference selection (TMC Account Request page)
- TMC is a 503A-only pharmacy (all prescriptions must be patient-specific)
- Serves 46 states including Puerto Rico
- Account manager provides email notifications on order status changes

Key UX characteristic: Account-manager-centric model with dedicated support. New portal (2024) suggests active investment in UX.

7.3 University Compounding Pharmacy (UCP) — San Diego

Portal platform: LifeFile (confirmed — "Lifefile Login" button on provider page). (UCP Provider Rx Form page)

- Prescribers submit via phone (direct pharmacist line), fax, or e-prescribe through their EMR.
- **Patient-side workflow:** Provider submits prescription → UCP contacts patient within 24 hours via text/email with a link to complete the order online (delivery address, billing, medications/allergies).
- **Quick Refill** available on website: patients can self-refill without calling. (UCP FAQs)
- **Auto-refill program** available via email enrollment.
- **503A only** — no office-use/bulk compounds to providers.
- Most prescriptions ship within 2–3 business days.

7.4 ScriptWorks Pharmacy (California, PCAB-accredited)

ScriptWorks offers a notable prescriber portal with several well-documented UX features: (ScriptWorks Prescriber Portal blog)

- **Practice-specific drug catalog** curated to the prescriber's patterns and patient needs
- Submit prescriptions online without requiring an EMR system

- Choose **patient-ship or provider-ship** at the order level
- **Adherence monitoring** — visibility into patient progress and medication timelines
- Direct pharmacist communication from within the portal
- **Structured order fields** to prevent errors associated with handwritten/faxed orders
- LegitScript-certified and HIPAA-compliant

This is one of the few pharmacies to explicitly describe a **customized practice-level catalog** as a portal feature.

7.5 CareFirst Specialty Pharmacy

Portal platform: No dedicated prescriber portal documented publicly. Prescriptions accepted via fax, phone, or e-prescribe (NCPDP# 3151266, searching "CareFirst Specialty Pharmacy" in prescriber's EMR). (CareFirst Prescribers page)

Prescribers can complete a prescriber form online for account registration, but form registration is not required to submit prescriptions. Pharmacist consultation available by phone.

7.6 My Practice Connect ERX Platform (Multi-Pharmacy Aggregator)

My Practice Connect is a notable aggregator platform that stands apart from single-pharmacy portals: (My Practice Connect ERX Platform)

- Single web-based portal with access to **multiple FDA-registered 503A and 503B pharmacies** simultaneously, including Empower Pharmacy, Wells Pharmacy, and others
- **Real-time inventory and pricing across pharmacies**
- Providers can place individual or bulk orders
- Automated refill reminders for patient compliance
- Order history and tracking across all connected pharmacies
- Secure messaging directly with pharmacy teams
- Interaction alerts for drug/allergy risks
- EPCS-compliant, HIPAA-compliant
- **No recurring monthly or annual fees**; multiple prescribers/locations at no extra cost

Connected pharmacies include: 1st Choice Compounding, Brooksville Pharmaceuticals, Empower Pharmacy, Grand Ave Pharmacy, Liquivida, Meta Pharmacy Services, Partell Pharmacy, PharmaLabs, Safe Chain Solutions, Wells Pharmacy, WP PharmaLabs, and others.

This aggregator model is a significant UX pattern — it removes pharmacy-specific portal fragmentation for providers who work with multiple compounding pharmacies.

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8. Cross-Platform UX Analysis

8.1 Catalog Architecture: Fixed SKU vs. Custom Build

Platform	Architecture	Notes
Olympia/DrScript (503B)	Fixed catalog	SKUs defined by concentration × volume; no customization
Empower 503B portal	Fixed catalog	Pre-manufactured batch SKUs; custom order forms per clinic
LifeFile (all 503A pharmacies)	Catalog + custom	Primary catalog of pre-formulated compounds; custom order option for off-catalog formulations
WellsPx3	Catalog with type-ahead	Medication selection from Wells formulary via type-ahead dropdown
Tailor Made Compounding	Unknown (proprietary)	New portal (Nov 2024), 503A patient-specific
ReviveRX	Likely catalog-based	Configuration discussed at onboarding; 503A model

8.2 Field Sequence by Portal Type

LifeFile standard order sequence:

1. Patient lookup/creation (name, DOB, address, email)
2. Order type (ERX)
3. Delivery type (pickup vs. ship)
4. Delivery address (copy patient or copy doctor)
5. Medication selection (search/browse catalog → select strength → select vial size)
6. Quantity (in mL)
7. Refills
8. Day supply
9. Directions/SIG (preloaded options or free text)
10. Injection supply add-on (checkbox)
11. Billing profile selection
12. Review and submit

WellsPx3 (P2P EHR-integrated) sequence:

1. Navigate to Medications in patient chart → select Wells Pharmacy
2. Click "New RX"
3. Patient info (auto-populated from EHR)
4. Last visit date
5. Visit method (in-office/telehealth)
6. Medication name (type-ahead dropdown)
7. Prescription details (auto-populate + manual completion)
8. Injection supplies (checkbox)

9. Notes to pharmacy (free text)
10. Shipping destination (patient/clinic)
11. Save and Ready to Sign → prescriber signs and sends

503B office-use catalog sequence (Olympia/Empower):

1. Login to portal
2. Browse catalog (Category/Route/Volume filters)
3. Select medication SKU
4. Enter quantity (units/vials)
5. Select patient or clinic shipping
6. Enter billing information
7. Review and confirm

8.3 Input Methods

Input Type	Use Case
Dropdown/selector	Medication name, strength, vial size, delivery type, billing profile
Type-ahead search	Medication name lookup from formulary (WellsPx3, LifeFile)
Pre-loaded SIG buttons	Common directions (LifeFile)
Free text	Custom SIG/directions, notes to pharmacy, custom compound ingredients
Checkbox	Include injection supplies, auto-refill enrollment
Auto-populate	Patient data from profile, address fields, prescription details from selected medication

8.4 Protocol Templates and Favorites

Feature	Platform	Implementation
MyRX List	LifeFile (all pharmacies)	Personal catalog of saved prescriptions for one-click reuse
Clone/Reorder	WellsPx3 (P2P integration)	Duplicates a previous prescription with all details
Custom Order Forms	Empower Pharmacy (503B)	Clinic-branded filtered sub-catalog; created by Empower team
Practice-Specific Catalog	ScriptWorks	Curated catalog matching prescriber's typical patterns
Multi-Order Management	LifeFile (all pharmacies)	Batch same-medication orders across multiple patients; effectively a protocol template

Feature	Platform	Implementation
Account Setup Preferences	ReviveRX	Ordering preferences configured at onboarding
Prescription Templates	SiCompounding platform	Prescribers create templates with one or multiple prescriptions; available for multi-patient batching (SiCompounding Release Notes)

8.5 Titration Schedule Handling

No compounding pharmacy portal currently reviewed has a dedicated titration schedule builder. The standard approaches across platforms:

Approach	Platform	Notes
Free-text SIG	All (LifeFile, WellsPx3, DrScript)	Write complete titration instructions in the SIG field: "Weeks 1-4: inject 0.25mg; Weeks 5-8: inject 0.5mg; Weeks 9-12: inject 1mg..."
Sequential prescriptions	All	Write one Rx per dose level; patient receives separate fills
Multi-vial concentration selection	All catalog-based	Order vials at different concentrations for different phases; each is a separate SKU
Multi-Order batch	LifeFile	Prescribe same dose to many patients simultaneously
"Save and Link Later"	LifeFile	Write sequential dose prescriptions for same patient, save drafts, then link and submit together for one shipment

The **most common real-world approach** for a 10-week semaglutide escalation (e.g., 0.25mg × 4 weeks, 0.5mg × 4 weeks, 1mg × 2 weeks) is to:

1. Issue a multi-vial prescription for the starting dose with a free-text SIG describing the escalation
2. OR issue monthly prescriptions at each dose level
3. The SIG conveys the protocol to the patient; the pharmacy fulfills each Rx as written

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9. Key Findings and Design Implications

9.1 The LifeFile Monopoly Problem

LifeFile dominates 503A compounding prescriber portals, but the platform has known usability issues. Strive Pharmacy's trainer explicitly warned: **"LifeFile is finicky. I do not want you to worry about it."** This is a significant signal. Despite being the market-leading platform, LifeFile's UX generates enough friction

that pharmacies build training infrastructure around managing provider frustration.

9.2 No Dedicated Titration UX Exists

The absence of a native titration schedule builder across all platforms is a significant gap. Clinicians managing GLP-1 weight management protocols, hormone optimization, or peptide therapy typically have multi-week dose escalation plans. The current state — writing paragraph-length SIG instructions or placing sequential single-dose prescriptions — is error-prone and cumbersome.

9.3 EHR Embedding Reduces Friction

WellsPx3's EHR-embedded (iframe) workflow is a notably better UX than requiring providers to leave their EHR and log into a separate portal. Patient data auto-populates, medication history is visible in context, and color-coded Wells prescriptions are visually distinct. The downside: it requires EMR-level integration agreements.

9.4 Catalog-Based 503B Portals Are Simpler

503B office-use portals (Olympia, Empower 503B) are functionally e-commerce stores. The ordering flow is closer to an Amazon checkout than a clinical prescription process. This simplicity is appropriate for the 503B model (no individual patient Rx required), but the tradeoff is zero customization.

9.5 Multi-Order / Batch Capabilities Are Underserved

For clinics running high-volume GLP-1 or hormone optimization programs, the ability to prescribe the same protocol to many patients simultaneously is critical. LifeFile's Multi-Order Management feature addresses this, but the UX is complex enough that Strive dedicates webinar time to teaching it. A cleaner batch-prescription experience would serve this use case well.

9.6 The Aggregator Pattern (My Practice Connect) Points to Future Direction

The existence of a multi-pharmacy aggregator (My Practice Connect) that normalizes ordering across Empower, Wells, and others suggests provider demand for a single-pane-of-glass interface. The pharmaceutical equivalent of a marketplace or GPO portal. No single pharmacy portal meets this need.

9.7 Input Hybrid is Standard

Every modern portal reviewed uses a hybrid of dropdowns (for catalog items) and free text (for SIG, notes, custom compounds). Pure free-text entry (fax/phone) is still widely supported as a fallback, reflecting the diversity of provider technical sophistication.

Sources

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- Olympia Medication Directory
- Olympia DrScript Portal
- IPC / Olympia Partnership Page

- 503 Success LLC
- WellsPx3 Home
- WellsPx3 About
- Wells Physician Resources
- Wells Order Management Training PDF
- Wells EPCS Training PDF
- Wells Pharmacy: Order New Rx — YouTube
- Wells Pharmacy: Managing Medications — YouTube
- WPNScripts Portal
- LifeFile Website
- LifeFile Provider Portal
- LifeFile RXinsider Profile
- FxMedSupport LifeFile-Cerbo Integration
- Strive Sessions Ep. 2: LifeFile Walkthrough — YouTube
- Strive LifeFile Webinar Blog
- South End Pharmacy LifeFile Tutorial — YouTube
- Empower Pharmacy Portal Login
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- Empower Custom Order Form Program
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- Empower Semaglutide/Cyanocobalamin Product Page
- Empower FAQs
- ReviveRX Website
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- ReviveRX Partners
- Belmar Portal Login
- Belmar How to Write a Compounded Prescription
- Belmar Submit a Prescription
- Tailor Made Compounding Join Us
- Tailor Made Compounding Legacy Portal
- Tailor Made Compounding Account Request
- University Compounding Pharmacy
- UCP FAQs
- UCP Provider Rx Forms
- ScriptWorks Prescriber Portal Blog
- CareFirst Specialty Pharmacy Prescribers
- My Practice Connect ERX Platform
- SiCompounding Release Notes
- Strive Pharmacy Homepage

- GobyMeds Semaglutide Dosing

Section 2: Structured Sig Building for Compounded Medications

Compiled: April 7, 2026

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1. NCPDP Structured Sig Standard

1.1 What is the SIG?

The term "Sig" (from Latin *signatura*, meaning "let it be labeled") refers to the patient directions on a prescription — the instructions for use conveyed from prescriber to pharmacist and ultimately printed on the medication label. A complete sig typically includes: action verb, dose, dose units, dose form, route of administration, frequency/timing, duration, and optionally an indication.

Source: JAMIA Evaluation of NCPDP Structured and Codified Sig Format

1.2 The NCPDP SCRIPT Standard

The NCPDP SCRIPT Standard defines XML-based transaction messages for electronic prescription exchange. Key transactions include NEWRX (new prescription), REFREQ (refill request), CANRX (cancellation), and RXHREQ (medication history query). Every NEWRX message includes a Sig element.

Version timeline relevant to sigs:

Version	Mandated	Key Sig Changes
SCRIPT 10.5	~2012	Formally incorporated Structured and Codified Sig Format v1.0
SCRIPT 10.6	2013	Introduced RxNorm drug coding; Structured Sig codification optionally available

Version	Mandated	Key Sig Changes
SCRIPT 2017071	Jan 1, 2020 (CMS mandate)	Sig field expanded from 140 to 1,000 characters; compound prescriptions supported (up to 25 ingredients); IV/wound care instructions added
SCRIPT 2023011	Effective 2028 (CMS mandate)	Further enhancements; electronic prior authorization

Sources: NCPDP SCRIPT Standard Guide (Intuition Labs, 2026); NCPDP press release on v2017071

1.3 Structured and Codified Sig (S&C SIG) Format

The S&C SIG format, developed by NCPDP starting in 2004 and released in 2006 (v1.0 in 2008), structures and codifies individual components of e-prescription directions into discrete XML fields. NCPDP requires the free-text direction generated from S&C SIG fields to be displayed to the prescriber and included in the e-prescription transmission. **Free text is always mandatory; structured fields are optional.**

13 Segments and Their Fields

Segment	Key Fields
Repeating Sig	Sig sequence position number; Multiple Sig modifier (AND/THEN/etc.)
Code System	SNOMED CT version; FMT (Federal Medication Terminologies) version
Sig Free Text String	The mandatory human-readable text string
Dose	Dose delivery method (action verb); Dose delivery method modifier; Dose quantity; Dose form; Dose range modifier
Dose Calculation	Dosing basis numeric value/units; Body metric qualifier/value; Calculated dose numeric value/units; Dosing basis range modifier
Vehicle	Vehicle name; Vehicle quantity; Vehicle unit of measure; Multiple vehicle modifier
Route of Administration	Route; Multiple route modifier
Site of Administration	Site; Multiple site modifier
Sig Timing	Administration timing; Rate of administration; Rate unit of measure; Time period basis; **Frequency numeric value**; **Frequency units**; Variable frequency modifier; **Interval numeric value**; **Interval units**; Variable interval modifier
Duration	Duration numeric value; Duration text
Maximum Dose Restriction	Max dose value/units; Variable max dose with duration
Indication	Indication precursor; Indication text; Indication value

Terminology systems used:

- **SNOMED CT** — For dose delivery method, vehicle, route, site, administration timing, frequency/interval units, duration, indication precursor
- **Federal Medication Terminologies (FMT) / NCI Thesaurus (NCIt)** — For dose form
- All other fields accept free text

Source: JAMIA Evaluation of NCPDP S&C Sig Format, 2011

1.4 Representational Coverage

A 2011 JAMIA study evaluating 20,161 ambulatory e-prescriptions found:

Capability	Result
Sigs fully representable by the Format's field structure	95% (CI 93–97%)
Administration timing terms mappable to SNOMED CT	Only 33%
Dose delivery method terms codifiable	60%
Route of administration terms codifiable	95%
Site of administration terms codifiable	70%
Indication terms codifiable	93%
Dose form terms codifiable (FMT)	84%

Key limitation: The format can *structurally* hold 95% of sigs, but codified vocabulary coverage for specific fields (especially timing and dose delivery method) is far lower, driving continued reliance on free text.

Source: JAMIA Evaluation of NCPDP S&C Sig Format, 2011

1.5 Adoption Rates

- In 2020, only **~10% of e-prescriptions** used S&C SIG structured fields (Intuition Labs, 2026)
- A 2022 JAMIA study of 3.8 million e-prescriptions (2019–2021) found **32.4%** transmitted with at least one S&C SIG component, but this was concentrated in 22 specific business accounts (JAMIA S&C SIG Implementation Outcomes, 2022)
- Quality issues persisted in **10.3%** of S&C SIG-formatted prescriptions

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2. Surescripts and Compounded Medication Sigs

2.1 Surescripts Network Overview

Surescripts operates the dominant U.S. electronic prescribing network. In 2025, **1.39 million prescribers** used e-prescribing via Surescripts, as did virtually all pharmacies. The network transmitted over 27.2

billion health intelligence exchanges in 2024.

Source: Surescripts 2025 Annual Impact Report

2.2 How Surescripts Handles Compounded Medication Sigs

Prior to SCRIPT 2017071: No version of NCPDP SCRIPT prior to 2017071 was designed to convey compound prescription information electronically. Prescribers had to "shoehorn" compound information into standard fields, causing confusion at the pharmacy.

With SCRIPT 2017071 (mandated Jan 1, 2020):

- Supports transmission of compound ingredients and quantities for **up to 25 active ingredients** in one electronic prescription
- Sig field expanded from **140 to 1,000 characters** to accommodate complex compound directions
- Additional support for: IV administration information, wound care information, flavoring, number of packages, manufacturer/lot/expiration data

Critical limitation: Even with 2017071, the compound sig itself remains **primarily free text**. There is no structured compound-specific sig format — the 1,000-character free text field is the mechanism used for complex compound directions. A 2019 Surescripts/NCPA CE article noted: "hopefully there will be very few instances in which a prescriber will actually use all 1,000 characters available to write a sig, because that would cause a different type of problem at the pharmacy end."

Sources: Surescripts/NCPA SCRIPT 2017071 CE Article, 2019; NCPDP SCRIPT v2017071 Testing Tool announcement

2.3 Surescripts Sig IQ

Surescripts has deployed **Sig IQ**, a machine learning technology that:

- Translates free-text Sigs into the Structured & Codified Sig format automatically
- Operates on new e-prescriptions, renewal messages, and Medication History responses
- In 2025, augmented **12.7% of all E-Prescribing transactions**
- Augmented **60% of Medication History for Ambulatory and Reconciliation responses** (up from 52% in 2024)

This is a significant development: rather than requiring prescribers to use structured sig builders, Sig IQ applies NLP/ML post-transmission to generate structured representations from free text.

Source: Surescripts 2025 Annual Impact Report

2.4 Practical Limitations for Compound Sigs via Surescripts

1. **No NDC for compounds:** Compounded medications have no NDC numbers, which breaks standard pharmacy system auto-matching. Surescripts/SCRIPT 2017071 uses the ingredient list (up to 25 components) as the compound identifier instead.
2. **Free text sig dominates:** Complex compound sigs (titration schedules, multi-phase dosing) are invariably entered as free text — the S&C SIG structured fields cannot reliably express titration or conditional dose escalation.
3. **Character limits:** Even at 1,000 characters, very complex titration sigs may still be truncated or require supplemental pharmacist notes.

4. **Interoperability gaps:** A 2024 scoping review found that "prescribing compounded medications using e-prescribing systems is a challenge due to the inconsistencies in drug names and interoperability issues between both computer systems." (Healthcare scoping review, 2024)

5. **Sig character limits in EHR systems:** Some EHR systems (e.g., Enterprise Health) warn that "SureScripts limits the number of characters in the sig, so you may need to put the complex sig in the Pharmacist Note field instead." (Enterprise Health Documentation)

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3. Standard Frequency Codes

3.1 Core Latin-Derived Frequency Abbreviations

These are the foundational pharmacy sig frequency codes derived from Latin phrases:

Code	Latin Origin	Meaning	Structured Data Mapping
QD (or qd)	*quaque die*	Every day / once daily	Frequency: 1 per day
BID (or bid)	*bis in die*	Twice a day	Frequency: 2 per day
TID (or tid)	*ter in die*	Three times a day	Frequency: 3 per day
QID (or qid)	*quater in die*	Four times a day	Frequency: 4 per day
QHS (or qhs)	*quaque hora somni*	Every night at bedtime	Administration timing: bedtime
PRN (or prn)	*pro re nata*	As needed	Variable frequency; requires indication
QOD (or qod)	*quaque altera die*	Every other day	Interval: 48 hours
QW (or qw)	—	Every week / once weekly	Frequency: 1 per week
BIW	—	Twice a week	Frequency: 2 per week
TIW (or tiw)	—	Three times a week	Frequency: 3 per week
Q2W	—	Every 2 weeks	Interval: 14 days
QM	—	Every month / monthly	Frequency: 1 per month
STAT	*statim*	Immediately / now	One-time, urgent

3.2 Extended Frequency / Interval Codes

Code	Meaning	Notes
QAM	Every morning	Paired with QPM
QPM	Every evening	—
QH	Every hour	—

Code	Meaning	Notes
Q2H	Every 2 hours	Interval: 2 hours
Q3H	Every 3 hours	Interval: 3 hours
Q4H	Every 4 hours	Common for acute pain
Q6H	Every 6 hours	Interval: 6 hours
Q8H	Every 8 hours	Interval: 8 hours
Q12H	Every 12 hours	Interval: 12 hours
Q24H	Every 24 hours	Functionally equivalent to QD
Q2-3H	Every 2–3 hours	Variable interval
Q4-6H	Every 4–6 hours	Variable interval, common PRN
Q2D	Every 2 days	= QOD
Q3D	Every 3 days	Interval: 72 hours
Q4W	Every 4 weeks	Common for injectables
5X/D	Five times a day	e.g., antiviral eye drops
AC	Before meals	Administration timing
PC	After meals	Administration timing
ATC	Around the clock	Scheduled, no PRN

Sources: Denali Rx Sig Abbreviations; PharmaClik Rx SIG Codes PDF; Pharmacy Sig Codes Quia reference

3.3 ISMP Error-Prone Abbreviations (Do Not Use)

The Institute for Safe Medication Practices (ISMP) maintains a list of abbreviations that should NOT be used due to misinterpretation risk. Updated April 2024:

Problematic	Reason	Preferred
QD (once daily)	Can be misread as QID (4x/day)	Write "daily" or "once daily"
HS (at bedtime)	Can be confused with "half-strength"	Use **QHS**
U (units)	Mistaken for "0"	Write "units"
µg (micrograms)	Confused with mg	Write "mcg"
nanog (nanograms)	Abbreviation error-prone	Spell out "nanograms"
NAS (intranasal)	Error-prone	Spell out "intranasal"
/ (slash for "per")	Misread as "1"	Use "and" or "per"

Problematic	Reason	Preferred
nightly / **HS**	Confused with other meanings	Use **QHS**

Source: ISMP 2024 Error-Prone Abbreviations update; PSQH summary

3.4 How Frequency Codes Map to Structured Data (NCPDP S&C SIG)

In the NCPDP S&C SIG Timing Segment, frequency is captured as:

- **Frequency Numeric Value** — the number (e.g., "2" for BID)
- **Frequency Units** — the time period (e.g., "day" for per day)
- **Variable Frequency Modifier** — for ranges (e.g., "3 to 4 times per day")
- **Interval Numeric Value + Interval Units** — for every-X-hours patterns (e.g., "8" + "hours" for Q8H)
- **Administration Timing** — for event-based timing (e.g., "bedtime" for QHS, "with meals" for AC)

The separation of frequency (events-per-unit-time) from interval (time-between-events) allows both patterns to be captured, though in practice the JAMIA 2022 study found only **33% of administration timing terms** in real prescriptions could be mapped to SNOMED CT codes.

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4. Complex Sig Handling

4.1 The Challenge of Multi-Phase and Titration Sigs

Modern pharmacy practice frequently requires complex sigs that encode:

- **Titration schedules** — dose escalation over defined intervals
- **Conditional dose changes** — "increase if tolerated"
- **Multi-phase therapy** — distinct phases with different doses/frequencies
- **Site rotation** — instructions for topical or injectable medications

These patterns are extremely common in compounded medications, particularly:

- GLP-1 receptor agonist compounds (semaglutide, tirzepatide) — requiring weekly dose escalation
- Hormone replacement therapy (HRT) — e.g., topical gels with rotating sites
- Pain management compounds — with PRN and scheduled components
- Titration-sensitive medications — ketamine, naltrexone LDN formulations

4.2 Example Complex Sigs and Structural Analysis

Example 1: GLP-1 Titration

""Inject 0.25mg SubQ weekly for 4 weeks, then increase to 0.5mg weekly""

- NCPDP handling: Requires **two Repeating Sig segments** connected by the "THEN" Multiple Sig Modifier
- Phase 1: Dose=0.25mg, Route=SubQ, Frequency=1/week, Duration=4 weeks
- Phase 2: Dose=0.5mg, Route=SubQ, Frequency=1/week, Duration=ongoing

• Challenge: The duration-based transition trigger and the dose change are representable in theory, but real-world S&C SIG adoption makes this almost always a free-text sig

Example 2: Oral Liquid Titration

""Take 0.1mL by mouth nightly, titrate up by 0.1mL every 3–4 days as tolerated up to 0.5mL""

• Requires: Starting dose, titration increment, variable interval, conditional modifier, maximum dose restriction

• NCPDP representation: The Maximum Dose Restriction segment captures the 0.5mL ceiling; the titration instruction ("increase by 0.1mL every 3–4 days as tolerated") has **no native structured representation** and falls into free text

• The 2011 JAMIA study identified "Increase dose quantity or frequency if tolerated" as a sig concept that was not representable, appearing in ~0.06% of study sigs

Example 3: Topical with Site Rotation

""Apply 1 click to inner forearm daily, rotate sites""

• Route/Site: Topical to forearm (captured in Site of Administration segment)

• "Rotate sites" instruction: **Not representable** in S&C SIG structured fields — must be free text

• Dose form "click" (pump actuations): May require custom mapping

Example 4: Extended Titration Protocol

""Take 0.1mL nightly weeks 1–2, then 0.2mL nightly weeks 3–4, then 0.3mL nightly thereafter""

• Requires 3 Repeating Sig segments with THEN modifiers

• Each segment needs: Dose + Duration

• In theory representable; in practice, the temporal linkage and conditional escalation push most prescribers to free text

4.3 NCPDP's Mechanism for Complex Sigs: Repeating Sig Segments

The S&C SIG format handles multi-step instructions through the **Repeating Sig Segment** with a **Multiple Sig Modifier** field that accepts:

• **THEN** — sequential phases ("take X for 3 days, THEN take Y for 3 days")

• **AND** — concurrent instructions ("take X in morning AND Y at bedtime")

• Numeric ranges within a single segment ("take 1–2 tablets")

Key limitation identified in research: The need to repeat the entire Sig segment for compound expressions creates ambiguity when combining numeric range repeats with multi-part AND/THEN instructions. There are no grouping/parenthetical rules, making complex titrations difficult to represent unambiguously.

Sources: JAMIA Evaluation of NCPDP S&C Sig Format, 2011; ADL Data Systems - ABCs of SCRIPT Standard

4.4 Do Any Systems Support Structured Titration Sigs?

Current state: No mainstream e-prescribing system fully structures titration/multi-phase sigs as discrete machine-readable data in a standardized way. The approaches observed in practice:

1. **Free text fallback (most common):** Complex titration sigs are entered as 1,000-character free text. The prescriber writes the full protocol in plain English. DoseSpot, Epic, and most EHRs handle complex compound sigs this way.
2. **Template-based "pre-built sigs":** Some compounding pharmacy systems (e.g., SiCompounding) allow prescribers to create **templates** that can contain one or multiple prescriptions, enabling reusable titration protocols as stored text templates. (SiCompounding Release Notes)
3. **Repeating S&C SIG (theoretical):** NCPDP's standard technically supports multi-phase sigs through repeating segments, but this is rarely implemented by EHR vendors due to interface complexity.
4. **Pharmacist Note field:** For sigs that exceed character limits or contain instructions requiring pharmacist interpretation (not patient-facing), the NCPDP SCRIPT standard provides a separate Notes field.
5. **Oncology compounding software (BD Cato, etc.):** Specialty compounding workflow software for IV/oncology medications handles multi-phase regimens through standardized protocol templates that calculate doses automatically from weight/BSA — but these are internal compounding workflow tools, not prescribing sig builders. (JMIR Human Factors evaluation of BD Cato workflow software)

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5. Sig Component Databases

5.1 First Databank (FDB) MedKnowledge

What it is: FDB is one of the two dominant drug database vendors in the U.S. FDB MedKnowledge is integrated into thousands of healthcare IT systems including EHRs, pharmacy management systems, and e-prescribing platforms.

Relevance to sigs:

- Provides **compound-drug clinical screening** — enables drug-drug interaction and contraindication screening for multi-ingredient compounded drugs at both the compound level and the individual ingredient level
- Provides **drug product nomenclature** including routes, dose forms, and strength descriptors that feed sig builder dropdown menus
- Offers **min/max dose modules** supporting dosing range validation in sig builders
- Drug vocabulary includes cross-references to NLM RxNorm
- Does **not** assign NDC codes to compounded medications (compounds are not FDA-registered products)
- NDC codes from FDB have been integrated into RxNorm since 2011

Sources: FDB MedKnowledge Drug Database; FDB MedKnowledge Framework Datasheet; First Databank Wikipedia

5.2 Medi-Span (Wolters Kluwer)

What it is: Medi-Span is the other major drug database provider, owned by Wolters Kluwer. Its primary classification system is the **Generic Product Identifier (GPI)**.

The GPI System:

- 14-character hierarchical classification system
- 7 levels of granularity: drug group → class → subclass → base → name → dose form → dose strength
- Unlike NDC (static number), GPI characters encode clinical meaning at each level
- Enables grouping, sorting, searching, and matching across drug products
- Used in retail pharmacy systems for finding all brand/generic options for a prescribed product
- Used by PBMs and payers for formulary development

Relevance to compounded medications and sigs:

- GPI can identify drugs at any level of the hierarchy — this allows matching compounded drugs by ingredient/class even without an NDC
- Supports medication reconciliation by grouping drugs with the "same ingredient, different sigs"
- Helps overcome different medication spellings from different providers/pharmacies

Limitation for compounded medications: GPI is designed around marketed drug products. Custom compounded formulations (unique strength/base combinations) often lack GPI entries and require manual entry or custom codes within the dispensing system.

Sources: Medi-Span GPI page (Wolters Kluwer); PAAS National - Medi-Span GPI overview

5.3 RxNorm

What it is: RxNorm is a standardized drug nomenclature system maintained by the National Library of Medicine (NLM). It assigns **RxCUI** (RxNorm Concept Unique Identifiers) to drug concepts defined by combinations of: active ingredient(s) + strength + dose form.

Key RxNorm concept types:

Term Type	Description	Example
IN	Ingredient	Metformin
SCD	Semantic Clinical Drug	Metformin 500 mg Oral Tablet
SBD	Semantic Branded Drug	Metformin 500 mg Oral Tablet [Glucophage]
SCDF	Semantic Clinical Drug Form	Metformin Oral Tablet
GPCK	Generic Pack	Metformin 500 mg Tablet [100 per pack]
BPCK	Branded Pack	Metformin 500 mg Tablet [Glucophage, 100 per pack]

In NCPDP SCRIPT: The standard allows either an NDC in the "Drug Product Code" field or an RxNorm RxCUI in the "Drug Database Code" field. In practice, most pharmacy workflows use NDC codes for dispensing/billing but may carry RxCUI for clinical decision support.

Compounded medications and RxNorm:

- Compounded medications typically **do not have RxCUI codes** because they are not FDA-registered finished drug products

- RxNorm contains only "clinically significant active ingredients" of marketed drugs
- A 2011 JAMIA paper proposed that "linking compounded products to Unique Ingredient Identifier (UNII) codes in the FDA Substance Registration System to create an indirect mapping to RxNorm at the ingredient level" as a workaround (Mapping Partners Master Drug Dictionary to RxNorm)
- In practice, e-prescribing systems transmit compound ingredients using individual ingredient-level RxCUIs (or NDCs for the base ingredients if commercially available) rather than a single compound-level code

NCPDP's position: NCPDP strongly opposes replacing NDC with RxNorm for pharmacy transactions, noting that "RxNorm lacks specificity and rarely maps to an NDC in a one-to-one relationship" and that "pharmacy claims systems are specifically coded to utilize [NDC] identifiers." (NCPDP Letter to ONC on RxNorm, 2021)

Sources: RxNorm Technical Documentation (NLM); NDC vs RxNorm Guide (Intuition Labs, 2025)

5.4 Compounded Medication Coding — No NDC Problem

This is one of the fundamental challenges in building structured sigs for compounded medications:

The core problem:

- Standard medications are identified in pharmacy systems by NDC number
- NDC numbers are issued by FDA only to registered finished drug products
- Compounded medications are **custom preparations** — they have no NDC
- Without an NDC, the standard auto-matching and drug identification workflows in pharmacy systems fail

Coding approaches used in practice:

Approach	Description	Used By
Ingredient-level NDCs	Use NDC codes for individual bulk drug ingredients; send compound as list of ingredients with quantities	NCPDP SCRIPT 2017071 compound support
HCPCS miscellaneous J-codes	J3490 (unspecified drug, non-oral), J3590 (unspecified biologics), J7999 (compounded drug)	Medicare billing for compounded drugs
Custom local drug codes	Pharmacy creates internal database entry with custom code	Most compounding pharmacy management systems (PK Software, SiCompounding)
UNII codes	FDA Substance Registration System Unique Ingredient Identifiers for bulk chemicals	Research systems; not widely adopted in e-prescribing
Pharmacy-specific compound databases	Compounding pharmacy maintains a formulary database of their standard compounds with custom IDs	Specialty compounding pharmacies

Source: Retinal Physician - Coding Q&A on Compounded Medications, 2021

6. Sig Building UI Patterns

6.1 Overview of Approaches

EHR and e-prescribing systems have evolved several UI patterns for sig construction. The approaches exist on a spectrum from fully structured to fully free text:

Pattern	Description	Advantages	Limitations
Pure free text	Single text field for entire sig	Maximum flexibility; no constraints	Error-prone; no structured data; downstream rekeying burden
Structured sig builder (dropdown/pick-list)	Series of dropdowns for each sig component	Standardized output; reduced keystrokes	May not accommodate complex sigs; SNOMED CT vocabulary gaps
Hybrid (builder + free text)	Dropdowns for standard components + optional free text appendage	Balances standardization and flexibility	Free text may contradict structured fields
Template-based (pre-built sigs)	Library of complete sig strings selectable by prescriber	Fast; consistent; reduces variation	Limited to pre-built options; customization requires manual edit
Auto-complete / NLP-assisted	Free text with intelligent suggestions based on drug/frequency patterns	Natural workflow; faster than dropdowns	Quality depends on NLP accuracy; may still produce free text output

Source: JAMIA Quality and Variability of Patient Directions, 2018

6.2 Structured Sig Builder Design (Dropdown-Based)

Most ePrescribing-enabled EHRs provide Sig builders that work as follows:

1. **Sequential component selection:** Prescriber selects options from dropdowns in order: action verb → dose amount → dose units → route → site (if applicable) → frequency → duration → auxiliary instructions
2. **Auto-concatenation:** The system concatenates selected values into a free text string displayed to the prescriber for review
3. **Favorites/recently used:** Sig builders typically allow adding frequently used sigs to a "favorites" list for one-click retrieval
4. **Hidden structured data:** In systems with S&C SIG support, the dropdown selections simultaneously populate both the free text field AND the structured SNOMED CT-coded fields — but the prescriber only sees the text output

The critical gap noted by experts: Even though prescribers create sigs using individual component fields (dose, route, frequency), most EHRs only populate the free text Directions field from those selections — not the separate S&C SIG structured fields. Vendors must create "behind-the-scenes tables" to map sig elements to S&C SIG codes, which is costly and has low prioritization due to lack of mandate and user demand. (LinkedIn - Tony Schueth on Structured Sig, 2016)

6.3 Template-Based Sig Building

Template-based approaches are particularly valuable for compounding pharmacies:

In compounding pharmacy systems (e.g., SiCompounding):

- Prescribers can create **prescription templates** containing one or multiple prescriptions
- Templates encode complete sig text including complex titration instructions
- Reused for patients on the same protocol without re-entering the sig
- Reduces transcription errors for complex compounded medications

In EHR systems (Epic, Cerner, etc.):

- "Preference lists" or "order sets" contain pre-built medication orders with standard sigs
- For common compounded formulations (e.g., standard HRT protocols), these templates provide consistent sig text
- Smart text / dot phrases allow rapid insertion of standard sig text

Source: SiCompounding Release Notes

6.4 Validation for Completeness

Surescripts/NCPDP guidance (Surescripts Best Practices via ChartMaker) specifies that complete sigs must include all of:

1. **Action** verb (take, inject, apply, instill, inhale)
2. **Dose** quantity
3. **Dose units**
4. **Route** of administration
5. **Frequency**
6. **Auxiliary information** (duration, indication, PRN qualifier)

Required element order: action → dose → dose units → route → frequency → auxiliary

Recommended UI validation rules:

- If prescriber selects PRN frequency, **require** entry of indication/condition before allowing submission
- If prescriber selects "take as directed," **require** specification of the direction source (e.g., "as per package insert," "as per prescriber")
- **Expand Latin abbreviations** to plain English automatically (e.g., convert "BID" to "twice daily" in the displayed sig)
- **Do not allow truncation** across multiple fields — if the sig exceeds character limits, flag it rather than silently truncating
- For acute treatments, **require** duration of therapy entry

Common sig deficiencies identified in research:

- Missing dose/dose form (especially for topical products): Topical sigs had the highest rate of quality-related events (QREs) in e-prescriptions — lacking explicit dose amounts and site specifications (JAMIA Quality and Variability study, 2018)
- Missing indication for PRN medications

- Conflicting information between free text and structured fields
- Abbreviations that could be misinterpreted (QD misread as QID)

6.5 DoseSpot Compound Prescribing Workflow

DoseSpot (a major e-prescribing platform used by many telehealth and specialty practices) provides a representative example of compound prescribing UI:

1. Prescriber clicks **"Compound"** button in the prescription entry interface
2. **Search for each ingredient** and select from database
3. Specify **strength and dispense unit** for each ingredient
4. Add ingredients one at a time using a "+" button
5. **Name the compound** and click "Build Compound"
6. System displays a standard prescription details screen — the sig is entered in the same fields as any standard medication (with pharmacist notes section for complex instructions)
7. Submit as standard e-prescription

Limitation: The sig entry step is effectively free text for complex compounds. There is no compound-specific structured sig builder for titration schedules.

Sources: DoseSpot Prescribing Compound Medications guide (via Healthie); DoseSpot ePrescribing Process guide

6.6 PK Software (PCCA's Compounder Rx) — Dedicated Compounding Software

For pharmacies (not prescribers), PK Software's "The Compounder Rx" is one of the dominant dedicated compounding pharmacy management systems. It:

- Tracks patients, prescriptions, formulas, chemicals/ingredients, quality assurance, clinical screening
- Provides clinical drug screening (drug-drug interactions, contraindications) at the ingredient level for compounded products
- Supports OBRA 90 compliance for compounded products
- Integrates barcode scanning with scales for gravimetric verification during compounding
- Does **not** provide a prescriber-facing structured sig builder — it receives prescriptions and processes them

Source: PK Software Products

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7. Summary and Implications

7.1 Key Findings Summary

Topic	Key Finding
NCPDP S&C SIG adoption	Only 10–32% of e-prescriptions use any structured sig fields; free text dominates

Topic	Key Finding
Compound sigs in SCRIPT 2017071	Supported (up to 25 ingredients, 1,000-char sig) but sig itself is free text
Titration/multi-phase sigs	No standardized structured representation; NCPDP Repeating Sig segments theoretically work but are rarely implemented; free text is universal
Frequency code standardization	Codes (QD, BID, etc.) are widely standardized but ISMP discourages many Latin abbreviations; SNOMED CT vocabulary coverage of timing terms is only 33%
Compounded med coding	No NDC; no RxCUI; use ingredient-level codes + custom local identifiers or HCPCS J-codes for billing
FDB / Medi-Span role	Provide drug vocabulary, dose form, route, and interaction data that feed sig builders; neither assigns codes to custom compounds
Sig builder UI	Dropdown builders are standard in EHRs but populate only free text, not structured fields; template-based sigs are the best practical approach for complex compounds
Validation	Completeness validation is UI-enforced (require action+dose+route+frequency+PRN indication) but structured data validation is largely absent
Surescripts Sig IQ	ML-based post-transmission conversion of free text → structured sig; augments 12.7% of transactions in 2025

7.2 The Practical Reality for Compounded Medication Sigs

Building a system that handles compounded medication sigs must confront:

1. **The free text reality:** Despite the existence of structured sig standards, the vast majority of compounded medication sigs — especially titration protocols — are and will be transmitted as free text. Any system must be capable of parsing, validating, and working with free text sigs.
2. **No drug code for compounds:** Systems cannot rely on NDC or RxCUI to identify compounded medications. Drug identity must be derived from ingredient names and strengths. This creates interoperability friction.
3. **Titration sigs are unsupported by current standards:** There is no standardized machine-readable format for expressing "increase dose by X every Y days up to Z." This must either be: (a) free text that humans parse, (b) a proprietary template system, or (c) a custom multi-phase sig data structure built outside the NCPDP standard.
4. **Character limits matter for complex protocols:** Even at 1,000 characters (SCRIPT 2017071), complex multi-month titration protocols may exceed limits. Systems should validate and warn.
5. **Sig quality is a documented patient safety problem:** Studies consistently show 10–16% of e-prescription sigs contain quality issues. Validation rules, template libraries, and post-transmission NLP (like Sig IQ) are the industry's responses.

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Section 3: Medication Configuration Data Models for Compounding Pharmacies

Date: April 7, 2026

Scope: How compounding pharmacy databases structure medication data — covering data representation, 503A vs 503B catalog structures, data standards, formulary design patterns, and multi-ingredient compound modeling.

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1. **Compounded Medication Data Representation**
2. **503A vs 503B Catalog Structure**
3. **Data Standards for Compounded Medications**
4. **Formulary Database Design Patterns**
5. **Multi-Ingredient Compound Modeling**
6. **Summary: Key Design Implications**

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1. Compounded Medication Data Representation

1.1 Core Conceptual Structure

A compounded medication is fundamentally a *recipe* — a set of ingredients in specified amounts, combined via a prescribed process, producing a final preparation with a defined dosage form, strength, and route of administration. Unlike a commercial NDC product (which is a fixed, pre-approved item), a compounded medication is a *configurable* combination whose representation must capture:

Dimension	Examples
Active ingredient(s)	Testosterone Cypionate, Anastrozole, Semaglutide
Salt / ester form	Testosterone *cypionate* vs. testosterone *propionate*; prazosin *HCl* vs. prazosin base
Concentration / strength	200 mg/mL, 0.5 mg/mL, 2.5%, 10 mg/g
Dosage form	Injectable solution, cream, gel, troche, capsule, sublingual drop, nasal spray, suppository, pellet
Route of administration	IM, SubQ, topical, oral, sublingual, intranasal, intravaginal, rectal
Total quantity	10 mL vial, 30 g tube, 60 capsules
Beyond-use date	Date/time after which the preparation must not be used

Dimension	Examples
Inactive ingredients / base	PLO gel, cottonseed oil, benzyl alcohol, anhydrous SuspendIt

1.2 The Relationship Between Core Attributes

In compounding, these attributes form a dependency chain:

```
Active Ingredient (moiety)
■■ Salt/Ester Form (affects molecular weight and potency factor)
■■ Concentration (mg/mL, %, mg/g — unit depends on dosage form)
■■ Dosage Form (solution, cream, troche, etc.)
■■ Route of Administration (IM requires injectable; topical requires non-oral form)
■■ Total Quantity (dispensed volume or weight)
■■ BUD (depends on sterility category, ingredients, storage)
```

Salt form is a critical data point, not merely cosmetic naming. PCCA's documentation on API calculations (PCCA API Calculations guide) shows that salt form determines the actual weight of chemical needed:

- 1 mg amlodipine (base) = 1.39 mg amlodipine *besylate* (MW factor 567/409)
- 1 mg prazosin (base) = 1.095 mg prazosin *HCl* (salt factor)
- 1 mg estradiol (base) = 1.033 mg estradiol *hemihydrate* (fixed conversion factor)

Accordingly, a well-designed database must store both the **active moiety** (therapeutic reference) and the **precise ingredient/salt form** (what is physically weighed). This mirrors the FDA's naming policy for salt-containing drugs (FDA Salt Naming Guidance), which distinguishes between:

- `active_moiety` — the pharmacologically active molecule (e.g., "amlodipine")
- `precise_ingredient` — the specific form used (e.g., "amlodipine besylate")
- `strength_moiety` — amount expressed in terms of moiety (e.g., 1 mg amlodipine)
- `strength_salt` — equivalent salt strength (e.g., 1.39 mg besylate)
- `conversion_factor` — molecular weight ratio or lot-specific COA factor

1.3 Standard Entity-Relationship Model

No single universally adopted ER standard exists for compounding pharmacy databases, but the dominant patterns converge on a **Master Formulation Record (MFR)** as the central entity. The MFR is required by both USP \<795\> (nonsterile) and USP \<797\> (sterile) (USP \<795\> MFR Requirements, University of Illinois):

Core entities and relationships:

```
FORMULATION (Master Formulation Record)
■■ formula_id (PK)
■■ name
■■ strength_display (e.g., "200 mg/mL")
■■ dosage_form_id (FK → DOSAGE_FORM)
■■ route_id (FK → ROUTE)
■■ total_quantity
■■ total_quantity_unit
■■ bud_hours_room_temp
■■ bud_hours_refrigerated
■■ bud_hours_frozen
■■ bud_basis (USP default / stability study / bracketed)
■■ version
```

```
■■ created_at
■■ updated_at
■■ [References: stability data, USP chapter, source formulation]

FORMULATION_INGREDIENT (junction table — one row per ingredient per formulation)
■■ formulation_ingredient_id (PK)
■■ formula_id (FK → FORMULATION)
■■ ingredient_id (FK → INGREDIENT)
■■ is_active (boolean)
■■ role (active / base / preservative / diluent / vehicle / excipient)
■■ strength_numerator_value
■■ strength_numerator_unit (mg, g, mL, %)
■■ strength_denominator_value
■■ strength_denominator_unit (mL, g, capsule)
■■ quantity_per_batch
■■ quantity_unit
■■ sort_order

INGREDIENT (master table)
■■ ingredient_id (PK)
■■ name (e.g., "Testosterone Cypionate")
■■ moiety_name (e.g., "Testosterone")
■■ rxnorm_cui (RxCUI for the ingredient)
■■ rxnorm_pin_cui (RxCUI for precise ingredient / salt)
■■ unii (FDA Unique Ingredient Identifier)
■■ cas_number (Chemical Abstracts Service)
■■ salt_form (e.g., "cypionate", "hydrochloride", "besylate")
■■ molecular_weight
■■ conversion_factor (moiety → salt)
■■ is_controlled (boolean, DEA schedule)
■■ dea_schedule
■■ is_hazardous (NIOSH designation)
■■ usp_monograph_ref
■■ ndc_source (for 503B reporting: source NDC of the API)

DOSAGE_FORM
■■ dosage_form_id (PK)
■■ name (e.g., "Injectable Solution")
■■ code (SNOMED CT / HL7 / FDA SPL code)
■■ is_sterile (boolean)
■■ calculation_method (weight-based / volume-based / % w/v / % w/w)
■■ default_unit
■■ usp_chapter (795 or 797)

ROUTE_OF_ADMINISTRATION
■■ route_id (PK)
■■ name (e.g., "Intramuscular")
■■ code (SNOMED or FDA SPL route code)
■■ abbreviation (IM, SubQ, PO, SL, IN, TOP, IVag, PR)
```

1.4 How Multi-Ingredient Compounds Are Handled

The **Testosterone Cypionate 200 mg/mL + Anastrozole 0.5 mg/mL** combination in a single vial is the canonical example of a multi-ingredient injectable compound. This is a well-documented clinical combination (patent US10201549B2, Google Patents; University of Maryland testosterone cypionate review), and Vios Compounding offers it commercially.

In a database, this compound is modeled with **one FORMULATION row** and **multiple rows in FORMULATION_INGREDIENT**:

```
FORMULATION row:
formula_id: F001
name: "Testosterone Cypionate / Anastrozole Injectable"
dosage_form: "Solution for Injection"
route: "Intramuscular"
total_quantity: 10 mL
bud: per USP <797> or stability study

FORMULATION_INGREDIENT rows:
[1] formula_id: F001, ingredient: "Testosterone Cypionate"
is_active: true, role: "active"
strength: 200 mg / 1 mL

[2] formula_id: F001, ingredient: "Anastrozole"
is_active: true, role: "active"
strength: 0.5 mg / 1 mL

[3] formula_id: F001, ingredient: "Benzyl Alcohol"
is_active: false, role: "preservative"
strength: 9 mg / 1 mL (0.9%)

[4] formula_id: F001, ingredient: "Cottonseed Oil"
is_active: false, role: "vehicle"
strength: qs to 1 mL
```

This **junction table pattern** is the standard approach confirmed across multiple implementations:

- HL7 FHIR R6 Medication resource uses `ingredient[]` as a repeating BackboneElement
- Medplum's formulary guide uses `MedicationKnowledge.ingredient[]` with `isActive` and `strength` per ingredient
- Back4app's compound pharmacy template uses an `ingredients: Array>` in MedicationRecipe

---

2. 503A vs 503B Catalog Structure

2.1 Regulatory Distinction

The Drug Quality and Security Act (DQSA) of 2013 created two distinct categories of compounding pharmacies with fundamentally different operational and database models (NCBI/National Academies, Regulatory Framework for Compounded Preparations):

Feature	503A Pharmacy	503B Outsourcing Facility
Regulatory oversight	State Board of Pharmacy	FDA (federal)
Prescription requirement	Required (patient-specific)	Not required (may compound for "office use")
Batch size	Small/individual; limited anticipatory compounding	Large batch production

Feature	503A Pharmacy	503B Outsourcing Facility
Distribution	Patient-specific; home use	Hospitals, clinics, surgical centers
Manufacturing standard	USP \<795\>, \<797\>, \<800\>	FDA cGMP
FDA registration	Not required	Required (voluntary)
Product reporting	Not required	Biannual to FDA (June and December)
NDC numbers	Not applicable / cannot assign	Can assign (encouraged); reported to NDC Directory
Beyond-use dating	Shorter (USP default or extended by study)	Longer (cGMP + stability testing = expiration dates)
BUD determination	Professional judgment + USP guidelines	Rigorous stability testing per cGMP

Sources: SCA Pharma, DrugPatentWatch, MediVera, TechTarget

2.2 503A Catalog Structure: Patient-Specific Formulations

A 503A pharmacy's "catalog" is not a catalog in the traditional sense — it is a **library of Master Formulation Records** that pharmacists can draw from to fulfill patient-specific prescriptions.

Key characteristics of 503A data structure:

1. **Formulation library** (MFR database): Each unique formulation has an MFR describing ingredients, quantities, processes, BUD, and quality controls. PCCA provides members with over 9,600 tested formulations (PCCA Formulary), and CompoundingToday's database contains over 3,600 formulas (CompoundingToday).
2. **No fixed SKUs**: Unlike 503B, a 503A product is not pre-manufactured. The "product" is the MFR + the resulting prescription-specific preparation. Each preparation gets a unique Compounding Record (CR) linked to the MFR.
3. **High customization**: Prescribers can specify any concentration, any base, any dosage form. The MFR must accommodate variable strengths (e.g., PCCA's "bracketed" FormulaPlus formulas cover a range of API concentrations with a single stability study).
4. **Prescription linkage**: The data model must link Formulation → Compounding Record → Prescription → Patient. Each CR records lot numbers and expiration dates of every ingredient used.
5. **No NDC codes**: 503A compounds are patient-specific and cannot be assigned NDC numbers. Identification uses prescription/RX numbers and internal compound lot numbers.

503A catalog data model summary:

```
503A Data Hierarchy:
MFR (template)
→ Prescription (patient-specific order)
→ Compounding Record (actual batch preparation)
→ Label (RX number, ingredients, BUD, storage)
```

2.3 503B Catalog Structure: Pre-Made Batches / Fixed Products

503B outsourcing facilities produce standardized batch products that are more akin to commercial pharmaceutical products. Their catalog structure is fundamentally different.

Key characteristics of 503B data structure:

1. **Fixed product catalog with NDC-like identifiers**: 503B facilities are strongly encouraged (though not required) to assign NDC numbers to their compounded products (FDA 503B Product Reporting guidance, FDA CDER Direct 503B Product Reporting). When assigned, an NDC has the standard 10-digit, 3-segment format (labeler-product-package).
2. **Biannual product reporting to FDA**: Every 503B must report all compounded products to FDA in June and December. Required fields per FDA include:
   - Active ingredient(s) and strength per unit
   - Source NDC of the active ingredient (bulk API or finished drug)
   - Dosage form
   - Route of administration
   - Package description
   - Number of individual units produced
   - NDC of the final product (if assigned)
3. **Marketing category**: 503B products in the NDC Directory carry the category "Outsourcing Facility Compounded Human Drug Product (Exempt from Approval Requirements)."
4. **Fixed SKU structure**: Once an NDC is assigned to a 503B product, the active ingredient, strength, dosage form, and route cannot change without a new NDC (FDA CDER Direct tutorial).
5. **Multi-ingredient reporting per ingredient**: For a compound with multiple active ingredients, each ingredient is reported separately (each with its own source NDC). A product with two active ingredients appears with two ingredient entries.
6. **Expiration dates, not BUDs**: 503B uses true stability-tested expiration dates (like commercial drugs), not the conservative USP-default BUDs of 503A (Olympia Pharmaceuticals, Epicur Pharma BUD vs Expiration Date).

503B catalog data model summary:

```
503B Data Hierarchy:
Product Catalog Entry (NDC assigned)
■■ Product NDC (10-digit)
■■ Nonproprietary Name
■■ Active Ingredient(s) [one row per ingredient in FDA reporting]
■■ Dosage Form
■■ Route
■■ Package Size / Description
■■ Expiration Date (stability tested)
■■ Lot Number
→ Batch Production Record (cGMP)
→ Distribution Record (shipped to healthcare facility X)
```

2.4 Database Design Implications

Design Dimension	503A	503B
Product identity	RX number + MFR reference	NDC code (optional but standard)

Design Dimension	503A	503B
Catalog type	MFR library (templates)	Fixed SKU catalog
Prescription linkage	Mandatory (1:1 patient)	Optional (1:many facilities)
Batch tracking	Per-prescription CR	Batch lot numbers with full cGMP records
Strength variability	High (any strength per Rx)	Low (fixed per batch)
BUD/expiration	Calculated per BUD rules; shorter	Stability-tested; longer
Reporting	State board only	FDA biannual + NDC directory
Ingredient source tracking	Lot/expiry per CR	Source NDC required in FDA reporting

3. Data Standards for Compounded Medications

3.1 NDC Codes and 503A Compounds

503A pharmacies cannot assign NDC numbers. The NDC system is designed for commercially manufactured drugs registered under FDA Sections 510/505. A 503A compound is patient-specific and exempt from FDA approval requirements; it receives no NDC (NCBI regulatory framework).

Instead, 503A pharmacies identify compounds using:

- **Internal prescription (RX) number** — unique per patient-prescription
- **Internal compound lot/batch number** — tied to the Compounding Record
- **MFR formula reference number** — links to the Master Formulation Record

In e-prescribing systems (e.g., Tebra EMR), compounded prescriptions are often transmitted by selecting a commercially available drug as a placeholder to trigger DEA controls, with the actual compound described in the Sig/Notes field (Coastal Pharmacy prescribing guide).

3.2 503B Outsourcing Facilities and NDC Numbers

503B facilities can and should assign NDC numbers to their compounded products, though it is not technically required. FDA has:

- Created a marketing category: "Outsourcing Facility Compounded Human Drug Product (Exempt from Approval Requirements)"
- Added a searchable "Compounded Products" tab to the NDC Directory
- Encouraged assignment of NDCs to all compounded products to ensure traceability (FDA 503B NDC guidance)

When assigned, 503B NDCs:

- Use the same 10-digit, 3-segment labeler-product-package format
- Do **not** constitute FDA drug approval

- Are listed in the NDC Directory with active ingredient(s), dosage form, route, package description, and reporting period
- Must remain stable: same active ingredient and strength must be maintained for the same NDC

3.3 RxNorm as the Standard for Ingredient Coding

RxNorm, maintained by the National Library of Medicine, is the de facto standard for medication coding in U.S. healthcare systems (RxNorm Overview, NLM; RxNorm Term Types, NLM). Its term type hierarchy is directly applicable to compounding database design:

RxNorm TTY	Name	Description	Example
IN	Ingredient	Active moiety / USAN name	Testosterone
PIN	Precise Ingredient	Salt or isomer form	Testosterone Cypionate
MIN	Multiple Ingredients	Two or more ingredients in one preparation	Testosterone / Anastrozole
SCDC	Semantic Clinical Drug Component	Ingredient + Strength	Testosterone 200 MG/ML
SCDF	Semantic Clinical Drug Form	Ingredient + Dose Form	Testosterone Injectable Solution
SCD	Semantic Clinical Drug	Ingredient + Strength + Dose Form	Testosterone 200 MG/ML Injectable Solution
DF	Dose Form	Standalone dose form concept	Injectable Solution

Key application to compounding databases:

- Store `rxnorm_in_cui` (ingredient/moiety) and `rxnorm_pin_cui` (precise ingredient/salt) as separate fields
- Multi-ingredient compounds are identified by `rxnorm_min_cui` (Multiple Ingredients)
- Strength is expressed per UCUM units of measure (mg, mL, g) with numerator/denominator pattern

Medplum's FHIR-based formulary implementation (Medplum formulary docs) uses RxNorm as the preferred code system for each ingredient in `MedicationKnowledge.ingredient[].itemCodeableConcept`.

3.4 PCCA Formulary Database Structure

PCCA (Professional Compounding Centers of America) maintains the industry's largest proprietary compounding formulary with over 9,600 formulations (PCCA formulary). Key structural features:

Formula organization:

- Each formula is a combination of **active pharmaceutical ingredients (APIs)** + **base/vehicle ingredients** + **excipients** in specific quantities
- Formulas are tied to **PCCA-brand bases and ingredients** — substituting a non-PCCA ingredient invalidates the BUD (PCCA Standard)

• Formulas carry a **"BUD Study" designation** when supported by stability-indicating analytical testing per the new USP \<795\>/\<797\> requirements (PCCA FormulaPlus changes)

FormulaPlus™ (bracketed formulas):

• Tested at low and high API concentrations, establishing stability across a *range* of strengths

• Approximately 225 FormulaPlus formulas, with ~20 new studies added annually

• Each study costs up to $50,000 and includes: stability-indicating assay, USP \<51\> antimicrobial effectiveness testing, container-closure testing

Regulatory compliance tracking:

• Formulas are tracked against FDA's 503A Bulk Drug Substances list; when substances are removed (e.g., tranilast in 2019), PCCA removes or flags affected formulas (PCCA blog on FDA ruling)

• Formulas include hazardous drug flags per NIOSH designation and reference USP \<800\> for handling requirements

3.5 USP \<795\> and \<797\>: Data Modeling Implications

USP \<795\> (nonsterile compounding) and **USP \<797\>** (sterile compounding) define the Master Formulation Record (MFR) and Compounding Record (CR) as the core documentation objects. These directly define the data schema for compounding software. As of November 2023, both chapters have been revised with significantly stricter BUD extension requirements.

Required MFR fields per USP \<795\> (University of Illinois USP 795 MFR guide, USP General Chapter \<795\> PDF):

Field	Database Column
Name, strength/activity, dosage form	`formula_name`, `strength_display`, `dosage_form_id`
Identities and amounts of all components	`FORMULATION_INGREDIENT` table
Relevant component characteristics (particle size, salt form, purity grade, solubility)	`ingredient_characteristics` in ingredient table
Calculations to determine/verify quantities and API concentrations	`calculation_notes`, `conversion_factor`
Compatibility and stability information with references	`stability_reference`, `stability_source`
Equipment required	`equipment_list`
Detailed mixing instructions / process steps	`FORMULATION_STEP` table
Sample labeling: generic name, concentration of each API, BUD, storage, RX number	`label_template`
Container used in dispensing	`container_type`

Field	Database Column
Packaging and storage requirements	`storage_conditions`, `container_closure`
Physical description of final preparation	`appearance_description`
Quality control measures and expected results	`qc_procedures`, `qc_expected_results`
Assigned BUD (or BUD extension references)	`bud_hours_room_temp`, `bud_hours_refrigerated`, `bud_hours_frozen`, `bud_reference`

Required Compounding Record (CR) fields — these are the *instance* data tied to each preparation:

Field	Database Column
MFR reference	`mfr_formula_id` (FK)
Names and quantities of all components used	`COMPOUNDING_RECORD_INGREDIENT`
Source, lot numbers, expiration dates of each component	`ingredient_lot_number`, `ingredient_expiry`
Total quantity compounded	`total_quantity_compounded`
Date and time of preparation	`prepared_at`
Assigned prescription/control number	`rx_number`
Assigned BUD	`bud_date`, `bud_time`
Personnel signatures (preparer, QC checker, verifying pharmacist)	`prepared_by`, `qc_by`, `verified_by`
Quality control results (pH, visual inspection, filter integrity)	`qc_results`
Description of final preparation	`final_appearance`
Deviations from MFR	`deviations`

USP \<797\>-specific additions for sterile preparations:

- Sterilization method (autoclave, filter sterilization, terminal sterilization) — `sterilization_method`
- ISO classification of compounding environment — `iso_class`
- CSP category (Category 1, 2, or 3 per revised USP \<797\>) — `csp_category`
- Extended BUD justification: must include documented stability-indicating analytical testing, USP \<51\> AET, container-closure testing (Fagron Academy BUD extension analysis)

---

4. Formulary Database Design Patterns

4.1 Ingredient Master Table

The ingredient master is the foundational reference table. Based on RxNorm structure, USP requirements, and FDA naming conventions, a well-designed ingredient master includes:

```
CREATE TABLE ingredient (
ingredient_id SERIAL PRIMARY KEY,

-- Identity
name VARCHAR(255) NOT NULL, -- "Testosterone Cypionate"
moiety_name VARCHAR(255), -- "Testosterone"
synonyms TEXT[], -- alternate names

-- Coding / cross-references
rxnorm_in_cui VARCHAR(20), -- RxNorm Ingredient CUI
rxnorm_pin_cui VARCHAR(20), -- RxNorm Precise Ingredient CUI
unii VARCHAR(10), -- FDA Unique Ingredient Identifier
cas_number VARCHAR(20), -- Chemical Abstracts Service number
ndc_source VARCHAR(12), -- Source NDC (for 503B reporting)

-- Chemical properties
salt_form VARCHAR(100), -- "cypionate", "hydrochloride", "besylate"
molecular_formula VARCHAR(100), -- "C27H40O3"
molecular_weight DECIMAL(10,4), -- in g/mol
conversion_factor DECIMAL(10,6), -- moiety-to-salt weight factor

-- Regulatory
is_controlled BOOLEAN DEFAULT FALSE,
dea_schedule VARCHAR(10), -- "III", "IV", "V"
is_hazardous_drug BOOLEAN DEFAULT FALSE, -- NIOSH hazardous drug list
niosh_hazard_category VARCHAR(50),
is_bulk_substance BOOLEAN DEFAULT FALSE, -- bulk API vs. finished product
bulk_list_status VARCHAR(50), -- "503A Category 1", "503B approved"

-- Quality
usp_monograph_ref VARCHAR(255),
purity_grade VARCHAR(50), -- "USP", "NF", "pharmaceutical grade"

-- Timestamps
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);
```

Notes on CAS numbers: CAS numbers uniquely identify chemical substances including specific salt forms. Testosterone cypionate has CAS 58-20-8; testosterone propionate has CAS 57-85-2. Storing CAS allows cross-referencing to external chemical databases, detecting duplicates, and linking to safety data.

4.2 Concentration and Strength Representation

The **numerator/denominator ratio pattern** is used universally in pharmacy data standards:

Concentration Type	Expression	Numerator	Denominator
mg/mL (injectable, solution)	200 mg/mL	200 mg	1 mL
% w/v (topical liquid)	2%	2 g	100 mL

Concentration Type	Expression	Numerator	Denominator
% w/w (topical cream/gel)	5%	5 g	100 g
mg/g (topical solid/gel)	10 mg/g	10 mg	1 g
mg/capsule (oral solid)	50 mg	50 mg	1 capsule
mcg/actuation (nasal spray)	50 mcg	50 mcg	1 actuation
mg/mL (sublingual drop)	1 mg/0.1 mL = 10 mg/mL	10 mg	1 mL
mg/pellet (implant)	100 mg	100 mg	1 pellet

FHIR Medication resource (R6) encodes strength as:

```
"ingredient": [{
"item": { "concept": { "coding": [{ "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
"code": "1001", "display": "Testosterone Cypionate" }] } },
"isActive": true,
"strengthRatio": {
"numerator": { "value": 200, "unit": "mg", "system": "http://unitsofmeasure.org", "code": "mg"
},
"denominator": { "value": 1, "unit": "mL", "system": "http://unitsofmeasure.org", "code": "mL"
}
}
}]
```

Source: HL7 FHIR R6 Medication

Unit normalization: Systems should store a `base_concentration_mg_per_mL` or `base_concentration_mg_per_g` as a computed/normalized field for cross-formulation queries and dose calculations, even if the display representation uses %, mg/g, etc.

4.3 Dosage Form Taxonomy

Dosage forms used in compounding span both sterile and non-sterile preparations. Based on FDA SPL terminology and compounding practice:

Category	Dosage Forms	Sterile?
Injectables	Solution for injection, Suspension for injection, Emulsion for injection	Yes
Topical	Cream, Gel, Ointment, Lotion, Foam, Transdermal patch, Spray	No
Oral solid	Capsule, Tablet, Troche/Lozenge, Rapid-dissolve tablet	No

Category	Dosage Forms	Sterile?
Oral liquid	Solution, Suspension, Syrup, Elixir, Emulsion	No
Sublingual	Sublingual tablet, Sublingual drop, Sublingual troche	No
Mucosal	Suppository (rectal/vaginal), Enema, Pessary	No
Inhalation	Nasal spray, Nasal drops, Inhalation solution	No/Yes
Ophthalmic	Eye drops, Eye gel, Eye ointment	Yes
Otic	Ear drops	No
Implant	Pellet	Varies

In Compound Direct's software, dosage forms have associated (Compound Direct dosage forms docs):

- **Calculation method** (weight-based, volume-based, capsule-fill with packing statistics)
- **Default base ingredient** (only ingredients marked as "base" are available)
- **Default devices** (capsule molds, syringes, tubes)
- **Days till expiry** (BUD default for all formulas in that dosage form)
- **Route of administration** (linked at dosage form level)
- **Prefix/suffix** (e.g., "Topical", "Vaginal", "Otic" suffixes)

4.4 Route of Administration Coding

Routes are encoded using standardized code systems. The most common approaches:

SNOMED CT Route Codes (used in FHIR):

- Oral (PO): `26643006`
- Intramuscular (IM): `78421000`
- Subcutaneous (SubQ): `34206005`
- Topical: `6064005`
- Sublingual (SL): `37839007`
- Intranasal: `46713006`
- Intravenous (IV): `47625008`
- Vaginal: `16857009`
- Rectal: `37161004`

FDA SPL Route Codes (used in NDC directory and 503B product reporting) — available at FDA's SPL terminology page.

Abbreviations used in compounding practice: SubQ, IM, IV, PO, SL, TOP, IN, IVag, PR, IO (intraosseous for veterinary)

4.5 Representing a "Configurable Product"

The challenge of 503A compounding is that the same "formulation template" may be used to produce preparations with different strengths on different prescriptions. Two patterns handle this:

Pattern 1: Fixed MFR per exact formulation A separate MFR exists for each unique combination of strength + dosage form. If a prescriber orders testosterone cypionate at 150 mg/mL (vs. the standard 200 mg/mL), a new MFR is created or cloned. This is the strictest interpretation of USP requirements.

Pattern 2: Parent-child (hierarchical) model A parent record defines the formula's ingredient *ratios* (without a fixed absolute strength), and child records represent specific dose variations. This is the approach used in Medplum's FHIR implementation (Medplum formulary docs):

```
PARENT MedicationKnowledge:
name: "Testosterone Cypionate Injectable"
ingredient: [Testosterone Cypionate, Cottonseed Oil, Benzyl Alcohol]
relatedMedicationKnowledge.type: "child" → [child1, child2, child3]

CHILD MedicationKnowledge:
name: "Testosterone Cypionate 100mg/mL 10mL vial"
amount: 10 mL
relatedMedicationKnowledge.type: "parent" → [parent]

CHILD MedicationKnowledge:
name: "Testosterone Cypionate 200mg/mL 10mL vial"
amount: 10 mL
relatedMedicationKnowledge.type: "parent" → [parent]
```

Pattern 3: PCCA bracketed formulas PCCA uses a single FormulaPlus formula with a defined range of API concentrations (e.g., 5-20% for a cream formulation), validated by testing at the low and high ends. The database stores `min_api_concentration` and `max_api_concentration` fields, and the BUD is valid for any concentration within the range.

4.6 Beyond-Use Dating as a Data Field

The BUD is not a single value but a complex, multi-dimensional field derived from USP chapter requirements, sterility category, and storage conditions. Since the November 2023 revision of USP \<795\> and \<797\> (Fagron Academy analysis; Mississippi Board of Pharmacy BUD fact sheet):

For nonsterile preparations (USP \<795\>):

- Nonaqueous: up to 180 days (default) or extended BUD with stability study
- Aqueous: up to 30 days (default) or extended with study
- Extension requires: stability-indicating analytical testing + USP \<51\> AET (for aqueous) + container-closure testing at bracketed concentrations

For sterile preparations (USP \<797\>):

CSP Category	Room Temp	Refrigerated	Frozen
Category 1	12 hours	24 hours	—
Category 2	Per stability data (up to 90 days RT)	Up to 120 days	Up to 180 days

CSP Category	Room Temp	Refrigerated	Frozen
Category 3 (qualified facility)	Up to 90 days	Up to 120 days	Up to 180 days

BUD fields in database:

```
-- In FORMULATION table:
bud_room_temp_hours INTEGER, -- BUD at room temp (CRT)
bud_refrigerated_hours INTEGER, -- BUD at 2-8°C
bud_frozen_hours INTEGER, -- BUD at ≤-20°C
bud_basis VARCHAR(50), -- "USP_default" / "stability_study" / "bracketed"
bud_reference TEXT, -- citation for extended BUD
usp_chapter VARCHAR(10), -- "795" or "797"
csp_category VARCHAR(20), -- "Category1" / "Category2" / "Category3" (797 only)

-- In COMPOUNDING_RECORD table (per-preparation instance):
bud_date DATE,
bud_time TIME, -- for CSPs, BUD includes hour
bud_assigned_by INTEGER -- FK to pharmacist
```

503A vs 503B BUD distinction: 503A preparations have BUDs (conservative, determined from date of compounding). 503B preparations have **expiration dates** determined by comprehensive stability testing — essentially equivalent to commercial pharmaceutical expiration dates (Epicur Pharma BUD vs Expiration).

5. Multi-Ingredient Compound Modeling

5.1 Junction Table Pattern

The standard approach across all major implementations is the **junction (associative) table** pattern, where each row in `FORMULATION_INGREDIENT` represents one ingredient in one formulation:

```
FORMULATION (1) ■■■■■■■■■■■■■■■■ (many) FORMULATION_INGREDIENT (many) ■■■■■■■■■■■■■■■■
(1) INGREDIENT
```

This was validated in:

- HL7 FHIR R6: `Medication.ingredient[]` is an array of BackboneElements, each referencing a substance or medication (FHIR R6 Medication)
- FHIR US Core: `Medication.ingredient[].item` references Substance or Medication, with `isActive` and `strength` per ingredient (US Core Medication Profile)
- Medplum: `MedicationKnowledge.ingredient[]` with per-ingredient `isActive` and `strength` ratio (Medplum formulary)
- Back4app template: `MedicationRecipe.ingredients: Array>` (Back4app compound pharmacy)

Full junction table schema:

```
CREATE TABLE formulation_ingredient (
formulation_ingredient_id SERIAL PRIMARY KEY,
formula_id INTEGER NOT NULL REFERENCES formulation(formula_id),
ingredient_id INTEGER NOT NULL REFERENCES ingredient(ingredient_id),

-- Role classification
is_active BOOLEAN NOT NULL DEFAULT TRUE,
ingredient_role VARCHAR(50), -- "active", "base", "vehicle", "preservative",
```

```
-- "diluent", "excipient", "flavor", "colorant"

-- Strength (numerator/denominator ratio pattern)
strength_numerator_value DECIMAL(12,6),
strength_numerator_unit VARCHAR(20), -- mg, g, mL, mcg, IU, %
strength_denominator_value DECIMAL(12,6),
strength_denominator_unit VARCHAR(20), -- mL, g, capsule, actuation

-- Absolute quantity per batch
quantity_per_batch_value DECIMAL(12,6),
quantity_per_batch_unit VARCHAR(20),

-- Concentration normalization
concentration_mg_per_ml DECIMAL(12,6), -- normalized for queries
concentration_mg_per_g DECIMAL(12,6), -- for solid forms

-- Salt form calculation
expressed_as VARCHAR(50), -- "moiety", "salt", "ester"
potency_factor DECIMAL(10,6), -- COA-specific adjustment

-- Ordering
sort_order INTEGER,

-- QS (quantity sufficient) flag
is_qs BOOLEAN DEFAULT FALSE, -- "add to 100 mL" / "qs ad"

UNIQUE(formula_id, ingredient_id, ingredient_role)
);
```

5.2 Concentration Normalization Across Different Units

When a formula contains multiple ingredients in different concentration units, normalization is essential for:

- Dose calculations
- Compatibility checking
- Cross-formulation comparison

The normalization approach:

Input Expression	Normalization Target	Calculation
200 mg/mL	mg/mL	Direct
2% w/v	mg/mL	2% = 2 g/100 mL = 20 mg/mL
5% w/w	mg/g	5% = 50 mg/g
50 mcg/actuation	mcg/actuation	Direct (no volume normalization)
100 mg/capsule	mg per unit	Direct

Salt form normalization: Strengths are typically expressed in terms of the **active moiety** per FDA/USP naming policy. The physical weight to dispense is calculated as `moiety_strength / conversion_factor`. For example:

- Target: 200 mg testosterone cypionate per mL
- Conversion: testosterone cypionate IS the salt (no moiety conversion needed; cypionate ester is part of the active molecule per FDA naming exception for steroids)
- But for prazosin: target 1 mg/mL prazosin → need 1.095 mg/mL prazosin HCl

5.3 Ingredient Interactions and Compatibility

Compounding databases must handle ingredient incompatibilities, which determine whether ingredients can be combined in the same preparation. This is a distinct concern from drug-drug interaction (which is clinical/pharmacological), focusing instead on **physical and chemical compatibility** (PK Software, rxcmpd.com includes clinical drug screening at ingredient level).

Compatibility data model:

```
CREATE TABLE ingredient_compatibility (
compatibility_id SERIAL PRIMARY KEY,
ingredient_id_1 INTEGER REFERENCES ingredient(ingredient_id),
ingredient_id_2 INTEGER REFERENCES ingredient(ingredient_id),
compatibility_type VARCHAR(50), -- "physical", "chemical", "pH"
result VARCHAR(20), -- "compatible", "incompatible", "conditionally_compatible"
conditions TEXT, -- e.g., "compatible at pH < 6 only"
reference TEXT, -- source citation
CHECK (ingredient_id_1 < ingredient_id_2) -- prevent duplicates
);
```

PCCA's approach: The FormulaPlus program tests specific ingredient combinations for stability and compatibility. The formula database flags whether ingredients other than the specified PCCA-brand components have been validated — substitutions invalidate the tested BUD.

The **testosterone cypionate + anastrozole** combination in cottonseed oil is validated in published literature and patent filings (US10201549B2), but the compatibility is solubility-dependent: anastrozole (slightly water-soluble) must dissolve in the oil vehicle at the target concentration.

5.4 Complete Example: Testosterone Cypionate 200 mg/mL + Anastrozole 0.5 mg/mL in Cottonseed Oil

Formulation record:

Field	Value
formula_id	F-TC-ANAST-001
name	Testosterone Cypionate 200mg/mL + Anastrozole 0.5mg/mL Injectable
dosage_form	Solution for Injection
route	Intramuscular
total_quantity	10 mL (per vial)
usp_chapter	797
csp_category	Category 2

Field	Value
bud_refrigerated_hours	720 (30 days, per USP <797> default)
is_sterile	true
sterility_category	503A or 503B depending on compounder

Formulation ingredient rows:

Ingredient	Role	isActive	Strength	Quantity/10 mL
Testosterone Cypionate	active	true	200 mg/mL	2,000 mg
Anastrozole	active	true	0.5 mg/mL	5 mg
Benzyl Benzoate	preservative/solubilizer	false	0.1 mL/mL (10%)	1 mL
Cottonseed Oil	vehicle	false	qs ad 1 mL	qs to 10 mL

Note on 503B product reporting: For FDA reporting, this product would be reported as two separate ingredient entries (one per active ingredient), each with the source NDC of the API used. The testosterone cypionate would use a bulk testosterone cypionate API NDC; the anastrozole would use a bulk anastrozole API NDC.

6. Summary: Key Design Implications

6.1 The Core Data Model

A compounding pharmacy database requires, at minimum, these interconnected entities:

```
INGREDIENT_MASTER
(ingredient_id, name, moiety_name, rxnorm_cui, unii, cas_number,
salt_form, molecular_weight, conversion_factor, is_controlled,
is_hazardous, ndc_source_api)

DOSAGE_FORM_TAXONOMY
(dosage_form_id, name, code, is_sterile, usp_chapter,
calculation_method, default_unit)

ROUTE_OF_ADMINISTRATION
(route_id, name, snomed_code, fda_spl_code, abbreviation)

FORMULATION [MFR]
(formula_id, name, dosage_form_id→, route_id→,
total_quantity, total_quantity_unit, version,
bud_room_temp_hours, bud_refrigerated_hours, bud_frozen_hours,
bud_basis, bud_reference, usp_chapter, csp_category,
is_sterile, sterility_process, stability_reference,
is_bracketed, min_api_concentration, max_api_concentration,
source_formula_ref [PCCA/CompoundingToday formula ID],
pharmacy_type [503A/503B], ndc_product [503B only])
```

```
FORMULATION_INGREDIENT [junction table]
(formulation_ingredient_id, formula_id→, ingredient_id→,
is_active, ingredient_role,
strength_numerator_value, strength_numerator_unit,
strength_denominator_value, strength_denominator_unit,
quantity_per_batch_value, quantity_per_batch_unit,
concentration_mg_per_ml, expressed_as, potency_factor,
is_qs, sort_order)

COMPOUNDING_RECORD [per-preparation instance]
(record_id, formula_id→, rx_number [503A] / batch_number [503B],
patient_id [503A],
prepared_at, bud_date, bud_time,
prepared_by, verified_by,
total_quantity_compounded, appearance)

COMPOUNDING_RECORD_INGREDIENT [instance ingredients with lot tracking]
(cr_ingredient_id, record_id→, ingredient_id→,
quantity_used, unit, lot_number, expiration_date,
supplier_ndc [for 503B reporting])
```

6.2 Key Distinctions from Commercial Drug Databases

Aspect	Commercial Drug DB (NDC)	Compounding DB
Product identity	Fixed NDC code	Formula ID (MFR) + RX number (503A) or NDC (503B optional)
Ingredients	Single record	Junction table (one row per ingredient)
Strength	Single value field	Ratio (numerator/denominator with units)
Dosage form	Lookup from fixed list	Lookup with calculation method and BUD rules
Expiration	Fixed manufacturer date	Calculated BUD (503A) or tested expiry (503B)
Salt form	Embedded in name	Separate field with conversion factor
Batch/lot	Not applicable	Required (CR tracks per-ingredient lot)
Customization	None	Parent-child or bracketed range patterns

6.3 Applicable Standards and Reference Implementations

Standard/System	Purpose	URL
RxNorm (NLM)	Ingredient coding (IN, PIN, SCDC, SCD, MIN term types)	nlm.nih.gov/rxnorm

Standard/System	Purpose	URL
HL7 FHIR R6 Medication	Open interoperability schema for medications	build.fhir.org/medication
HL7 FHIR MedicationKnowledge	Formulary-level medication catalog	medplum.com/docs/medications/formulary
USP \<795\>	MFR/CR requirements for nonsterile compounding	USP-NF
USP \<797\>	MFR/CR requirements for sterile compounding + BUD rules	USP-NF
FDA 503B NDC directory	Product catalog for 503B outsourcing facilities	FDA NDC Directory
PCCA Formulary	9,600+ tested compounding formulas	pccarx.com/AboutUs/Glossary/CompoundingFormulas
CompoundingToday	3,600+ public formulas with BUD and source refs	compoundingtoday.com/formulation/Formula.cfm
SNOMED CT Routes	Standard route of administration codes	HL7/SNOMED
UCUM	Standard units of measure for concentrations	unitsofmeasure.org
FDA Salt Naming Policy	How to handle salt form in naming and strength	FDA guidance PDF

Research compiled April 7, 2026. All sources cited inline throughout the document.

Section 4: Protocol Template Systems in Functional Medicine

Compiled: April 2026

Executive Summary

Functional medicine clinics rely heavily on multi-medication treatment protocols that combine compounded pharmaceuticals, peptides, hormones, and supplements into coordinated care plans. Current EHR software provides highly variable support for these workflows: some platforms (Cerbo, CharmHealth) offer meaningful compounding prescription infrastructure, while others (Practice Better, Fullscript) are limited to supplement and lifestyle recommendations without true prescription capability. No mainstream functional medicine EHR currently offers native conditional logic (lab-triggered medication additions), and titration is universally handled through workarounds rather than purpose-built scheduling engines. The compounding pharmacy workflow — direct-to-patient shipping, pharmacy routing, and multi-medication bundling — remains a largely manual process across most platforms.

1. Protocol Definition and Storage in Functional Medicine Clinics

1.1 What Constitutes a "Protocol"

Functional medicine protocols are structured, multi-medication care plans designed around root-cause investigation rather than symptom management. Unlike single-drug prescriptions, these protocols bundle multiple compounded medications, peptides, hormones, and nutraceuticals into a unified clinical plan that may run for 12–24 weeks or longer. (Fullscript — Functional Medicine Protocols)

Common protocol archetypes in functional medicine clinics include:

Protocol Name	Typical Components
Weight Loss Protocol	Semaglutide + BPC-157 + Lipo-Mino injection (B vitamins + MIC)
Hormone Optimization — Male	Testosterone Cypionate + Anastrozole + HCG
Hormone Optimization — Female	Estradiol + Progesterone + DHEA + Testosterone
Pain Management / Recovery	BPC-157 + KPV peptide + Low-Dose Naltrexone (LDN)
Anti-Aging / Longevity	Sermorelin + NAD+ + Thymosin Alpha-1
Metabolic Optimization	Tirzepatide + Metformin (compounded) + Berberine
IV Therapy (Myers Cocktail)	Magnesium + Calcium + B vitamins + Vitamin C

(King's Pharmacy — Functional Medicine and Compounding)

1.2 How Multi-Medication Protocols Are Structured

Functional medicine protocols typically follow a three-phase framework — **Assessment → Intervention → Monitoring** — with each protocol defining:

- **Active ingredients** with specific doses and concentrations
- **Delivery method** (injection, sublingual, capsule, cream/gel, IV)
- **Frequency and timing** (e.g., weekly injection, nightly capsule)
- **Duration** (e.g., 12-week initial course, then maintenance)
- **Lab triggers or titration milestones** (e.g., recheck labs at week 8)

In practice, protocols are often stored as:

1. **Paper or PDF prescription templates** — Faxed to compounding pharmacies; providers fill in patient-specific fields by hand. Example: JohnsRx Compounded Semaglutide/Tirzepatide Prescription Template shows a pre-printed form with checkboxes for dose selection and pre-filled SIG language.
2. **EHR Chart Part templates** — In platforms like Cerbo, a "Chart Part" can bundle text, diagnoses, plan items, and prescriptions. The provider types a template name and the system inserts the entire protocol block into the encounter note.
3. **Supplement protocol templates** — In Fullscript or Practice Better, providers build reusable templates of supplement stacks that can be applied to patients with one click.
4. **Shared Google Docs / internal wikis** — Many smaller functional medicine clinics maintain protocol reference documents outside their EHR.

1.3 Patient Profile Differentiation (Male vs. Female, Weight-Based)

Protocol differentiation by patient profile is handled in current practice primarily through **clinical judgment at prescribing time** rather than automated branching. The provider selects a base protocol template and manually adjusts:

- **Sex-specific hormone substitutions** (testosterone for males vs. estradiol/progesterone for females)
- **Weight-based dosing** (e.g., semaglutide dose sometimes anchored to starting BMI)
- **Lab-informed adjustments** (e.g., anastrozole added only if estradiol is elevated; see Section 4)

No mainstream functional medicine EHR has a built-in "patient profile selector" that automatically serves a male vs. female variant of the same protocol. This is a manual workflow step.

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2. Existing Protocol Management Software

2.1 Practice Better

Category: Health coaching / wellness EHR **Primary Users:** Nutritionists, health coaches, naturopaths, integrative practitioners

Practice Better supports reusable **Protocol Templates** under My Practice > Protocol Templates. A protocol can contain food recommendations, supplement recommendations (via built-in database or Fullscript integration), lifestyle recommendations, and attached files. (Practice Better Help — Working with Protocol Templates)

Key features:

- Create named protocol templates with defined focus, duration, and notes
- Add supplement recommendations from the Practice Better database or Fullscript catalog
- Import templates from a linked Fullscript account
- Apply a template to a client with one click, then customize per patient before publishing
- Share protocols across providers within a practice (via the My Practice library)

Critical limitations:

- **No prescription medication support** — Practice Better protocols handle supplements and lifestyle only; they do not support Rx prescriptions or compounded medications.
- **No titration scheduling** — Protocol supplements have a dose and frequency, but there is no phased titration engine.
- **No conditional logic** — Cannot define "add medication X if lab value Y"
- Supplement recommendations generate a patient-facing "supplement chart," not a prescription

(Practice Better — Protocols and Recommendations)

Assessment: Suitable for supplement-only functional medicine practices (health coaches, nutritionists). Not suitable for compounding prescription management.

2.2 Cerbo EHR

Category: Clinical EHR for integrative and functional medicine **Primary Users:** Physicians, NPs, PAs at functional medicine and DPC practices

Cerbo is widely considered the most feature-complete EHR for functional and integrative medicine. Its **Chart Parts** system is its core templating mechanism: a Chart Part is a named template that can auto-populate text, diagnoses, lab orders, treatment plan items (prescriptions, supplements, IV therapies), and charges when the provider types the template name in an encounter note. (Cerbo — EHR Features)

Compounding prescription support:

- Supports **e-prescribing of compounded medications** via eRx and e-fax
- Custom compounded medications (bio-identical hormones, peptides) can be added to the medication list and formally prescribed
- **Preferred pharmacy pre-selection**: the patient's preferred compounding pharmacy can be set in advance, making it the default routing destination when sending a compounded prescription (Cerbo FAQ)
- Compounded medications are managed in the **Medications module** (not the Alternate Plan Items module, which is for supplements, IV therapy, and injections)

Protocol template support:

- Chart Parts can bundle multiple prescriptions, diagnoses, and plan items into one insertable template
- Custom **IV therapy protocols** are supported with structured dosing fields
- **Dosing profiles** for supplements, injections, and IV infusions save default dosing parameters
- Wellness Plan tracker for comprehensive long-term care plans shared via the Patient Portal

• **Supplement protocols**: supplements can be grouped into protocols within the Cerbo supplement database, with saved dosing instructions (Cerbo Alternate Plan Items)

Limitations:

• No native titration scheduling engine — multi-phase dose escalation is handled by creating separate prescriptions for each phase (or writing a complex SIG)

• No conditional logic (lab-triggered medication additions)

• Protocol sharing across providers is possible within the same practice via Chart Parts

(Cerbo — Functional Medicine EHR Guide by Meelio)

Assessment: Best-in-class for functional medicine compounding workflows. Handles compounding Rx, preferred pharmacy routing, and supplement/injection protocols well. Multi-medication protocol templates require manual construction via Chart Parts.

2.3 Elation EHR

Category: Primary care / DPC EHR **Primary Users:** Independent physicians

Elation's **Prescription (Rx) Templates** allow providers to save frequently used medication specifications (medication name, sig, quantity, unit, refills, days supply, NDC) as reusable templates. When prescribing, the provider searches for a medication and practice-created templates appear at the top of the dropdown, auto-filling all saved fields. (Elation Help — Prescription Rx Templates Guide)

Key features:

• Per-medication Rx templates with pre-saved sig, quantity, refills, and NDC

• Templates appear first in the medication search dropdown

• Managed from the Prescriptions Settings page

• Custom Sigs can be saved for reuse

Limitations:

• Rx templates are per-medication, not multi-medication protocol bundles

• No compounding-specific fields or compounding pharmacy routing

• Note templates (for visit documentation) are separate from prescription templates

• No titration scheduling, conditional logic, or auto-fill of multi-drug protocols

• Compounding support is functional but not documented as specialized

(Fullscript — Integrating Compounding Pharmacy mentions Elation as a platform used for EMR integration with compounding pharmacies)

Assessment: Adequate for saving individual prescription templates; not built for multi-medication protocol management or complex compounding workflows.

2.4 CharmHealth EHR

Category: Comprehensive EHR, including functional medicine **Primary Users:** Multi-specialty practices, integrative medicine

CharmHealth supports multiple template types including prescription templates, supplement templates, and consultation (SOAP) templates. (CharmHealth — Creating Templates)

Supplement/protocol template support:

- Create **Supplement Templates** via Settings > Templates > My Templates
- For supplement type "Compounded": select from a Preparations list with intake details, directions, dispensing information, and a **Weaning schedule** field — the closest analog to a titration schedule in the platform
- Templates can be created from the CharmHealth database, Fullscript, or WholeScripts (CharmHealth — Creating Supplement Templates)
- Apply a Fullscript supplement template directly from the Patient Dashboard (Medications > Supplements > Add from Templates)
- Multiple supplements can be transmitted to Fullscript simultaneously (CharmHealth — Fullscript Integration)

Prescription template support:

- General template framework supports prescription templates (SOAP, Diagnosis, Lab Orders, Prescriptions, Vaccination, Referrals)
- Rx templates can be created and reused

Limitations:

- No documented multi-drug compounding protocol template
- No conditional logic
- Fullscript integration handles supplement orders, but compounding Rx workflow is less documented
- "Weaning schedule" field in supplement templates suggests some titration note capability, but not a structured engine

Assessment: Functional but mid-tier. Fullscript integration is strong. Compounding-specific prescription workflows are less developed than Cerbo.

2.5 Fullscript

Category: Supplement dispensary platform **Primary Users:** All healthcare practitioners who recommend supplements

Fullscript is the leading professional supplement dispensary platform, used by over 100,000 healthcare providers serving more than 10 million patients. (Meelio — Supplement Protocol Management Guide)

Protocol template features:

- **Built-in template library** (made by Fullscript's Medical Advisory team) — evidence-informed templates for common conditions
- **Custom templates** — practitioners can create and save their own supplement protocol templates
- **Template sharing** — practitioners can share custom templates with colleagues in the same practice (Fullscript Support — Plan Building Tool)
- Applying a template adds all products to a draft plan, which the provider then customizes per patient before sending

- **Ingredient breakdown tool** calculates real-time ingredient totals as products are added or doses adjusted
- Plans include supplements, lab testing recommendations, resources (handouts, infographics), and a personalized message
- **Auto refill** and **refill reminders** are built in for patient adherence

Limitations:

- Supplements only — not a prescription platform; does not handle compounded medications, Rx drugs, or controlled substances
- No titration scheduling
- No conditional logic

(Fullscript — 8 Features to Improve Treatment Adherence)

Assessment: The gold standard for supplement protocol templates. Excellent template library, sharing, and dispensing workflow. Not applicable to compounding Rx prescriptions.

2.6 Wellevate

Category: Supplement dispensary platform (now integrated into Fullscript ecosystem) **Primary Users:** Practitioners who recommend professional supplements

Wellevate is an e-prescribing supplement platform that allows practitioners to create patient profiles, send supplement recommendations, and manage orders from a centralized platform. Practitioners can create electronic prescriptions with dosage instructions including quantity, frequency, duration, and daily timing. (GetApp — Wellevate 2026)

Note: As of 2025–2026, Wellevate has been substantially integrated into the Fullscript ecosystem. Cerbo (and other EHRs) list both Fullscript and Wellevate as supplement dispensary integrations.

Assessment: Functionally similar to Fullscript for supplement protocol recommendations; supplement-only platform.

2.7 Power2Practice

Category: EMR for functional, integrative, and anti-aging medicine **Primary Users:** Anti-aging, functional medicine, integrative practices **Endorsement:** American Academy of Anti-Aging Medicine (A4M)

Power2Practice is purpose-built for functional medicine with:

- Customizable note templates for functional medicine workflows
- Over 30 built-in IV therapy protocols
- Lab integrations with ZRT, Genova, Precision Analytical, Quest, and LabCorp
- Comprehensive e-prescribe system including EPCS (controlled substances) and compounded prescriptions
- Pharmacy integration for real-time prescription routing

(Power2Practice — Features) (EHRInPractice — Power2Practice)

Assessment: Solid option for anti-aging/functional medicine with built-in compounding Rx. Less documentation available than Cerbo on protocol template depth.

2.8 Salus EMR

Category: Functional medicine-specific EHR **Positioning:** "Next-generation functional medicine EHR"

Salus EMR explicitly includes a **Protocol builder for supplements, herbs, peptides, and lifestyle plans**, along with hormone and metabolic protocol support. Supports building custom supplement recommendations, tracking compliance, and updating protocols over time. (Salus EMR)

Assessment: Purpose-built positioning suggests strong protocol support, though market adoption is smaller than Cerbo or CharmHealth.

2.9 My Practice Connect ERX Platform

Category: Standalone compounding prescription e-prescribing platform (not a full EHR)

My Practice Connect is a centralized eRx platform designed specifically for the functional medicine / compounding pharmacy workflow. Key features: (My Practice Connect — ERX Platform)

- Access multiple FDA-registered 503A and 503B compounding pharmacies through a single portal
- View real-time inventory and pricing across pharmacies
- Secure messaging with pharmacy teams
- Place individual or bulk orders
- Automated refill reminders
- Access to over 20 IV nutrient protocols and wellness protocols (autism, insomnia, low libido, anxiety, fibromyalgia)
- Physician-formulated supplements and compounded medications shipped directly to patients' homes

Assessment: Best specialized tool for compounding Rx routing when the provider's EHR lacks deep compounding support. Acts as a middleware layer between EHR and multiple compounding pharmacies.

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3. Titration Phase Handling

3.1 Clinical Context of Titration

Titration — the stepwise escalation of medication dose over time — is a core feature of many functional medicine protocols. The primary driver is minimizing adverse effects during dose initiation while reaching therapeutic effect.

Semaglutide Standard Titration Schedule:

Phase	Weeks	Dose	Notes
Initiation	1–4	0.25 mg SQ weekly	Adjustment phase; GI tolerance
Escalation 1	5–8	0.5 mg SQ weekly	Therapeutic effect begins
Escalation 2	9–12	1.0 mg SQ weekly	Appetite suppression significant
Escalation 3	13–16	1.7 mg SQ weekly	(Wegovy protocol)
Maintenance	17+	2.4 mg SQ weekly (Wegovy) or 2.0 mg (Ozempic)	

(TrimRx — Semaglutide Dosage for Weight Loss: Complete Titration Guide) (NIH PMC — The Art and Science of Drug Titration)

Compounding pharmacies typically supply semaglutide in single multi-use vials (e.g., 5 mg/2 mL or 10 mg/2 mL) sized to accommodate multiple titration phases from a single prescription, reducing the number of separate Rx orders needed. (Strive Pharmacy Compounded Semaglutide FAQ via goodhormonehealth.com)

Low-Dose Naltrexone (LDN) Standard Titration:

Phase	Dose	Duration
Start	0.5 mg nightly	Week 1
Escalation	+0.5 mg every 5–7 days	Weeks 2–9
Target	4.5 mg nightly	Maintenance
Sensitive patients	Slower titration possible	Start at 0.1 mg

(Bateman Horne Center — Low Dose Naltrexone PDF) (TC Compound — LDN Titration Kits)

3.2 How Titration Is Represented in Current Software

None of the mainstream functional medicine EHRs have a dedicated titration scheduling engine. The two primary workarounds used in practice are:

Option A: Complex SIG on a single prescription The prescriber writes a narrative sig that encodes the entire escalation schedule:

> *""Inject 0.25 mg SQ weekly for 4 weeks, then 0.5 mg weekly for 4 weeks, then 1.0 mg weekly for 4 weeks. May increase dose as directed by prescriber.""*

This is the most common approach. The compounding pharmacy fulfills a single large vial containing enough medication for all phases. (JohnsRx Prescription Template PDF)

The sig field in EHRs (including Cerbo, Elation, and CharmHealth) supports free-text entry, so any titration schedule can be encoded in the directions. Elation even supports **custom SIGs** saved as reusable templates. (Elation Help — Prescription Rx Templates)

Option B: Multiple sequential prescriptions Some clinics write separate prescriptions for each titration phase:

- Prescription 1: Semaglutide 0.25 mg weekly × 4 weeks, no refills
- Prescription 2: Semaglutide 0.5 mg weekly × 4 weeks (sent at week 4)
- Prescription 3: Semaglutide 1.0 mg weekly × 4 weeks (sent at week 8)

This approach requires either **manual tracking** of when to send the next phase or use of calendar-based reminders. DrChrono, for example, supports sending multiple eRx orders simultaneously from a patient chart. (DrChrono Support — How Do I Send Multiple eRx Orders)

Option C: Titration kits from the compounding pharmacy

Several compounding pharmacies offer pre-packaged "titration kits" that contain multiple labeled vials (e.g., LDN 0.5 mg, 1.0 mg, 1.5 mg, 2.0 mg, 2.5 mg, 3.0 mg, 3.5 mg, 4.0 mg, 4.5 mg). A single prescription is written for the kit, and the pharmacy fulfills it as a boxed set with instructions. This is common for LDN. (TC Compound — LDN Titration Kits)

3.3 Summary: EHR Titration Support

Platform	Titration Mechanism	Notes
Cerbo	Complex SIG / multiple Rx	Chart Parts can template the SIG
Elation	Complex SIG / custom saved SIGs	Rx templates save per-medication sig
CharmHealth	Weaning schedule field in supplement templates	Supplement-only; Rx uses free-text sig
Practice Better	Not applicable	Supplement-only platform
Fullscript	Not applicable	Supplement-only platform
Power2Practice	Complex SIG / multiple Rx	Compounding Rx supported

4. Conditional Medications in Protocols

4.1 Clinical Use Cases

Functional medicine protocols routinely involve medications that are added conditionally based on lab results:

- **Anastrozole**: Added to male hormone optimization protocols only if estradiol > 30–40 pg/mL (too high estrogen from testosterone aromatization). Typical dose: 0.25–1 mg twice weekly. (RPC2B — Guide to Writing Compounded HRT Prescriptions)
- **B12 Injection**: Added if serum B12 < 400–500 pg/mL
- **Thyroid compound**: Added if TSH, Free T3, or Free T4 outside optimal functional range
- **DHEA**: Added if DHEA-S below functional threshold, with sex-specific dosing
- **HCG**: Added to TRT protocols for testicular preservation, sometimes conditional on testicular atrophy or fertility goals

4.2 Current Software Support for Conditional Logic

No mainstream functional medicine EHR currently offers native lab-triggered conditional medication logic in protocol templates.

What exists in the broader EHR space:

- **Clinical Decision Support (CDS) tools** in hospital-grade systems (Epic, Cerner) can create rules that fire alerts when lab values meet specific criteria. A systematic review found these tools improve appropriate medication adjustments when lab values are abnormal. (NIH PMC — Effect of Lab-Based CDS Tools on Medication Dosing)
- The **Event-Condition-Action model** ("ON event IF condition THEN action") is a standard CDS architecture, but implementation requires significant engineering effort and is typically only present in enterprise EHRs. (SNOMED CT CDS Guide)

How functional medicine clinics currently handle this:

Conditional medication decisions are made at the point of care by the clinician reviewing lab results. The workflow is:

1. Labs return and auto-populate into the EHR
2. Provider reviews lab panel at follow-up visit or via asynchronous chart review
3. Provider manually adds anastrozole (or adjusts dose) to the prescription list based on the estradiol value
4. The updated prescription is sent to the compounding pharmacy

Protocol templates may include a **note or reminder** in the template text (e.g., "If E2 > 35, add Anastrozole 0.5 mg twice weekly") but the trigger is human, not automated.

Cerbo has lab integrations with over 60 specialty and conventional labs, and lab results populate the patient chart automatically. However, there is no documented feature for auto-triggering a prescription order based on a lab value threshold. (Cerbo — Functional Medicine Labs)

Meelio AI (an AI assistant built on top of Cerbo) is emerging as a way to provide phase-aware documentation and protocol suggestions based on lab values — though this still requires clinician approval rather than automated prescribing. (Meelio — Cerbo EHR Functional Medicine AI Integration)

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5. Auto-Fill from Protocol Templates

5.1 Platform Capabilities

Feature	Cerbo	CharmHealth	Elation	Practice Better	Fullscript
Create custom templates	✓ (Chart Parts)	✓	✓ (note + Rx)	✓	✓
Auto-fill Rx from template	✓ (Chart Parts)	✓ (Rx templates)	✓ (per-drug Rx template)	× (no Rx)	× (no Rx)

Feature	Cerbo	CharmHealth	Elation	Practice Better	Fullscript
Auto-fill multi-drug protocol	✓ (via Chart Parts bundling)	Partial	×	×	×
Adjust dose within template	✓ (edit before finalizing)	✓	✓	✓	✓
Share across providers	✓ (practice-wide Chart Parts)	✓	✓ (practice-wide Rx templates)	✓	✓
Lab-based conditional auto-fill	×	×	×	×	×

5.2 Cerbo: Chart Parts — Deepest Multi-Medication Auto-Fill

Cerbo's **Chart Parts** system is the most capable native multi-medication protocol auto-fill currently available in the functional medicine EHR space. A Chart Part can encode:

- Free-text documentation blocks (assessment, plan narrative)
- Multiple diagnosis codes
- Multiple plan items (prescriptions, lab orders, supplement orders, IV therapy orders)
- Billing charges

When a provider types the Chart Part name in an encounter note, the entire bundle is inserted and individual items appear as "suggested actions" to accept or dismiss. This effectively functions as a one-click protocol template that populates multiple prescriptions simultaneously.

The provider can then edit any individual item (dose, sig, quantity) before finalizing — supporting per-patient dose adjustment within the protocol template. (Cerbo — EHR Features)

5.3 Fullscript: Supplement-Specific Template System

For supplement protocols, Fullscript is the most polished platform:

- **Made by Fullscript** templates in the Knowledge Center are pre-built condition-specific supplement stacks
- **Custom practitioner templates** are saved from the Templates section and shared practice-wide
- Applying a template opens a draft plan with all products pre-populated, including manufacturer-suggested dosing
- Providers can adjust individual doses, add resources, and write patient messaging before sending (Fullscript Support — Plan Building Tool)

5.4 Practice Better: Reusable Supplement Protocols

Practice Better allows creating protocol templates that are applied per client:

- Click "Create for..." next to any template, select the client, and a draft protocol opens
- Can be customized per client before publishing to the client portal

- Generates a patient-facing supplement chart with dose and frequency summary
- Integrates with Fullscript for direct supplement fulfillment (Practice Better Help — Working with Protocol Templates)

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6. Compounding-Specific Workflow in EHRs

6.1 The Compounding Prescription Workflow

Compounded medications differ fundamentally from standard retail pharmacy prescriptions in their workflow:

1. **Prescription creation**: Provider writes the compound spec (active ingredient, strength/concentration, quantity, delivery form, SIG)
2. **Pharmacy routing**: Prescription is routed to a specific compounding pharmacy (503A individual patient or 503B outsourcing facility) — not the patient's local pharmacy
3. **Patient shipping**: The compounding pharmacy ships directly to the patient's home address (not dispensed locally)
4. **Multi-medication bundling**: When a patient is on multiple compounds, the pharmacy may bundle them in one monthly shipment

The required fields on a compounding prescription differ from standard Rx: (Symphony Pharmacy — Writing a Prescription for a Compounded Medication)

- Patient identification (name, DOB, **shipping address**)
- Generic name of ALL active ingredients
- Strength/dose of each ingredient (in mg, %, or g)
- Formulation desired (cream, gel, capsule, injectable, sublingual)
- Quantity and days supply
- Directions for use
- Medical necessity statement (increasingly required)
- Refills

6.2 Patient Shipping Address in EHRs

Patient shipping address is a critical but often-overlooked field for compounding prescriptions. Key platform behaviors:

- **Cerbo**: Patient demographic fields include address. Compounding pharmacy prescriptions sent via e-fax include patient information including address. Patient can update preferred pharmacies and associated address information via the Patient Portal. The system pre-populates the patient's preferred compounding pharmacy. (Cerbo — EHR Features)
- **Elation**: Patient address is required before sending an eRx. The system stores patient address in demographics and includes it in prescription transmissions. (DrChrono — Multiple eRx Orders)
- **My Practice Connect**: Explicitly positions compounded medications as "shipped directly to patients' homes," with the platform coordinating the delivery logistics. (My Practice Connect — ERX Platform)

- **Empower Pharmacy**: One of the largest compounding pharmacies in the U.S., ships via UPS ground or overnight with tracking. Patient address is captured at prescription time. (Empower Pharmacy — Shipping)

6.3 Pharmacy Selection

Functional medicine clinics typically work with 1–3 preferred compounding pharmacies with which they have established relationships, pricing agreements, and familiarity with formularies. The major compounding pharmacies used by functional medicine practices include Empower Pharmacy, Strive Pharmacy, Wells Pharmacy Network, Tailor Made Compounding, and Hallandale Health.

EHR pharmacy selection support:

- **Cerbo**: Nationwide pharmacy database; provider can set a short-list of preferred pharmacies including compounding pharmacies. The patient's preferred pharmacy (including their compounding pharmacy) is accessible and pre-populated when creating a new Rx. (Cerbo — EHR Features)
- **General EHRs with eRx**: Surescripts network (used by most EHRs for eRx) includes some compounding pharmacies that accept electronic prescriptions; others require e-fax. Many compounding pharmacies accept eFax prescriptions as the primary transmission method.
- **My Practice Connect**: Routes prescriptions to multiple 503A and 503B pharmacies from a single portal. Providers can view real-time inventory and pricing across pharmacies before choosing where to route. (My Practice Connect — ERX Platform)

6.4 Multi-Medication Bundle Orders

When a patient is on a protocol involving multiple compounded medications (e.g., Testosterone Cypionate + Anastrozole + HCG), there are two shipping models:

Model 1: Separate prescriptions, same pharmacy, bundled shipment The provider writes separate Rx for each compound but routes all to the same compounding pharmacy. The pharmacy consolidates into one monthly shipment. Wells Pharmacy Network's "Wellness Bundles" explicitly supports this model — pairing commonly co-prescribed compounds (e.g., Naltrexone + Sermorelin; Semaglutide/Metformin + Sermorelin) into one consistent monthly shipment. (Wells Pharmacy Network — Wellness Bundles)

Model 2: Multi-ingredient single compound The compounding pharmacy formulates multiple active ingredients into a single preparation (e.g., Testosterone + Anastrozole cream, or Semaglutide + B12 + Glycine injectable). This reduces the number of separate Rx orders and simplifies patient administration.

EHR support for multi-Rx simultaneous ordering:

- **DrChrono**: Documented support for sending multiple eRx orders simultaneously from a patient chart using a batch workflow. (DrChrono — Multiple eRx Orders)
- **Cerbo**: Chart Parts can include multiple Rx items that are inserted simultaneously into the encounter note, then processed as a batch.
- **General workflow**: Most EHRs allow multiple prescriptions to be written during a single encounter and transmitted together or sequentially.

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7. Emerging and Adjacent Tools

7.1 PioneerRx and Liberty Software

These pharmacy management systems are used by compounding pharmacies themselves (rather than by prescribers) to process and manage compounded orders. Integration between EHRs (Elation, Cerbo) and these systems is cited as a key workflow improvement tool for functional medicine practices. (Fullscript — Integrating Compounding Pharmacy)

7.2 AI-Assisted Protocol Documentation

Meelio AI is an AI assistant purpose-built for functional medicine that listens to consultations and automatically generates care plans populated with supplement protocols from custom templates. It can handle phase-aware documentation for multi-stage protocols (e.g., Shoemaker Protocol for CIRS) and surfaces protocol-specific AI recommendations based on lab results — though clinician approval is still required. (Meelio — Cerbo EHR Functional Medicine AI Integration)

7.3 DTP (Direct-to-Patient) Dispensing Platforms

Develop Health and similar platforms are building direct-to-patient dispensing infrastructure that integrates with EHRs to automate the compounding prescription journey: prescribing → coverage check → fulfillment → home delivery. These platforms handle patient shipping addresses automatically and coordinate cold-chain logistics for temperature-sensitive compounds. (Develop Health — DTP Dispensing)

8. Key Gaps and Implications for Software Design

Based on this research, the following critical gaps exist in current functional medicine software:

Gap	Current State	Implication
True multi-medication protocol bundles	Workarounds via Chart Parts (Cerbo) or manual bundling	A "protocol template" that instantiates multiple Rx simultaneously with one click would save 10+ minutes per encounter
Titration scheduling engine	Complex SIG text or multiple separate Rx	A visual titration timeline that automatically generates scheduled prescriptions per phase would reduce errors
Conditional lab-triggered medications	Entirely manual; clinician reviews and decides	Even a soft alert ("Estradiol is 52 pg/mL — consider adding Anastrozole per your protocol") would improve workflow
Patient shipping address on Rx	Captured in demographics but not always surfaced in Rx workflow	Dedicated "ship to patient" toggle with address confirmation would reduce shipping errors
Multi-pharmacy price comparison	Only My Practice Connect offers this	EHR-native pharmacy price/inventory comparison would simplify pharmacy selection

Gap	Current State	Implication
Protocol variant logic (male/female)	Manual template selection	Protocol templates with branching (select patient sex → load appropriate variant) would reduce selection errors
Cross-provider protocol sharing	Available in Cerbo (Chart Parts), Fullscript, Practice Better	Practice-wide protocol libraries with version control are a meaningful workflow improvement

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Section 5: Progressive Disclosure UX in Healthcare — Medication Ordering

Medication Ordering Interface Design — Best Practices for CompoundIQ

Research compiled April 2026

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1. EHR Medication Ordering UX

1.1 Epic CPOE and the "Order Composer"

Epic's Computerized Provider Order Entry system is the most widely deployed EHR platform in the U.S. hospital market. Its medication ordering workflow is anchored around several core UX patterns:

The Order Composer is the term Epic uses for the modal interface where a clinician finalizes a medication order's details — dose, route, frequency, duration, priority, and special instructions. When a user selects a medication from search results or a preference list, the Order Composer opens to allow confirmation or modification of each field. Epic's order friction analytics track the number of field changes made within the Order Composer before signing, generating a "Friction per Order" (FPO) score. This score measures how much effort each order requires. According to research published in the AMIA Annual Symposium Proceedings, Epic's vendor-recommended benchmark FPO is **5 for preference list medication orders** and **1 for order set orders**, reflecting the expectation that orders from pre-built sets require almost no modification at all.

SmartSets (Order Sets) are Epic's pre-built bundles of related orders for specific clinical scenarios (e.g., "Acute Pain Management," "Admission Orders — Cardiology"). In one academic medical center analysis of 5.8 million inpatient orders, 46.6% of orders came from order sets, and order set orders had an average FPO of 0.41, compared to 2.11 for preference list medication orders — a **5x reduction in ordering friction** (AMIA 2024). SmartSets can be customized by individual users, departments, or specialties, and can

include pre-checked defaults that pre-populate the Order Composer.

Preference Lists / Favorites are the "frequently used" medication configurations that clinicians save for quick access. Epic's University of Iowa documentation describes how preference lists and user panels reduce ordering time by surfacing a clinician's most-used orders at the top of search results without requiring full-text search.

Order Sentences are pre-written complete medication orders (e.g., "Acetaminophen 1000mg PO Q6H PRN pain"). Rather than configuring each field, a clinician can select the matching sentence and sign with minimal changes. However, as analyzed by EHR UX researcher Gregory Schmidt, order sentences have significant UX problems: for commonly ordered drugs with many permutations (e.g., insulin, warfarin), the list of available sentences becomes overwhelming and visually similar items become dangerous to confuse. Schmidt recommends showing only the medication name in initial search results, then revealing sentences only after the user selects a specific drug.

Epic-specific speed features include:

- Ctrl+O keyboard shortcut to jump directly to the order field
- Voice commands via Dragon Medical One integration to queue orders without mouse navigation
- User Panels: clinician-created groups of their most common orders, accessible in one click
- Copy Previous Orders: reuse a prior order's settings as a starting point
- Real-Time Prescription Benefits (RTPB): shows cost and formulary status as the order is built, integrated within the ordering workflow (Epic open.epic interface documentation)

UX Pain Point: The AMA-commissioned multi-center study "62 clicks to order Tylenol" found a **9x difference in task completion time** and an **8x difference in click count** for identical tasks across Epic and Cerner implementations at different hospitals. Error rates for ordering oral prednisone taper reached 50% at one institution versus 16.7% at the best-performing site. This demonstrates that while Epic's underlying architecture supports efficient ordering, local configuration dramatically affects real-world usability.

1.2 Cerner/Oracle Health PowerChart

Cerner Millennium's PowerChart is the clinician-facing EHR interface for ordering in Oracle Health environments. Its medication ordering UX uses several analogous but distinctly named patterns:

PowerPlans are Cerner's equivalent of Epic's SmartSets — pre-built order sets for specific clinical scenarios. A 2024 JMIR Formative Research study of PowerPlan usage found that the 5 most utilized order sets across a health system were Acute Pain Management (325,664 uses), Acute Behavioural Disturbance (63,354), VTE Prophylaxis (51,023), Healthy Newborn Protocols (26,229), and Postoperative Nausea and Vomiting (25,458). A systematic review found 36 published studies on PowerPlans (Cerner) and SmartSets (Epic), with 30 of 36 reporting favorable outcomes, though only half measured clinical variation impact.

Order Sentences in PowerChart work similarly to Epic's: when a user searches for a medication, they are presented with a list of pre-built order sentences. In the Cerner PowerChart intro to ordering tutorial, the interface presents "pre-built sentences" for commonly ordered medications (e.g., "Acetaminophen 1000mg Q6H") and falls back to blank field-by-field entry when no sentence matches. The "Missing Required Details" button in Cerner's order entry form counts how many required fields are unfilled and

navigates the user sequentially through each one — a form of progressive disclosure by necessity rather than design.

UX icons and contextual indicators in Cerner are extensive — the CST Cerner Help documentation lists dozens of distinct icons for: allergy conflicts, drug interactions, formulary status, non-formulary flags, soft/hard stop indicators, prescription vs. ambulatory vs. inpatient order type, and PowerPlan membership. This density of visual information provides contextual awareness but can contribute to cognitive load for less experienced users.

New Order Entry (NOE) is Cerner's primary search interface. It provides a unified search across medications, labs, imaging, and other orders, with type-ahead suggestions. Quick Orders provide a favorites-style shortcut panel for the most common orders by department.

1.3 Athenahealth (athenaOne)

Athenahealth's athenaOne system focuses primarily on ambulatory/outpatient practice workflows. Its medication ordering UX emphasizes role-based differentiation and exception-based surfacing of information:

Role-specific views mean that physicians see different interfaces than medical assistants, billers, or care managers. The athenaOne efficiency blog highlights that "workflows surface tasks that require attention, rather than having users sift through all available data" — a form of progressive disclosure at the workflow level rather than the field level.

Formulary check inline: The athenahealth athenaPractice roadmap documentation shows that before selecting a medication to prescribe, providers can see the number and types of contraindications at a glance and drill down for details — a tiered disclosure from summary indicators to detail views.

e-prescribing integration: athenaOne connects to Surescripts for pharmacy-direct prescribing. The athenaOne Base Service description notes that Order-specific questions are maintained globally and prompt additional information for specific orders — only showing supplemental fields when the order type requires them.

API-based order creation is documented at docs.athenahealth.com: orders are created via API for open encounters and reviewed and signed by providers in the athenaOne UI. This enables pre-population of order details from external systems.

1.4 Key Lessons for CompoundIQ

Pattern	EHR Implementation	CompoundIQ Application
Order sentences / pre-built orders	Show common compound configurations upfront	"Favorites" library of full compound configurations by prescriber
Progressive field revelation	Missing Required Details button; dynamic fields	Show base fields first, reveal advanced fields (diluent, BUD, special packaging) conditionally
Friction measurement	Epic's FPO score; track field changes	Track how often prescribers modify defaults; use data to tune smart defaults

Pattern	EHR Implementation	CompoundIQ Application
Order sets	Disease-specific or patient-type bundles	Compound-type order sets (e.g., "Pain Management Compounding Set")
Role-based views	Athena's role differentiation	Prescriber view vs. pharmacist review vs. technician preparation view
Inline contextual warnings	Formulary check, allergy alerts inline	Ingredient interaction warnings, BUD warnings, and route-dosage consistency checks

2. Progressive Disclosure Pattern

2.1 Definition and Core Principles

Progressive disclosure is a UX design technique that defers advanced features and information to secondary interface components, presenting users with only the content essential for their immediate task. As defined by the Interaction Design Foundation, it is a method to "defer advanced features and information to secondary UI components... to improve usability for novice and experienced users."

The pattern was formally articulated by Tidwell in *Designing Interfaces* (2005) and popularized through Jakob Nielsen's research in 2006. It rests on Sweller's Cognitive Load Theory (1988), which established that human working memory can only process a limited number of information elements simultaneously. By revealing complexity in stages, progressive disclosure ensures that the working memory available is directed at the clinical decision being made, not at navigating the interface.

Core principles:

1. **Prioritize information hierarchy** — Identify which fields are essential for every order vs. which are only needed conditionally. Show only essential fields initially.
2. **Contextual revelation** — Reveal additional fields based on prior selections (e.g., if Route = "IV", reveal IV-specific fields like infusion rate, diluent). If Route = "topical", hide IV fields entirely.
3. **Preference persistence** — Remember a user's chosen complexity level across sessions (e.g., a clinician who always configures duration should have that field pre-expanded).
4. **Graduated complexity** — Match the disclosed information to the user's expertise and the order's complexity; an order sentence for a common drug should require zero additional configuration.
5. **Smooth transitions** — Expanding sections should animate naturally; collapsing them should not lose data.
6. **Reversibility** — Users must be able to expand to full detail and collapse back to simplified view at any time.

Measured impact:

- Interfaces applying progressive disclosure achieve **30–50% faster initial task completion** while maintaining **70–90% feature discoverability** for advanced capabilities (UX/UI Principles research synthesis)

- Progressive interfaces improve completion rates **25–40%**, reduce abandonment **30–50%**, and increase feature adoption **40–60%** (UX Collective)

2.2 How to Reveal Fields Only When Prior Selections Make Them Relevant

This is the core mechanic for medication ordering UX. The principle is **conditional field display**: a field only appears when its parent condition is met.

In medication ordering, the decision tree looks like:

```
1. Drug selection (required, always visible)
↓
2. Dose + units (required, appears after drug selection)
→ If drug is pediatric-indicated: show weight-based dosing calculator
→ If drug requires renal adjustment: show kidney function flag
↓
3. Route (required, filtered to valid routes for selected drug)
→ If Route = "IV": reveal Infusion Rate, Diluent, Concentration fields
→ If Route = "topical": reveal Application Site, Coverage Area
→ If Route = "oral": hide all IV fields
↓
4. Frequency (required)
→ If Frequency = "PRN": reveal PRN Indication field ("give only for...")
→ If Frequency = "custom": reveal Days/Times selector
↓
5. Duration (contextual — show for outpatient, optional for inpatient continuous)
→ If Duration = "as needed": reveal Stop Criteria
↓
6. [Advanced section — collapsed by default]
→ Specific patient instructions
→ Prescriber-to-pharmacist notes
→ Priority / STAT flag
→ Brand substitution preference
```

Examples of this pattern in healthcare applications:

- The Czech pilot mHealth study on UX for older adults (PMC) found that replacing an open text field for medication selection with a multi-select list plus an "other — please specify" option significantly improved usability for the medication ordering task
- The Hannah Dym UX prototype at Brigham and Women's / Partners HealthCare organized the prescribing interface around the **patient's medical problem (indication)** rather than the drug name as the primary navigation axis — a progressive disclosure pattern that starts with "why" before "what", and was found to statistically reduce clicks and errors
- Athenahealth's pre-prescribing contraindication summary shows indicator counts (not full lists) until the prescriber clicks — a collapsed/expanded disclosure pattern

2.3 Cognitive Load Reduction in Medical Ordering

The Betsy Lehman Center for Patient Safety documents that medication prescribing is one of the "critical times" where interruption must be minimized and cognitive load carefully managed. Recommendations include dedicated interruption-free zones, cognitive job aids (like pre-filled order sets), and well-designed CDS tools that do not interrupt for low-priority alerts.

The AMA study on EHR usability found that physicians in emergency departments dedicate **43% of their time to data input** vs. only 28% to direct patient care (Journal of Emergency Medicine, 2013 —

documented in ScienceDirect), accumulating nearly **4,000 mouse clicks during a busy 10-hour shift**. Reducing cognitive load through progressive disclosure directly reduces this burden.

At a large academic medical center, hospitalists spent **47 minutes of every shift (20.2% of their EHR time) placing orders** — a target for systematic optimization (AMIA 2024).

3. Speed Optimization

3.1 Click Count Optimization Studies

The "62 clicks" benchmark: The AMA-commissioned study published in JAMIA (2018) measured EHR tasks at four hospitals using Cerner and Epic. It found:

- **9x difference in task completion time** for common tasks across implementations of the same vendor
- **8x difference in click count** for the same tasks
- Up to **62 clicks to order Tylenol** at the highest-friction site
- **Error rates up to 50%** for prednisone taper orders — meaning that click-heavy workflows don't just slow users down, they cause dangerous mistakes

The study explicitly attributed these differences to local configuration decisions, not vendor platform limitations, demonstrating that interface design choices are the primary driver of efficiency.

Order Friction (FPO) benchmarks from Epic log data (AMIA 2024):

- **Target for medication orders from order sets**: FPO ≤ 1 (approximately 1 field change before signing)
- **Target for medication orders from preference lists**: FPO ≤ 5
- **Actual median for preference list medication orders**: FPO = 2.11
- **Intervention result**: Changing blank default frequency fields to "Once" for 22 lab order types reduced average CPO from 1.82 to 0.15 — a **92% reduction in order changes** per order

Interpretation for CompoundIQ: Every blank required field that forces a prescriber to make a selection is one "click" of friction. Setting evidence-based smart defaults for the most common compound configurations can reduce the interaction cost of routine orders by an order of magnitude.

"10 clicks per order" as historical benchmark: An early CPOE reference from a hospital implementation that required an average of 10 mouse clicks per order, or 1–2 minutes per order, was cited as a baseline. For sub-30-second workflows, this must be reduced by at least 60–80%.

3.2 Keyboard-First Navigation Patterns

Experienced clinical users strongly prefer keyboard navigation for speed. Key patterns:

- **Tab-advance through fields**: Enterprise Health's prescribing documentation (Enterprise Health docs) describes using the Tab key to advance from medication name to form, sig, duration, and quantity — cursor advances automatically, eliminating mouse movement between fields
- **Auto-trigger on selection**: When the medication is selected from autocomplete, the form should immediately focus the first empty required field (dose) without any additional click
- **Smart phrases / text expansion**: Epic's SmartText, Dragon Medical One commands, and similar systems allow typing abbreviations that expand to full medication orders (e.g., ".metop25" →

"Metoprolol 25mg PO BID")

- **Keyboard shortcuts for common actions**: Ctrl+K for universal search (used in Healthie), Ctrl+O to open order field in Epic. In complex medication ordering, shortcuts to "Accept default and advance", "Mark STAT", and "Add to favorites" should all be keyboard-accessible
- **Tab navigation optimization**: Fields that are contextually hidden (not relevant given prior selections) should be skipped entirely in the Tab order — a user should never have to Tab through a field they cannot interact with

3.3 "Happy Path" Optimization

The happy path is the most common ordering sequence — the 80% case that must be fastest.

Evidence-based approach to identifying the happy path:

1. Analyze order log data: What are the 20 drugs accounting for 80% of orders for a given prescriber type or specialty?
2. For each high-volume drug, what is the most common dose/route/frequency combination?
3. Set those combinations as the pre-populated defaults in preference list order sentences
4. Measure FPO after change to validate default accuracy

Illustrative example from compounding practice (SAT133 Diabetes Discharge Order Set case study, Journal of the Endocrine Society PMC): A teaching hospital found that 55% of insulin-prescribed discharge patients were missing insulin supplies. They redesigned the order set from 20 fragmented items to 2 top-level choices ("Patient uses insulin pen" vs. "Patient uses vial and syringe"). Selecting "insulin pen" automatically pre-selected the pen, pen needle, glucometer, testing strips, lancets, and alcohol swabs — collapsing a multi-step process into a single binary selection plus review. This is the pure "happy path" pattern: one meaningful choice produces a complete, validated order.

For CompoundIQ, happy path means:

- Compound Drug selected → system auto-populates the most common base, vehicle, strength, and quantity for that compound type
- Prescriber sees pre-filled form, reviews, may modify specific fields, and signs
- Average order time for routine compounds should approach **15–20 seconds** (similar to an Epic order set order)
- Custom/novel compounds should still be achievable in **< 90 seconds** with full field revelation

3.4 Time-to-Complete Benchmarks

Workflow Type	Benchmark Time	Source
Pen/paper prescription	~15–30 sec (gold standard for speed)	Gregory Schmidt analysis
Epic order set order (well-optimized)	~30–60 sec with review	AMIA FPO benchmark data
Epic preference list order (average)	~60–120 sec	FPO = 2.11, ~2 field changes

Workflow Type	Benchmark Time	Source
Epic standalone search + configure	~2–5 min	Up to 62 clicks in worst case
CPOE order-to-pharmacy verification (pre-CPOE)	115 min	Am J Health Syst Pharm 2009
CPOE order-to-pharmacy verification (post-CPOE)	3 min	Same study — 97% reduction
Medication documentation (CPOE vs. paper, simulation)	14:40 vs 12:12 min for 6–11 drugs	Methods Inf Med 2023 — 20% longer with CPOE but 100% quality score vs 66.7%

Key takeaway: CPOE adds time at the point of documentation but compresses total cycle time dramatically (order → pharmacy → administration). The documentation time can be compressed back toward pen-and-paper speeds through smart defaults and order sets without sacrificing completeness or safety.

4. Auto-Complete, Smart Defaults, and Favorites

4.1 Type-Ahead Search for Medications

Type-ahead (autocomplete) search is the universal standard for medication name entry across all EHRs. Best practices:

- **Frequency-ranked results**: The Enterprise Health prescribing module explicitly notes that autocomplete results are ranked by frequency of use, not alphabetically, so the most-commonly ordered drugs for a given partial string appear first
- **Two-stage result display**: Show drug name first (compressed list with more unique results), then reveal order sentences or variants after drug selection — Schmidt's recommended approach to prevent overwhelming users
- **Fuzzy/phonetic matching**: Critical for handling misspellings, since 17.4% of free-text medication entries contain misspellings including common drugs like "Glimepiride," "Glipizide," and "Humalog"
- **FHIR ValueSet/$expand** for standardized terminology: The Stack Overflow EHR medication autocomplete solution references FHIR's ValueSet/$expand operation, already deployed in production for real-time autocomplete against a live drug database as the user types
- **Brand/generic equivalence**: Systems should match both brand names and generics as equivalent, since prescribers may know either
- **First DataBank (FDB) integration**: FDB's MedKnowledge database is the industry standard drug library (FDB CPOE page), used by Enterprise Health, athenahealth, and other systems. It provides not just drug names but contextual dose/frequency recommendations based on patient demographics

4.2 Pre-Populating Dose, Frequency, and Quantity

When a medication is selected, the system should immediately propose appropriate defaults for downstream fields:

- **Context-aware defaults**: FDB's patient-centric medication ordering uses **patient context** (weight, age, renal function, indication) to generate smart medication order strings. For a pediatric patient, the suggested dose differs from an adult default without any prescriber calculation required (FDB CPOE description)
- **Duration → Quantity auto-calculation**: Enterprise Health automatically calculates total quantity from duration + sig fields (Enterprise Health docs), eliminating manual math and a frequent source of dispensing errors
- **Indication-driven defaults**: The Brigham/JAMA prototype study showed that organizing prescribing around indication (the medical problem) rather than drug name — with the system suggesting drug regimens from problem context — reduced error rates from 29.7% (vendor systems) to 5.5%, a **5x improvement**, while also being faster and preferred by clinicians
- **Renal/hepatic dose adjustment flags**: FDB provides built-in adjustments that surface contextually only when patient labs indicate the need

4.3 Favorites / QuickList Patterns

"Favorites" are personally curated lists of frequently used medication configurations. They represent the ultimate "happy path" for experienced prescribers.

Epic Preference Lists and User Panels: Clinicians can save any combination of orders as a named panel for one-click retrieval. Epic efficiency documentation from the University of Iowa describes creating preference lists, adding SmartSets to favorites, and building user order panels from any set of orders in the manage orders view.

"Remembered Prescriptions" (Enterprise Health): The Remembered Prescriptions feature allows prescribers to type a custom name (e.g., "BOB" or "Heart Disease") to save a complete prescription configuration. On future encounters, typing the drug name surfaces both "new" and named remembered versions in the autocomplete. Selecting the remembered version populates all fields instantly. This is the "favorites" pattern made maximally efficient.

Cerner Favorites: PowerChart's Quick Orders provide a favorites panel. The Cerner ordering intro video shows a dedicated "favorites" tab adjacent to the order search window, providing immediate access to saved configurations.

What makes a good favorites implementation:

- Favorites should appear in autocomplete results alongside standard drug matches, visually distinguished
- Favorites should be editable: selecting a favorite should load its settings but allow modification before signing
- Favorite usage should be tracked: items used frequently auto-surface, items unused for months should quietly de-prioritize or offer to archive
- Favorites should be shareable or templatable: a senior pharmacist or physician should be able to create a "clinic favorites" set that new prescribers inherit

4.4 Order Sets — Pre-Built Bundles

Order sets represent the highest level of automation in medication ordering. Rather than composing one prescription at a time, a prescriber selects a clinical scenario and reviews/approves a pre-configured bundle.

Published data on order set impact:

- CPOE with pre-printed order sets reduced chemotherapy problem-order rates from **30.6% (handwritten) to 12.6% (preprinted) to 2.2% (CPOE with order sets)** — a 14x error rate improvement (PubMed 2013)
- AHRQ recommendation: Order sets should "minimize the amount of order details a user must complete, utilize default values where appropriate, and standardize placement of similar elements" (CapMinds CPOE design guide)
- NIH StatPearls notes that order sets require investment to build but pay off in adoption, with specialty-specific sets generating highest use rates (NCBI Bookshelf CPOE article)
- A systemic review of 36 studies on Epic SmartSets and Cerner PowerPlans found the majority (30/36) reported favorable outcomes for clinical efficiency and safety (JMIR Formative Research 2024)

For CompoundIQ, equivalent structures might include:

- "Pain Management Compound Set" (common topical NSAID compounds)
- "Hormone Therapy Set" (standard HRT compound combinations)
- "Pediatric Compound Set" (common flavored oral compounds for children)
- "Wound Care Compound Set" (common topical antibiotics + antifungals)

Each set would pre-select compounds, strengths, vehicles, BUD ranges, and packaging — requiring only patient-specific confirmation before ordering.

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5. Structured Data vs. Free Text

5.1 The Core Tension

Structured data fields (dropdowns, radio buttons, numeric inputs with units) enable clinical decision support, drug interaction checking, dose validation, automated dispensing, and analytics. Free text is faster to enter for experienced users but bypasses all safety checks.

The tension is real: when the structured interface is poorly designed, prescribers work around it with free text — but free text creates safety gaps that may not be visible until a patient event occurs.

5.2 Error Rates: Structured vs. Free Text

Research data on free-text medication order rates and errors:

- **9.3% of hypoglycemic medications** were entered as free-text in a large ambulatory EHR, despite 75% of those entries having exact matches in the structured drug dictionary (AMIA 2012, Zhou et al.)
- **17.4% of free-text entries contained misspellings** (up to 22.8% for nurses, 13.5% for physicians)
- Free-text entries were **missing dose data 45% of the time**, frequency 43%, and duration 79%

- **92 drug-drug interaction alerts were not triggered** for 84 patients due to free-text entries bypassing the CDS engine
- **196 patients had duplicate medications** (identical drug entered as both free-text and coded) — with potential for dangerous dose stacking

Free-text CPOE orders as workarounds (Applied Clinical Informatics 2021):

- Clinicians described **8 reasons for using free-text orders**: poor usability of structured entry, limited functionality (can't express "hold for dialysis" in structured fields), complex navigation, structured order systems being too slow to update or modify, lack of visible confirmation that orders were seen, need to communicate urgency, insufficient context in structured fields, and alert fatigue from CDS in structured workflows
- **5 risks they identified**: missed orders, increased nursing workload, outdated orders cluttering the chart, conflicting information between free-text and structured fields, and bypassed safety checks
- **Clinicians' unanimous recommendation**: Improve structured order entry to eliminate the need for workarounds, rather than trying to eliminate free text by policy

5.3 When to Force Structured Selection vs. Allow Free Text Override

Scenario	Recommended Approach	Rationale
Drug name	Structured (autocomplete)	Enables DDI checking, allergy alerts, dose range validation
Dose + units	Structured with numeric input + unit dropdown	Enables dose range checking; free text bypasses CDS
Route	Structured (dropdown filtered by drug)	Wrong route errors are serious; limits to valid options
Frequency	Structured (validated list)	Enables scheduling logic, missed-dose tracking
Duration	Structured with free-text fallback	Commonly missing in free text; structured enables auto-refill, stop dates
Patient instructions / sig details	Hybrid: structured options + "custom" free text	Broad variation; structured covers 90%, free text covers edge cases
Special dispensing notes to pharmacist	Free text in dedicated field	Communication intent; cannot be fully structured
Prescriber clinical rationale	Optional free text	Context and audit, not operational; should not be required

5.4 Hybrid Patterns — Structured Fields with Custom Override

The best-practice design is a **layered input model**:

1. Offer the most common structured options first (e.g., dose radio buttons: "250mg / 500mg / 1000mg / Custom")
2. If user selects "Custom," reveal a free-text/numeric entry field with inline validation

3. Validate free-text input on blur: does the entry match a valid dose range? Flag if not, but don't hard-block
4. Track which selections require "Custom" most often — use frequency data to add those values to the structured list

This is the pattern used by Enterprise Health's prescribing module: free-text entry is allowed but clearly labeled as "uncoded," with an explicit warning that "drug/allergy interaction warnings cannot work since it is an unrecognized medication."

5.5 Handling Edge Cases That Don't Fit Dropdown Options

Practical strategies:

1. **"Other — please specify" option**: Always include this at the bottom of any dropdown for valid but uncommon options. Flag these orders for pharmacist review.
2. **"Suggest this option"**: When a prescriber types a value that doesn't match any structured option, offer to submit it to the formulary team for consideration.
3. **Fuzzy match before free text**: Before accepting free text, run a fuzzy match against the structured list and prompt "Did you mean [X]?" to catch near-misses
4. **Free-text surveillance**: Track free-text usage patterns. If 20% of orders for a specific compound include a free-text note saying "do not use preservatives," that note should become a structured toggle field
5. **Inline validation messages** (Applied Clinical Informatics 2016): Adding inline validation for missing fields (e.g., "Provider information required" before allowing order submission) reduced error rates from 20.27% to 12.96% in one outpatient CPOE study — a **36% error reduction** simply from prompting for completeness
6. **Hard stops for safety-critical fields**: For high-risk parameters (e.g., concentration of a controlled substance compound), use hard stops that require resolution before submission. For administrative fields, use soft stops (warning, but allow override with documented reason).

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6. Case Studies

6.1 Brigham and Women's Hospital — CPOE Pioneering Case

Brigham and Women's Hospital (BWH) deployed one of the first CPOE systems in the U.S. in 1993 and has published extensively on its outcomes.

Published results (PubMed, Bates et al. 1999):

- Non-missed-dose medication errors fell **81%** — from 142 per 1,000 patient-days to 26.6 per 1,000 patient-days
- Non-intercepted serious medication errors (those with potential for patient injury) fell **86%**
- Improvements were seen across all error categories: dose errors, frequency errors, route errors, substitution errors, and allergy errors

CPOE at BWH included (PubMed, Bates et al. 2001): required fields, pick lists (structured dropdowns), order sets, standard scales (e.g., insulin sliding scale), drug-drug and drug-allergy interaction alerts,

and adjunct features (pharmacy system integration, online reference access). Physicians entered **85% of orders** electronically.

Key UX lesson: The combination of structural features (required fields + pick lists) with enhanced workflow features (order sets + scales) and automated safety checks produced outcomes no single feature could achieve alone.

6.2 Indications-Based Prescribing Prototype — JAMA Study

The Brigham/JAMA indications-based CPOE prototype study compared a newly designed prescribing interface against two leading commercial EHR vendors in a usability study with 32 physicians, residents, and physician assistants across 8 clinical scenarios.

Results:

- **Error rate**: 5.5% (prototype) vs. 29.7% (vendor system) — a **5.4x error reduction**
- Efficiency and satisfaction both significantly outperformed vendor systems ($p < 0.001$)
- With only **2 minutes of training** on the prototype, all clinicians quickly adopted the new workflow and overwhelmingly preferred it
- Orders included indication **100% of the time** in the prototype vs. rarely in vendor systems
- The prototype automatically filtered drug suggestions based on patient-specific allergies and contraindications

Key design innovation: The interface was organized around the **clinical indication** (why you're prescribing) rather than the drug name (what you're prescribing). Selecting an indication from the problem list surfaced a ranked list of drugs of choice with evidence-based defaults, pre-filtered for the patient's contraindications. This is progressive disclosure from clinical context to drug configuration.

6.3 Chemotherapy Order Set Study

PubMed study (2013) comparing three sequential methods for oncology ordering:

Method	Problem Order Rate	Harm-Capable Error Rate
Handwritten	30.6%	4.2%
Preprinted order sets	12.6%	1.5%
CPOE with embedded order sets	2.2%	0.1%

Order sets reduced problem orders by **14x** compared to handwritten. CPOE did not eliminate all errors and introduced new error types (novel to electronic systems), requiring continued vigilance.

6.4 CPOE Optimization with Order Friction Data — Columbia University

AMIA 2024 study at a large academic medical center using Epic's Order Friction data:

Intervention: Changed default frequency from blank to "Once" for 22 high-volume lab orders (urine, stool, nasal swabs — 15,104 orders/month)

Result: Average changes per order reduced from **1.82 to 0.15** — a **92% reduction in ordering friction** (all 22 orders significant at $p<0.01$)

Lesson: A single configuration change (setting a smart default) eliminated ~25,000 manual field entries per month across the health system. The friction data methodology provides a replicable approach to identifying and fixing high-impact defaults.

6.5 CPOE Errors from Free Text — FDA Study

The FDA's CPOE Medication Safety (CPOEMS) study analyzed CPOE safety across multiple sites and found:

- **Substantial variability** in CPOE displays, functionality, and workflow — between sites, within sites, and even between two different drugs in the same system
- **Drug search problems**: Difficulty finding medications, wrong item selection from similar-looking results, excessive results including non-medications
- **Autocomplete risks**: Despite convenience, autocomplete introduces vulnerability including selecting the wrong drug and overlooking the correct drug
- **Display problems**: Items not visible in dropdown lists without scrolling; inconsistent numerical ordering of doses; brand/generic inconsistency
- **Adjacency errors**: Similar-sounding drugs listed next to each other in ordered lists is a source of wrong-drug errors

6.6 Insulin Discharge Order Set UX Redesign

Case study published in the Journal of the Endocrine Society (PMC 2023) using Figma prototyping:

Problem: 55% of insulin-prescribed discharge patients were missing correct insulin supplies. Prior order set presented 20 separate items for insulin pens, syringes, glucometers, testing strips, lancets, and alcohol pads as a flat list requiring individual selection.

Redesign: Collapsed to 2 choices: "Patient uses insulin pen" or "Patient uses vial and syringe." Each selection automatically pre-populated all associated supplies, default quantities, and referral orders.

Method: Multi-stakeholder collaborative prototyping in Figma with residents, diabetes educators, attendings, pharmacists, and endocrinologists.

Key lesson: The redesign applied progressive disclosure at the macro level (one decision → complete bundle) and at the micro level (supplies automatically populated based on device choice). The result was both faster for prescribers and more complete for patients.

6.7 Mayo Clinic — Barcode Medication Administration

While not a prescribing UX case study, Mayo Clinic's BCMA implementation demonstrates the magnitude of UX-driven safety improvements: adopting barcode medication administration cut reported medication administration errors by **43.5%** with harm events dropping **55.4%**. The mechanism was workflow redesign (nurses must scan both patient wristband and medication), not algorithm change.

6.8 Mobile-Responsive Medication Ordering

Mobile EHR adoption has grown substantially: Empeek research reports that 78% of physicians now use a certified EHR, and mobile-first EHR development is a growing priority. Key mobile-specific findings:

- **64% of nurses** report improved care coordination with mobile healthcare apps
- Barcode scanning on mobile devices can **decrease medication error risk by up to 74.2%** (NDC drug label scanning in e-prescribing workflows)
- Mobile order entry requires progressive disclosure even more than desktop, as screen real estate is dramatically reduced
- Voice dictation is the most efficient input modality for mobile (free-form notes or structured field navigation by voice command)
- Thinkitive's mobile-first EMR design guide identifies key principles: minimize taps, reduce cognitive load, avoid screen clutter, use AI predictive UI to suggest relevant data

Mobile-specific progressive disclosure patterns:

- Accordion sections: expand only the section being actively edited
- Bottom sheets: slide up a configuration panel over the main order summary
- Steppers/wizards: one main decision per screen with clear progress indication
- Contextual keyboard types: numeric keypad for dose, unit selector for route

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7. Synthesis: Implications for CompoundIQ

7.1 Recommended UX Architecture for Medication Ordering

Based on this research, the optimal UX architecture for CompoundIQ's prescription ordering flow should follow a **three-tier progressive disclosure model**:

Tier 1: Quick Order (< 20 seconds)

- Favorites / recent orders surfaced immediately
- Full compound configuration pre-filled; prescriber reviews summary card
- "Approve & Sign" completes the order
- Designed for the 80% of recurring orders that match an established pattern

Tier 2: Guided Order (30–90 seconds)

- Autocomplete search for compound or active ingredient
- Upon selection, drug-specific defaults populate dose/strength/vehicle/route/frequency
- Fields with smart defaults are shown but pre-filled — user only changes deviations
- Conditional fields appear based on selections (e.g., IV delivery triggers diluent/rate fields)
- Advanced options collapsed by default but clearly accessible ("Advanced Options ▼")

Tier 3: Custom Order (1–3 minutes)

- Full field-by-field configuration
- Structured fields for all safety-critical parameters with clear validation
- Free-text fields for pharmacist notes and special instructions in dedicated, clearly labeled areas
- Inline validation prevents submission with safety-critical fields blank

7.2 High-Priority Design Principles

Principle	Implementation
Default to the most common answer	Analyze order history to set smart defaults; every blank field is friction
Show only what's needed now	Conditional field display based on prior selections; hide irrelevant fields
Make common things fastest	Favorites library at the top level; order sets for common compound types
Validate inline, not at submission	Show errors as fields are completed, not after the full form is filled
Never lose user input	If a field is hidden by a prior selection change, preserve but collapse its data
Track what gets changed	Use order friction data to identify defaults that consistently get overridden
Keyboard-first	Full ordering possible without mouse; Tab advances through all visible fields
Structured first, free text as escape valve	Structured fields with "Custom" option; free text flagged for pharmacist review
Match mobile patterns	Accordion sections, single-action screens, bottom sheets for complex inputs

7.3 Specific Feature Recommendations

1. **"Favorites / QuickList"**: Allow prescribers to save named compound configurations from any completed order. Surface in autocomplete. Most important for specialty prescribers who order the same compounds repeatedly.
2. **Compound Type Order Sets**: Pre-build full configurations for common compound categories (pain, HRT, pediatric, wound care). Allow pharmacist administrators to maintain these sets; allow prescribers to customize personal versions.
3. **Indication-First Search Option**: Offer the ability to search by condition/indication ("pain management," "hormone replacement") to surface recommended compound options, reducing the cognitive demand of recalling specific formulations.
4. **Auto-Calculate Quantity from Sig × Duration**: Automatically compute total compound quantity (e.g., grams of topical cream) from application instructions × days supply, eliminating a common source of dispensing error.
5. **BUD/Stability Contextual Display**: When a compound configuration is selected, surface the expected Beyond-Use Date range as an inline advisory (not an alert), so prescribers can inform patients of storage requirements during the ordering step.
6. **Smart Defaults Tuning Dashboard**: Build a pharmacist-facing dashboard showing which defaults get overridden most often and by how much — the same approach as Epic's Order Friction data — to support continuous improvement of pre-filled values.
7. **Inline Drug-Compound Interaction Checking**: Surface ingredient-level interaction warnings (e.g., "selected vehicle inactivates Drug A") as contextual inline text, not as interruptive modal alerts, to reduce alert fatigue.

8. **Mobile-First Tier 1 Experience**: The favorites/quick-order tier should be fully functional on tablet. For clinic-based compound prescribers using tablets, the experience should be optimized for touch, with bottom-sheet configuration panels and large tap targets for approval.

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Section 6: Regulatory Requirements for Electronic Compounding Prescriptions

Research compiled: April 2026

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1. Required Fields on a Compounding Prescription

1.1 Federal Requirements — 21 CFR 1306

The DEA specifies distinct field requirements based on controlled substance schedule. These apply to all prescriptions, including compounded preparations containing controlled substances.

Schedule II Controlled Substances (21 CFR § 1306.05(a)):

#	Required Field
1	Date of issuance
2	Patient's full name
3	Patient's address
4	Drug name
5	Drug strength
6	Dosage form
7	Quantity prescribed
8	Directions for use
9	Practitioner's name
10	DEA registered address of practitioner

#	Required Field
11	Practitioner's DEA registration number
12	Practitioner's signature (no stamps)

Schedule III–V Controlled Substances (21 CFR § 1306.21(a)): Same 12 fields plus:

#	Additional Field
13	Number of refills authorized (maximum: original + 5 refills)

Note on Opioid Treatment (X-waiver): Prescriptions for Schedule III–V narcotic drugs approved for detoxification or maintenance treatment must also include the practitioner's DEA "X number" per 21 CFR § 1306.05(b).

Note on NPI: The NPI (National Provider Identifier) is not explicitly required by DEA regulations under 21 CFR 1306, but is required by many state boards of pharmacy and payers. Most platforms collect it as a standard field.

1.2 Non-Controlled Substance Prescriptions

Non-controlled prescriptions for compounded medications are governed by state pharmacy boards, not DEA. The standard required fields mirror the DEA structure but the DEA registration number is replaced by the prescriber's state license number. Core fields common across virtually all states: patient name/DOB/address, prescriber name/address/credentials, medication name, strength/dosage form, quantity, directions, refills, date.

1.3 Compounding-Specific Required Fields

Compounded prescriptions carry **additional requirements** beyond a standard prescription, driven by federal law (Section 503A of the FD&C Act) and state boards:

(a) "Clinically Necessary" Notation

Under Section 503A of the FD&C Act, for a 503A pharmacy to qualify for exemptions from FDA approval requirements, the prescription must be for an **identified individual patient** and must indicate that the compounded product is clinically necessary for that patient. The Brookings Institution's analysis notes that "the prescriber must include a notation on the prescription that the compounded product is needed as compared to the approved version (e.g., compound without dye, patient allergy)."

Per FDA's final guidance on Section 503A (December 2016), a prescription for a compounded drug product must indicate that the compounded product is "based on the receipt of a valid prescription order or a notation, approved by the prescribing practitioner, on the prescription order that a compounded product is necessary for the identified patient." This effectively creates a required field on every compound prescription: a reason or notation that compounding is clinically necessary for this specific patient.

(b) Formulation Details (Strength, Dosage Form, Ingredients)

Unlike a manufactured drug prescription where the drug name (brand or generic) implies a complete formulation, a compounded prescription must specify:

- The active ingredient(s) by name and strength/concentration
- The dosage form (cream, capsule, oral suspension, injectable, etc.)
- Any special instructions for preparation (e.g., specific base, vehicle, flavorings)

The ingredient list requirement is at the **pharmacy level** (internal compounding records under USP 795/797), not necessarily on the face of the prescription. However, in practice, the prescriber must specify the formulation clearly enough that the pharmacy can compound it, which requires listing the active ingredients and their concentrations. For multi-ingredient compounds, all ingredients and their quantities must be communicated.

(c) "Essentially a Copy" Differentiation

Both federal law and state boards (notably California since October 2025) require documentation that the compound is **not essentially a copy** of a commercially available product, or if it is similar, why clinical differences justify compounding. California's October 2025 regulations place an affirmative duty on the pharmacist to verify and document a "clinically significant difference" — but this starts with prescriber documentation on the prescription.

1.4 Summary: Required Fields for a Compounding Prescription (UI Checklist)

Field	Controlled Substance	Non-Controlled	Compounding-Specific
Patient full name	Required	Required	—
Patient date of birth	State-dependent	State-dependent	—
Patient address	Required	Required	—
Date of issuance	Required	Required	—
Drug/compound name	Required	Required	—
Active ingredient(s) + strength	Required	Required	Critical for compounding
Dosage form	Required	Required	Must specify (not implied)
Quantity prescribed	Required	Required	—
Directions for use (sig)	Required	Required	May need extended text
Refills authorized	Sched III–V only	State-dependent	—
Prescriber name	Required	Required	—
Prescriber address	Required	Required	—
Prescriber DEA number	Required (CS only)	N/A	—

Field	Controlled Substance	Non-Controlled	Compounding-Specific
Prescriber NPI	State/payer requirement	State/payer requirement	—
Prescriber signature	Required	Required	—
Clinical necessity notation	Not required by DEA	N/A	**Required for 503A**
Reason compounding is necessary (vs. commercial product)	Not required by DEA	N/A	**Required for 503A compliance**

2. EPCS: Electronic Prescribing of Controlled Substances

2.1 Federal Framework — 21 CFR Part 1311

The DEA's EPCS regulations were established by interim final rule published March 31, 2010, effective June 1, 2010, and are codified at 21 CFR Parts 1300, 1304, 1306, and 1311. EPCS applies to all Schedule II–V controlled substances, including compounded preparations containing them.

Key points:

- EPCS is **not mandatory** at the federal level from DEA's perspective — practitioners may still write and manually sign paper prescriptions (DEA EPCS FAQ)
- However, many states now mandate EPCS for some or all controlled substances (see Section 5)
- An electronic controlled substance prescription is only valid if both the prescriber application and the receiving pharmacy application meet all 21 CFR Part 1311 requirements
- Applications must be certified by a DEA-approved third-party auditor before use (DEA Approved Certification Processes)

2.2 Two-Factor Authentication Requirements

Per 21 CFR § 1311.115 and confirmed by the DEA EPCS FAQ, two-factor authentication (2FA) is required for signing all EPCS prescriptions. 2FA uses two of three factor categories:

Factor Category	Examples	Technical Requirements
Something you **know**	Password, PIN	Cannot be used alone — username + password = one factor only
Something you **have**	Hard token, cryptographic device	Must meet FIPS 140-2 Security Level 1
Something you **are**	Biometric (fingerprint, iris, facial)	False match rate ≤ 0.001 (1 in 1,000); meets FIPS-201 PIV

Important: A username + password combination is considered **one factor** (both are "something you know") and does not satisfy 2FA requirements. An additional factor — hardware token or biometric — is always required (Imprivata).

Identity proofing must be conducted at NIST SP 800-63-1 Assurance Level 3 or higher, by a federally approved Credential Service Provider (CSP) or Certification Authority (CA) (DEA EPCS FAQ). Remote identity proofing at IAL2 is acceptable and was reinforced by DEA guidance.

2.3 Which Compounded Medications Are Controlled?

Drug	Schedule	DEA Citation
Testosterone (all forms, including compounded injectable, cream, gel, pellet)	Schedule III (anabolic steroid)	21 CFR § 1308.13
Ketamine (including compounded lozenges, nasal spray, injectable)	Schedule III	21 CFR § 1308.13(c)(7)
Buprenorphine (compounded formulations)	Schedule III	21 CFR § 1308.13
Codeine (products with < 90mg/dosage unit)	Schedule III	21 CFR § 1308.13
Diazepam, alprazolam, clonazepam (compounded)	Schedule IV	21 CFR § 1308.14
Tramadol (compounded)	Schedule IV	21 CFR § 1308.14
Testosterone cypionate, enanthate (most compounded androgens)	Schedule III	DEA drug scheduling

All common telehealth compounding categories — testosterone replacement therapy (TRT), ketamine mental health treatment, and many hormone optimization compounds — involve **controlled substances requiring EPCS compliance**.

2.4 How EPCS-Certified Systems Handle Compound Prescriptions

EPCS-certified systems like Surescripts handle compound prescriptions through the NCPDP SCRIPT Standard v2017071 (mandatory since January 2020), which added compound prescription support allowing **up to 25 active ingredients** per electronic prescription (Surescripts press release, May 2021). Prior versions (v10.6 and earlier) had no mechanism for compound prescriptions, leading prescribers to "shoehorn" compound information into electronic prescriptions in ways that caused pharmacy confusion (Surescripts CE article).

For compounded controlled substances sent via EPCS:

- The prescription must still satisfy all DEA field requirements (21 CFR Part 1306)
- Two-factor authentication by the prescriber is required at signing
- The transmitted EPCS record must be retained electronically and cannot be converted to fax
- The pharmacy application must be separately DEA-certified (DEA EPCS FAQ)

Not all compounding pharmacies are connected to the Surescripts network or EPCS-certified. Many specialty compounding pharmacies that primarily serve telehealth platforms maintain their own proprietary intake systems alongside or instead of Surescripts integration.

2.5 Exemptions for Compounded Controlled Substances

There are no general exemptions from EPCS requirements specifically for compounded controlled substances. However, there are **exemptions from the e-prescribing mandate** for compounded medications in states that otherwise require EPCS:

- **Utah**: Exempts prescriptions for "a medication that requires compounding two or more ingredients" from the mandatory EPCS requirement (NABP Utah newsletter, February 2024)
- **Other states** with mandatory EPCS may have similar carve-outs — this is a state-by-state issue

The DEA itself does not create a compound-specific exemption: if a practitioner chooses to e-prescribe a compounded controlled substance, all 21 CFR Part 1311 requirements apply in full.

Fax exception for Schedule II narcotics for direct administration: Under 21 CFR § 1306.11(e), Schedule II narcotic prescriptions for compounded preparations intended for **direct administration to a patient** (e.g., infusion, injection in a clinical setting) may be faxed as the original. This is a narrow exception and does not apply to compound prescriptions dispensed directly to a patient.

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3. DEA UI Requirements (21 CFR 1311)

3.1 What 21 CFR 1311 Mandates for Application UI/UX

21 CFR § 1311.120 is the primary source for specific UI requirements imposed on electronic prescription applications. These are binding technical and design requirements for any EPCS-certified system.

(a) Prescriber Identity and DEA Linkage

- The application must link each prescriber by name to at least one DEA registration number
- If the prescriber has multiple DEA registrations (e.g., multi-state practice), the system must require selection of the correct DEA number for each prescription
- DEA registration must be valid and current before access controls can be set

(b) Logical Access Controls (Two-Person Rule)

- The system must support setting access controls by **individual name or role**
- **Two separate people** must be involved in setting or changing access controls for EPCS: one enters the data, one approves it — no single person can both grant and approve their own access (21 CFR § 1311.125)
- Access must be promptly revoked when a practitioner loses DEA registration or state licensure

(c) Required Pre-Signing Display to Prescriber

Before a controlled substance prescription can be signed, the application **must display all of the following** to the prescriber for review (21 CFR § 1311.120(b)(9)):

Required Display Element
Date of issuance
Full name of the patient
Drug name
Dosage strength and form
Quantity prescribed
Directions for use
Number of refills authorized (Schedule III–V)
Earliest fill date (for multiple prescriptions per § 1306.12(b))
Prescriber's name, address, and DEA registration number
The mandatory attestation statement (see below)

(d) Mandatory Attestation Statement

The review screen must display the following statement (or its **substantial equivalent**) per 21 CFR § 1311.140(a)(3):

> *""By completing the two-factor authentication protocol at this time, you are legally signing the prescription(s) and authorizing the transmission of the above information to the pharmacy for dispensing. The two-factor authentication protocol may only be completed by the practitioner whose name and DEA registration number appear above.""*

No additional keystroke acknowledgment is required — completing the 2FA protocol itself constitutes agreement. However, the system must not allow any other person to complete the protocol on the named prescriber's behalf.

(e) "Ready to Sign" Workflow

- The prescriber must **affirmatively indicate** that each prescription is ready to be signed (21 CFR § 1311.120(b)(10))
- After this indication, the system must **not permit alteration of DEA-required elements** without forcing a new review and re-indication of readiness
- Any alteration of required fields after digital signing **cancels the prescription**
- Multiple prescriptions for the same patient can be signed with a single 2FA invocation, provided each was individually indicated as ready while its full details were displayed

(f) Two-Factor Authentication at Point of Signing

The 2FA credential use at signing serves as the prescriber's legal digital signature. The system must (21 CFR § 1311.140(a)(4)):

- Prompt for 2FA after the prescriber indicates readiness to sign
- Complete the 2FA protocol while the prescription details remain displayed

- Timestamp the prescription when signing occurs
- Synchronize system time within **5 minutes of official NIST time**

(g) Digital Signature and Archiving Requirements

Per 21 CFR § 1311.120(b) and 21 CFR § 1311.145:

- The cryptographic module for signing must meet **FIPS 140-2 Security Level 1**
- Digital signature algorithm and hash function must comply with **FIPS 186-3** and **FIPS 180-3**
- The private key must be stored encrypted on a FIPS 140-2 validated module
- The digitally signed record must be electronically archived
- The system must not transmit a prescription unless the digital signing function has been used

3.2 Audit Trail Requirements

Under 21 CFR § 1311.120(b)(23)–(25) and § 1311.150, the system must maintain a comprehensive, tamper-evident audit trail capturing:

Auditable Events:

- Creation, alteration, indication of readiness for signing, signing, transmission, or deletion of any controlled substance prescription
- Setting or changing of logical access control permissions
- Failed transmission notifications
- Authentication events (successes and failures)
- Administrative changes (access grants, revocations, updates)

Required Data in Each Audit Record:

- Date and time of the event
- Type of event
- Identity of the person taking the action
- Outcome (success or failure)

Retention and Reporting:

- Records must be retained electronically for **at least two years** (21 CFR § 1311.305)
- The system must conduct **automated internal audits** at least daily and generate reports of security incidents
- If a compromise is identified, it must be reported to the application provider and **DEA within one business day**
- The system must generate monthly logs of controlled substance prescriptions issued, provided to the practitioner within 7 days after month-end; on-demand logs covering at least the previous 2 years must also be available

3.3 Third-Party Certification Requirement

Before any application can be used to transmit EPCS prescriptions, a **DEA-approved third-party auditor** must certify that the application meets all 21 CFR Part 1311 requirements. The DEA maintains a list of approved certification organizations. Re-certification is required every two years or whenever significant

changes are made to the application.

4. Compounded vs. Manufactured Medications — Regulatory Differences

4.1 Prescribing a Compounded vs. Manufactured Medication

The core legal difference is that compounded medications are **not FDA-approved** and receive no pre-market safety, efficacy, or quality review. This creates several downstream differences in the prescribing and transmission workflow:

Dimension	Manufactured Drug	Compounded Drug (503A)
FDA approval status	FDA-approved	Not FDA-approved
Drug identification	Standard NDC code	No NDC; formulation must be fully specified
Prescribing authority	Any licensed prescriber	Same, but with additional documentation requirements
"Clinically necessary" notation	Not required	Required (Section 503A)
Can be an "essentially a copy"	N/A	Restricted (must differ meaningfully from commercial product)
Patient-specific prescription	Not always required	Always required for 503A
Sig character limit (NCPDP SCRIPT)	140 chars (v10.6) / 1,000 chars (v2017071)	1,000 chars (v2017071 required for compounds)
E-prescribing standards	Full NCPDP SCRIPT support	NCPDP SCRIPT v2017071+ required (compound field support)

Per the Brookings Institution analysis, the FDA oversight asymmetry is significant: manufactured drugs face pre-market review; 503A compounds are primarily overseen by state boards of pharmacy with no routine FDA inspection or pre-market quality assessment.

4.2 Can Compounded Medications Be E-Prescribed via Surescripts?

Yes, since January 2020 — but with important limitations.

NCPDP SCRIPT Standard v2017071, which became mandatory for Medicare Part D on January 1, 2020, added compound prescription support for the first time. It allows for the drug name and quantity of up to 25 different ingredients in a single electronic prescription. Prior to this, no version of NCPDP SCRIPT was designed to handle compound prescriptions, causing widespread "shoehorning" of compound data into standard free-text fields (Surescripts CE article).

Practical limitations that still exist:

1. **Pharmacy connectivity**: Not all compounding pharmacies are connected to the Surescripts network. Many specialty compounding pharmacies serving telehealth platforms operate outside the standard Surescripts network and have proprietary or direct API integrations with prescribing platforms.
2. **NDC codes**: Surescripts and standard pharmacy systems rely on NDC codes to identify drugs. Compounded preparations have no NDC. While v2017071 added structured compound fields, many pharmacy systems still struggle with non-NDC prescriptions in their dispensing workflow.
3. **Formulary checking**: Real-time drug interaction checks, formulary lookups, and benefit checking tools built on NDC databases do not work for compounded medications.
4. **Controlled substance compound e-prescribing**: Possible under v2017071 with EPCS certification, but both the sending EHR and the receiving pharmacy system must be EPCS-certified. Not all compounding pharmacies have this capability.
5. **Sig complexity**: Compounded preparations often require elaborate dosing instructions (e.g., topical application techniques, injection volumes, titration schedules) that can exceed even the expanded 1,000-character sig field.

4.3 When Is Fax Still Required (and Why)?

Fax remains common or required for compounded prescriptions in several scenarios:

Scenario	Why Fax Is Used
Pharmacy not on Surescripts network	No electronic connection available
Compounding pharmacy uses proprietary intake	Surescripts integration not built or maintained
State compounding exemption from e-prescribing mandate	Some states (e.g., Utah) exempt compound prescriptions from EPCS mandate
Schedule II for direct administration	21 CFR § 1306.11(e) allows fax for Schedule II narcotics for direct patient administration
Technical system failure	E-prescribing regulations require fax fallback capability
Older EHR not updated to v2017071	Cannot generate structured compound prescription electronically

Important: Per DEA guidance (November 2021), any prescription generated by computer and faxed is a **paper prescription** and must be manually signed. Computer-generated faxes that are not manually signed are not valid. This means that "fax from a prescribing platform" workflows require a printed-and-signed step, unless the platform transmits as a true EPCS.

4.4 Limitations of E-Prescribing Standards for Compounds

The NCPDP SCRIPT standard analysis identifies several ongoing limitations relevant to compound prescriptions:

- **No standard compound drug codes**: Unlike manufactured drugs (which use NDC), there is no universal coding system for compounded drug formulations. This impedes clinical decision support and drug interaction checking.

- **Free-text Sig still dominates**: Despite structured Sig fields in v2017071, only ~10% of prescriptions used structured Sig as of 2020. Complex compound instructions are almost always free text.
- **Preparation instructions**: NCPDP SCRIPT has no standard mechanism for conveying preparation instructions to the pharmacist (e.g., "use propylene glycol 30% base" or "sterile filter required").
- **Base/vehicle specification**: There is no standard field for specifying the compounding vehicle or base, which is critical for topical compounds.
- **Beyond-use dates**: BUD requirements from USP 795/797 are not part of the prescription transmission standard; they are handled internally by the pharmacy.

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5. State Board of Pharmacy Regulations

5.1 State E-Prescribing Mandates

The national landscape is rapidly evolving toward mandatory e-prescribing. As of early 2024, 35 states require e-prescribing in some form by state law (NABP, January 2024):

Mandate Type	States
All prescriptions (CS and non-CS) mandated electronically	California, Delaware, Florida, Iowa, Michigan, Minnesota, New York
All controlled substances mandated electronically	Illinois, Rhode Island, South Carolina, and others
Certain CS only (often opioids)	Kansas, Maine, Virginia, Arizona, Tennessee, and others
No state e-prescribing mandate	Remaining ~15 states (as of 2024)

CMS Medicare Part D requirement: CMS requires that prescriptions for most controlled substances under Medicare Part D be sent electronically using NCPDP SCRIPT Standard v2017071 (Surescripts, 2021).

5.2 Compounding Exemptions from E-Prescribing Mandates

Some states with mandatory e-prescribing create specific exemptions for compounded preparations:

- **Utah**: Explicitly exempts "a medication that requires compounding two or more ingredients" from mandatory EPCS, with documentation requirement on the hard copy (NABP Utah newsletter, February 2024)
- Other states with blanket e-prescribing mandates may not have explicit compound exemptions, meaning compound prescriptions must be e-prescribed in those states unless another valid exemption (e.g., technical failure) applies

5.3 Wet Signature / Fax Still Required

Even under a state e-prescribing mandate, the following remain valid fallbacks when exceptions apply:

- Technical failure of the e-prescribing system (with documentation)
- Prescriber licensed in another jurisdiction (in some states)

- Research protocol prescriptions
- Some state-specific carve-outs for compounds (as above)

Schedule II controlled substances **without EPCS** require paper with a **wet (manual) signature** — computer-generated and computer-faxed prescriptions must still be manually signed (DEA, November 2021).

5.4 State-Specific Fields Required on Compounding Prescriptions

Beyond federal requirements, states add various fields or attestations:

State	Notable Additional Requirement
California (as of Oct 2025)	Pharmacist must verify and document "clinically significant difference" vs. commercially available product (Frier Levitt, January 2026); prescriber notation triggers this obligation
Florida (SB 860/HB 877, proposed 2026)	Active pharmaceutical ingredient (API)-sourcing documentation and certification requirements for certain compounded drugs, including weight-loss medications
Indiana (SB 282, proposed 2026)	Tightened bulk drug substance compounding, new medical spa registration regime
Most states	Require prescriber's state license number on prescription (NPI increasingly common)
Some states	Require prescriber DEA number even for non-controlled compounded substances if compound contains a controlled substance precursor

Workers' compensation programs in many states require additional fields for compound prescriptions billed to WC:

- Individual ingredient NDCs, quantities, and calculated reimbursement values (Arizona, Nevada, Texas)
- Prior authorization documentation for all compounds (Texas, Montana, Louisiana) (Optum state-by-state analysis, February 2025)

5.5 Interstate Compounding — 503A vs. 503B

The interstate shipping rules differ fundamentally between 503A and 503B:

503A Pharmacies (traditional patient-specific compounders):

- May not distribute compounded drugs outside their state **in excess of 5% of total prescriptions** unless the state has entered an MOU with FDA (NCBI / National Academies of Sciences analysis)
- Must be licensed in each state where they ship to patients
- Licensing requires separate applications, inspections, and compliance with each state's pharmacy law — typically 3–12 months per state (Newtropin, February 2026)
- State-by-state heterogeneity is substantial: some states prohibit sterile office stock compounding, some allow it with restrictions

503B Outsourcing Facilities:

- Can ship interstate without patient-specific prescriptions (Frier Levitt)
- Subject to FDA CGMP requirements and risk-based inspections
- States regulate 503B facilities inconsistently — some treat them as pharmacies (New York), some as manufacturers (Texas), some have distinct frameworks (California, New Hampshire) (FDLI, October 2022)
- 503B facilities must register in every state they ship to for patient-specific prescriptions if those states require pharmacy licensure

UI Design Implication: A compounding prescription platform serving multiple pharmacies must be aware of and potentially flag when a patient's state may not be serviced by a given pharmacy, particularly for 503A facilities.

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6. HIPAA and PHI Considerations

6.1 Coverage Status

All entities in the compounding prescription ecosystem are HIPAA covered entities or business associates:

- **Prescribing telehealth platform**: Covered entity (health care provider) and/or business associate depending on structure
- **Compounding pharmacy**: Covered entity — pharmacies that transmit PHI in connection with HIPAA standard electronic transactions (claims, eligibility, e-prescribing) qualify as covered entities under 45 CFR § 160.103
- **E-prescribing intermediaries** (e.g., Surescripts): Business associate
- **Technology vendors** (cloud hosting, EHR, API services): Business associates

6.2 Patient Data in Transit — Prescription Transmission

NCPDP SCRIPT e-prescribing transactions carry PHI and must be protected under the HIPAA Security Rule even though SCRIPT is not itself a named HIPAA standard transaction (Accountable, January 2025):

- All transmission channels must use encryption in transit (TLS 1.2+)
- Access to prescriptions in transit must be restricted to authorized parties
- Audit logs of all prescription transmissions must be maintained
- No PHI may be included in transmission metadata visible to unauthorized parties

For fax transmissions: HIPAA-compliant electronic fax (not plain fax) is standard; the fax must be sent to a verified secure recipient number with cover sheet protections.

6.3 Patient Shipping Address Handling

Shipping address is PHI when combined with patient name and treatment information. For compounding pharmacies shipping medications to patients:

Minimum Necessary Standard (45 CFR § 164.502(b)):

- Only the shipping address data necessary for delivery should be shared with shipping carriers and logistics vendors

- The carrier (UPS, FedEx, USPS) receives only: patient name, address, weight, dimensions — not health information
- **Nothing on the exterior of the package may reveal the nature of the medication or health condition** (e.g., no "Testosterone Pharmacy" return addresses, no medication names on labels)

Packaging and labeling:

- Exterior label: patient name and address only; neutral return address using pharmacy's general name, not a specialty name that implies a diagnosis
- Tracking information shared with platforms via API should be treated as PHI-adjacent data if it can be correlated back to a prescription

Special sensitivity for certain compound categories:

- Testosterone, ketamine, and hormonal compounds may be considered particularly sensitive PHI as they can reveal mental health conditions, gender-affirming care, or reproductive health status
- Some states (California, New York, Texas) have additional state privacy laws beyond HIPAA that may apply to certain medication categories

6.4 Minimum Necessary Standard for Compounding Orders

In the compounding prescription context, the minimum necessary standard applies to what data each entity receives:

Entity	Minimum Necessary PHI
Prescriber platform	Full prescription data + patient clinical history needed for safe prescribing
Compounding pharmacy	Full prescription data (all required fields) + shipping address
Shipping carrier	Name, shipping address, package dimensions — no clinical data
Analytics/reporting tools	De-identified or aggregated data where possible
Payment processors	Billing data only; no clinical details beyond diagnosis codes where required

6.5 Business Associate Agreement (BAA) Requirements

Under 45 CFR § 164.504(e), a BAA must be executed before any business associate can access PHI. The full chain requires BAAs between:

Relationship	BAA Required?
Telehealth platform ↔ Compounding pharmacy	**Yes** (pharmacy is BA of platform, or both are covered entities — treatment exception may apply; see below)
Telehealth platform ↔ EHR/e-prescribing software vendor	**Yes**
Telehealth platform ↔ Cloud infrastructure provider	**Yes**

Relationship	BAA Required?
Telehealth platform ↔ Surescripts (or other network)	**Yes**
Prescribing provider → Compounding pharmacy (for treatment)	**Possible exception** — under 45 CFR § 164.502(e)(1), disclosures from a covered entity to a healthcare provider for treatment purposes do not require a BAA (Holland & Hart, October 2023); however, the telehealth platform intermediary likely does require one
Compounding pharmacy ↔ Shipping vendor handling PHI	**Yes**
Compounding pharmacy ↔ API integration partners	**Yes**

A BAA must specify:

- Permitted uses and disclosures of PHI
- Required safeguards
- Breach reporting obligations and timelines
- Data return or destruction at end of relationship
- Right-to-audit provisions
- Flow-down obligations to subcontractors

Common platform error: Launching a telehealth prescribing platform without executing BAAs with every vendor in the stack. Per Compliancy Group, telehealth provision does not alter a covered entity's HIPAA obligations — all standard requirements apply.

6.6 Security Rule Technical Safeguards

For the prescribing platform and pharmacy systems:

- **Access controls**: Unique user IDs, automatic logoff, encryption of ePHI at rest and in transit
- **Audit controls**: Hardware, software, and procedural mechanisms to record and examine ePHI access
- **Integrity controls**: Electronic mechanisms to ensure ePHI is not improperly altered or destroyed
- **Transmission security**: Encryption for all ePHI transmitted over open networks

For EPCS-specific systems, the DEA audit trail requirements (Section 3.2 above) and HIPAA audit requirements overlap substantially — a single robust audit log system can satisfy both sets of requirements.

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7. Summary: Key UI Design Implications

The following table distills the regulatory requirements into actionable UI/UX design constraints for an electronic compounding prescription platform:

Design Area	Requirement	Regulatory Source
Required prescription fields	Must collect all 12–13 DEA fields for CS; clinical necessity notation mandatory for all compounds	21 CFR § 1306.05; FDA Section 503A guidance
Compound-specific fields	Active ingredient(s) + strength, dosage form, basis for clinical necessity, reason compounding vs. commercial product	FDA 503A guidance; state boards
Review screen before signing	Must display all required DEA fields + mandatory attestation statement — no signing without this screen	21 CFR § 1311.120(b)(9)–(10)
Mandatory attestation	The specific DEA attestation language must appear on the signing screen	21 CFR § 1311.140(a)(3)
Two-factor authentication	2FA required at signing for all CS; must use 2 of 3 factor types (know/have/are); password alone is insufficient	21 CFR § 1311.115
Post-signing immutability	Fields cannot be altered after digital signing; any alteration must cancel the prescription and require re-creation	21 CFR § 1311.120(b)(19)
Logical access controls	Two-person rule for granting prescribing access; cannot be set by one person alone	21 CFR § 1311.125
Audit trail	Every create/edit/sign/transmit/delete event must be logged with timestamp, user identity, and outcome	21 CFR § 1311.120(b)(23)–(25)
Prescription log	Monthly CS prescription logs delivered to prescribers within 7 days of month-end; on-demand 2-year lookback	21 CFR § 1311.120(b)
No DEA element alteration after ready-to-sign	Once prescriber clicks "ready to sign," DEA elements are locked; any change requires re-review	21 CFR § 1311.120(b)(10)
NCPDP SCRIPT v2017071+	Required for electronic compound prescriptions; supports up to 25 ingredients	CMS Final Rule 2018; Surescripts
State license validation	Must verify pharmacy is licensed in patient's state before routing compound prescription	503A state licensing requirements
PHI on packaging	Shipping integrations must not expose medication name or clinical information on exterior labels	HIPAA minimum necessary standard
BAA execution	Every vendor in the stack touching PHI must sign a BAA before go-live	45 CFR § 164.504(e)

Design Area	Requirement	Regulatory Source
Fax fallback	Platform must support fax output for pharmacies not on EPCS network; all faxed CS prescriptions require manual signature	21 CFR § 1306.05(d); DEA 2021 guidance
EPCS certification	Third-party DEA audit required before platform can transmit CS prescriptions electronically	21 CFR § 1311.300; DEA certification list
Time synchronization	System clock must stay within 5 minutes of NIST time	21 CFR § 1311.120(b)

Sources referenced throughout this document are cited inline. All federal regulatory citations are to the Code of Federal Regulations as published at law.cornell.edu/cfr and the DEA Diversion Control Division.

Section 7: Real-World Compounding Prescription Examples

Compiled: April 7, 2026 **Purpose:** Document the full complexity of compounded medication configurations for prescribing UI design — covering variants, dose ranges, formulations, protocols, and cascading data dependencies.

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1. **Semaglutide (Compounded)**
2. **Testosterone (Compounded)**
3. **Low-Dose Naltrexone (LDN)**
4. **BPC-157 (Body Protection Compound)**
5. **UI Data Field & Cascading Dependency Analysis**

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1. Semaglutide (Compounded)

1.1 Regulatory Context (Critical — Read First)

Compounded semaglutide existed primarily under FDA shortage-based authorizations. The timeline:

- **2022–Feb 2025:** Semaglutide on FDA drug shortage list → 503A pharmacies and 503B outsourcing facilities permitted to compound under shortage exemption.
- **Feb 21, 2025:** FDA declared semaglutide shortage resolved. All doses of injectable semaglutide (Ozempic/Wegovy) removed from shortage list.
- **April 22, 2025:** Enforcement discretion period ended for 503A compounding pharmacies ("essentially a copy" products). Source: Burr & Forman LLP
- **May 22, 2025:** Enforcement discretion period ended for 503B outsourcing facilities. Source: Alston & Bird
- **As of early 2026:** Legal status is in flux. Enforcement against mass-marketed compounded products has escalated. Warning letters, DOJ referrals, and enforcement actions have accelerated.
- **Important exception:** Compounders may still compound semaglutide that is **not essentially a copy** of an approved drug — for example, products meeting documented individual patient needs (different dose, different inactive ingredients due to allergic reaction, or documented medical necessity). Source: Burr & Forman LLP
- **Salt form controversy:** FDA has explicitly warned that semaglutide sodium and semaglutide acetate are unapproved salt forms different from the base semaglutide in approved drugs. Many pharmacies were using these salts; FDA stated it is "not aware of any basis" for compounding with salt forms.

1.2 Available Formulations (Injection)

The predominant compounded form is **subcutaneous injectable** in multi-dose vials.

Form	Notes
Subcutaneous injectable (primary)	Multi-dose vials, most common
Sublingual drops/ODT	Some pharmacies; absorption data limited
Oral capsule	Available; bioavailability lower than injection
Nasal spray	Offered by some providers
Lyophilized powder (reconstitutable)	Less common; requires bacteriostatic water

Sources: GoodRx, Eden Health

1.3 Concentrations and Vial Sizes

Common concentrations (single-agent semaglutide base):

Concentration	Typical Vial Sizes	Notes
2.5 mg/mL	1 mL, 2 mL, 2.5 mL, 3 mL	Most common; 3 mL vial = 7.5 mg total
5 mg/mL	1 mL, 2.5 mL, 4.5 mL	Higher concentration; fewer units per dose
1 mg/mL	1 mL, 2.5 mL	Lower concentration for microdosing protocols

Sources: Flow Wellness, Rivas Medical Weight Loss, Strive Pharmacy FAQ

Dose-to-unit conversion (U-100 insulin syringe):

Dose (mg)	At 2.5 mg/mL	At 5 mg/mL
0.25 mg	10 units	5 units
0.5 mg	20 units	10 units
1.0 mg	40 units	20 units
1.7 mg	68 units	34 units
2.4 mg	96 units	48 units

Source: Eden Health dose chart

Critical UI note: A "units" value is meaningless without the concentration. Selecting concentration must cascade into auto-calculated unit display.

1.4 Combination Formulations

Semaglutide + Cyanocobalamin (Vitamin B12):

- Most common combination; Empower Pharmacy is a major compounder
- Concentrations: 1/0.5 mg/mL or 5/0.5 mg/mL (sema/B12)
- Vial sizes: 1 mL, 2.5 mL
- B12 fixed at 0.5 mg/mL across all strengths
- Rationale: B12 for energy support and potential nausea mitigation during GLP-1 therapy
- Source: Empower Pharmacy — Semaglutide/Cyanocobalamin

Semaglutide + L-Carnitine:

- Combination for metabolic/fat-burning support
- Available as injectable and oral formulations
- Sources: Seattle Plastic Surgery, Good Day Pharmacy

Semaglutide + B12 + L-Carnitine (triple combination):

- Oral and injectable forms available
- Source: Fridays oral formulation

Semaglutide + Glycine + B12:

- Glycine added for nausea/GI management and muscle preservation
- Source: Strive Pharmacy

Semaglutide + NAD+:

- Available from select pharmacies for broader metabolic/longevity support
- Source: Mochi Health — GLP-1 Additives

Other additive options reported:

- Methylcobalamin (active B12 form)
- Vitamin B6 (anti-nausea)
- Glycine alone

1.5 Standard Titration Protocols

Standard Ozempic-style titration (diabetes/weight loss):

Phase	Dose	Duration
Initiation	0.25 mg/week	Weeks 1–4
Dose 2	0.5 mg/week	Weeks 5–8
Dose 3	1.0 mg/week	Weeks 9–12
Dose 4 (Wegovy weight loss)	1.7 mg/week	Weeks 13–16
Maintenance	2.4 mg/week	Week 17+

Source: Eden Health, Mountcastle Medical Spa

Microdosing / ultra-slow titration protocol:

Phase	Dose	Duration
Start	0.05 mg/week	Weeks 1–4
Step 2	0.1 mg/week	Weeks 5–8
Step 3	0.2 mg/week	Weeks 9–12
Continue	+0.1 mg every 4 weeks	Until effective dose

Source: Mountcastle Medical Spa

Month-by-month syringe kit protocol (pharmacy-packaged):

Month	Dose	Injections
Month 1	0.25 mg	4× weekly
Month 2	0.5 mg	4× weekly
Month 3	1.0 mg	4× weekly
Month 4	1.75 mg	4× weekly
Month 5+	2.4–2.5 mg (maintenance)	4× weekly

Source: VITAstir Clinic kit, Rock Ridge Pharmacy

1.6 How Pharmacies Package Titration Kits

Two primary approaches observed in the market:

Approach A — Single large vial with provider-directed dose escalation:

- Patient receives one multi-dose vial (e.g., 10 mg or 20 mg total)
- Provider gives written instructions on how to increase dose week-by-week
- Patient uses same vial throughout the protocol until empty
- Strive Pharmacy 5–20 mg vials described as "typically lasting 3 months"
- Source: Strive Pharmacy FAQ

Approach B — Monthly sequential vials/kits at different concentrations:

- Patient receives a new vial each month pre-filled at the month's target dose
- VITAstir Clinic packages "Month 1" through "Month 5+" home kits separately
- Some pharmacies offer prefilled syringes (rather than vials) per dose-increment
- Source: VITAstir Clinic, Rock Ridge Pharmacy

Approach C — Lyophilized powder vials (reconstitution required):

- Patient reconstitutes with bacteriostatic water before first use
- Requires separate mixing syringe (3–5 mL, 18-gauge) plus insulin syringe for injection

- Source: FormBlends Starter Kit

1.7 UI Data Fields — Semaglutide Cascade Dependencies

```
Drug: Semaglutide
■■ Formulation type
■■ Injectable (primary)
■ ■■ Single-agent vs. Combination
■ ■ ■■ [If combination] → Select additive(s): B12/Cyanocobalamin, Methylcobalamin,
■ ■ ■ L-Carnitine, Glycine, NAD+, B6
■ ■ ■■ [If combination] → Additive concentration (e.g., B12 0.5 mg/mL fixed)
■ ■■ Semaglutide concentration → [1 mg/mL | 2.5 mg/mL | 5 mg/mL]
■ ■■ Vial size → [1 mL | 2 mL | 2.5 mL | 3 mL | 4.5 mL | 5 mL]
■ ■ (Available sizes cascade from concentration selection)
■ ■■ Starting dose (mg) → auto-calculates units based on concentration
■ ■■ Titration protocol → [Standard 4-week steps | Microdose | Custom]
■ ■ ■■ If standard → dose escalation schedule displayed/confirmed
■ ■■ Reconstitution required? [Y/N] → [Y] triggers BAC water fields
■■ Sublingual
■■ Oral capsule
■■ Nasal spray

Required prescription context fields:
- Salt form confirmation: Must be semaglutide BASE (not sodium, not acetate)
- Clinical justification for compounding (post-shortage):
[Different dose | Allergy to ingredient | Other documented need]
- 503A vs 503B pharmacy designation
```

2. Testosterone (Compounded)

2.1 Salt Forms / Esters

Ester/Form	Half-Life	Primary Use	Notes
Testosterone Cypionate	~8 days	Injectable TRT (men)	Most common for TRT in US; IM or SubQ
Testosterone Enanthate	~5–7 days	Injectable TRT (men)	Slightly shorter than cypionate; IM or SubQ
Testosterone Propionate	~1–2 days	Injectable (frequent dosing)	Fast-acting; requires q2–3 day injections
Testosterone Undecanoate	~21 days	Long-acting injectable	Not commercially compounded in US typically
Testosterone Free Base	N/A (topical)	Cream, gel	No ester; used for topical/transdermal absorption

Sources: ExcelMale TRT Forum, MediVera Compounding

2.2 Dosage Forms

Form	Concentrations Available	Route	Notes
Injectable solution	20–200 mg/mL	IM or SubQ	Oil-based; multi-dose vials
Cream (transdermal)	1–250 mg/mL (1 mg/mL increments up to 20, then 30/40/50/100/150/200/250)	Topical	Pump dispenser; penetration enhancers
Gel	1–20 mg/mL	Topical	Similar to cream; various bases
Troche/Lozenge	2.5 mg, 5 mg, 10 mg, 100 mg, 200 mg	Sublingual/buccal	Dissolve in mouth
Pellet/Implant	75 mg/pellet (Testopel)	SubQ implant	q3–6 months; 2–6+ pellets at a time
Capsule (oral)	1 mg slow-release (women); variable	Oral	Lower bioavailability
Vaginal cream	20 mg/mL (30 mL dispenser)	Intravaginal	Women; 0.25–0.5 mL per use
Suppository	0.5 mg vaginal (with estriol)	Vaginal	Women only

Sources: Empower Pharmacy Cream, Empower Pharmacy Troche, Empower Pharmacy Vaginal Cream, NCBI/NIH Compounded BHT Table, Bayview Pharmacy Women

2.3 Injectable: Concentration Ranges and Vial Sizes

Testosterone Cypionate (single agent, Empower Pharmacy):

Concentration	Vial Sizes Available
20 mg/mL	5 mL
50 mg/mL	5 mL
100 mg/mL	5 mL
200 mg/mL	1 mL, 2.5 mL, 5 mL, 10 mL, 30 mL

Source: Empower Pharmacy — TC Injection

Testosterone Cypionate / Propionate blend:

- 160/40 mg/mL (200 mg/mL total) in grapeseed oil
- Vial sizes: 5 mL, 10 mL
- Source: Empower Pharmacy — TC/TP

2.4 Dose Ranges: Men vs. Women

Men (TRT — hypogonadism):

Route	Starting Dose	Typical Maintenance	Frequency
Injectable (IM) — cypionate	100–200 mg	50–200 mg	q1–2 weeks
Injectable (SubQ) — cypionate	50–100 mg	75 mg	Weekly
Injectable (IM) — propionate	25–50 mg	25–50 mg	q2–3 days
Transdermal cream	100–200 mg	100–200 mg	Daily (scrotal or other)
Troche/sublingual	100–200 mg	100–200 mg	Daily
Pellet	150–450 mg (2–6 pellets)	150–450 mg	q3–6 months

Sources: Balance My Hormones, MediVera, Testopel DailyMed

Women (HSDD / hormone therapy):

Route	Typical Starting Dose	Typical Maintenance	Frequency
Transdermal cream (1% = 10 mg/g)	1–5 mg	1–10 mg max	Daily
Transdermal cream (20% = 200 mg/g)	5 mg (0.25 g)	5–10 mg max	Daily
Vaginal cream (20 mg/mL)	5 mg (0.25 mL)	5–10 mg	Daily
Sublingual troche	0.5–5 mg	1–10 mg	Daily
Slow-release capsule	1 mg	1–5 mg	Daily
Vaginal suppository (with estriol)	0.5 mg	0.5–1 mg	Daily

Sources: DUTCH Test, Empower Vaginal Cream, Dr. Oracle 1% Cream, Bayview Pharmacy

2.5 Combination Formulations

Testosterone + Anastrozole (injectable or separate):

- Commonly prescribed when aromatization (conversion to estrogen) is a concern in men
- Available as testosterone/anastrozole pellets (Wells Pharmacy)
- Also as oral anastrozole prescribed separately alongside injectable testosterone
- Rationale: Anastrozole (aromatase inhibitor) blocks testosterone→estrogen conversion

Testosterone + Nandrolone Decanoate:

- Nandrolone Decanoate / Testosterone Cypionate / Testosterone Enanthate injection
- 60/70/70 mg/mL in grapeseed oil, 5 mL vial
- Indications: hypogonadism + catabolic state, anemia, muscle wasting
- Source: Empower Pharmacy — Nandrolone/TC/TE

Testosterone + Progesterone (cream, women):

- Common in BHRT for perimenopausal/postmenopausal women
- Combined topical cream with custom ratios
- Source: NCBI/NIH BHT Table

Testosterone + DHEA (cream):

- Available as combined topical for women
- Source: Bayview Pharmacy Women's Formulations

Testosterone + Clomiphene/Anastrozole/DHEA/Progesterone (oral capsule combo — men):

- "Test Booster" — Clomiphene + DHEA + 7-Keto-DHEA + Progesterone + Anastrozole capsule
- Stimulates endogenous testosterone production rather than replacing externally
- Source: Strive Pharmacy Test Booster

2.6 Oil Base Options for Injectables

Oil Base	Viscosity	Inflammatory Potential	Notes
Cottonseed Oil	Medium-high	Low-moderate	FDA-approved brand (Depo-Testosterone/Pfizer) uses this
Sesame Oil	Medium	Low-moderate	FDA-approved brand (Delatestryl/enanthate) uses this; allergy risk
Grapeseed Oil	Low	Very low	Most common in compounded TRT; less viscous, easier to inject
MCT Oil (Miglyol 812N)	Very low	Very low	Pharmaceutical-grade; oxidatively stable; increasingly preferred by compounders
Ethyl Oleate	Very low	Low	Sometimes used in blends to lower viscosity
Castor Oil	Very high	Low	Used for very long-acting esters (undecanoate); slow release

Sources: ExcelMale TRT Forum, Marek Health MCT vs Seed Oil, Carie Boyd Pharmaceuticals

Why oil choice matters clinically:

- Viscosity affects injection comfort (MCT/grapeseed easiest)
- Some patients have sesame or cottonseed oil allergies

- Oil type affects release rate and site reactions
- Compounding pharmacies can accommodate any preferred oil with a prescription note

---

2.7 Preservatives in Injectable Testosterone

FDA-approved brand (Depo-Testosterone — Pfizer):

- Benzyl alcohol: **9.45 mg/mL** (same for both 100 mg/mL and 200 mg/mL formulations)
- Benzyl benzoate: 0.1 mL/mL (100 mg/mL) or 0.2 mL/mL (200 mg/mL)
- Cottonseed oil base
- Source: Pfizer Depo-Testosterone label

Compounded formulations:

- Benzyl alcohol (0.9% = ~9 mg/mL) standard preservative for multi-dose vials
- Some compounders offer benzyl alcohol-free single-dose vials for sensitive patients
- Benzyl benzoate sometimes included as solubilizer/co-solvent (not preservative)
- **Contraindication:** Benzyl alcohol associated with "gasping syndrome" in neonates/premature infants; patients with benzyl alcohol hypersensitivity need preservative-free formulations
- Source: Pfizer label, ExcelMale Forum

2.8 UI Data Fields — Testosterone Cascade Dependencies

```
Drug: Testosterone
■■ Dosage Form
■■ Injectable
■ ■■ Ester(s): [Cypionate | Enanthate | Propionate | Cypionate+Propionate blend]
■ ■■ Concentration: [20 | 50 | 100 | 200 mg/mL] (cascade from ester + compound)
■ ■■ Vial size: [1 mL | 2.5 mL | 5 mL | 10 mL | 30 mL] (cascade from conc.)
■ ■■ Oil base: [Grapeseed | Sesame | MCT | Cottonseed | Ethyl Oleate]
■ ■■ Preservative: [Benzyl alcohol 0.9% | Benzyl alcohol-free]
■ ■■ Route: [IM | SubQ]
■ ■■ Combination: [None | + Anastrozole | + Nandrolone | + Enanthate blend]
■ ■■ Dosing: dose (mg) + frequency
■
■■ Cream / Gel (Topical)
■ ■■ Patient sex: [Male | Female] → changes available concentration ranges
■ ■ ■■ Female: 1–20 mg/mL typically (free base in cream base)
■ ■ ■■ Male: 100–250 mg/mL typically (scrotal or body application)
■ ■■ Concentration: [1–250 mg/mL] in 1 mg/mL increments (30 mL dispenser)
■ ■■ Base vehicle: [Standard cream | Lipoderm | Anhydrous | PLO gel]
■ ■■ Application site: [Arm | Thigh | Abdomen | Scrotal | Vaginal]
■ ■■ Combination: [None | + Progesterone | + DHEA | + Estriol]
■ ■■ Dosing: dose (mg) + frequency + volume per application
■
■■ Troche / Sublingual
■ ■■ Strength: [2.5 | 5 | 10 | 100 | 200 mg per troche]
■ ■■ Patient: male vs. female drives strength selection
■
■■ Pellet / Implant
■ ■■ Individual pellet = 75 mg testosterone (fixed)
■ ■■ Number of pellets: [2–12+] depending on patient weight and T level
```

```
■ ■■ Total dose = number × 75 mg
■ ■■ Combination: [None | + Anastrozole pellet]
■ ■■ Frequency: q3–6 months
■
■■ Capsule / Suppository
■■ Strength and release: [immediate | slow-release]
■■ Patient: female use primarily

Patient sex is a top-level cascade driver:
Male → enables higher cream concentrations, larger injectable doses, pellet counts
Female → restricts cream to ≤10–20 mg/mL max, injectable rarely used,
troche/capsule/vaginal cream emphasized
```

3. Low-Dose Naltrexone (LDN)

3.1 Dose Classification System

Category	Dose Range	Mechanism	Clinical Uses
Ultra-Low-Dose (ULDN)	0.001 mg – < 0.5 mg (often 0.001–0.1 mg)	Binds filamin-A; attenuates μ-opioid Gs-coupling	Opioid adjunct; reduce tolerance; pain amplification
Very Low-Dose (VLDN)	0.001–1 mg	Possible immune/opioid modulation	Experimental; methadone detox adjunct (0.125–0.25 mg)
Low-Dose (LDN)	1–5 mg (classic 0.5–4.5 mg)	TLR4 antagonism + transient opioid blockade → endorphin rebound	Fibromyalgia, MS, Crohn's, cancer, autoimmune
Moderate Dose	< 25 mg	Mixed antagonism	Less common
Standard Dose	50 mg	Full opioid receptor blockade	Alcohol/opioid use disorder

Sources: PMC — LDN Review 2018, Fusion Specialty Pharmacy ULDN, Accurate Clinic ULDN Handout

3.2 Available Compounded Forms

Form	Notes
Capsule	Most common; immediate-release preferred; Avicel or similar filler
Oral liquid (suspension)	1 mg/mL typical; enables 0.1 mg increments with syringe
Sublingual drops	1 dropper bottle; each drop = 0.5 mg (UK standard)
Troche/lozenge	Dissolve under tongue; allows splitting (e.g., 4 mg troche → 1/2/3/4 mg doses)
Transdermal cream	~0.5 mg/mL; very low bioavailability (~1.1%); rarely used

Form	Notes
Suppository	Available from some pharmacies
Eye drops	Niche use reported

Sources: McMahan Pharmacy, Restorative Compounding, Bare Compounding, MayaScript Pharmacy, LDN Research Trust — Liquid/Sublingual

3.3 Liquid Concentrations and Dosing Precision

Concentration	Volume per dose	Precision	Notes
1 mg/mL	Variable (0.1 mL increments = 0.1 mg increments)	High	Most common in UK; baby oral syringe
2 mg/mL	Variable	Medium	Less common

Sources: LDN Research Trust Fact Sheet, LDN Research Trust — Liquid Dosing

3.4 Common Capsule Strengths

Standard capsule strengths available from compounding pharmacies:

`0.5 mg | 1.0 mg | 1.5 mg | 2.0 mg | 2.5 mg | 3.0 mg | 4.5 mg`

Also available as custom strengths (e.g., 0.25 mg, 6 mg) upon prescription.

Sources: McMahan Pharmacy, LDN Research Trust Fact Sheet, WellPharma Pharmacy

3.5 Standard Titration Protocols

Protocol A — Classic 3-step (most common in practice):

Step	Dose	Duration
1	1.5 mg	2 weeks
2	3.0 mg	2 weeks
3	4.5 mg	Ongoing maintenance

Source: WellPharma Maryland, Rock Ridge Pharmacy

Protocol B — Very slow 0.5 mg increment titration (LDN Research Trust/Dr. Goldstein):

Step	Dose	Duration
1	0.5 mg	Week 1

Step	Dose	Duration
2	1.0 mg	Week 2
3	1.5 mg	Week 3
4	2.0 mg	Week 4
...	+0.5 mg/week	Until optimal dose found
Target	3.0–4.5 mg	Ongoing

Note: Some patients find their optimal dose between 1.5–3.0 mg and do not need to reach 4.5 mg. Patient must be allowed to stop escalating at their "sweet spot." Sources: LDN Research Trust — Titration, Town & Country Compounding

ULDN Protocol (opioid adjunct):

- Start at 0.001 mg (1 mcg) twice daily
- Titrate as tolerated

Source: Accurate Clinic ULDN Handout

3.6 Titration Kit Packaging

Pharmacies have developed specific titration kit formats:

Town & Country Compounding Kit (example):

- Color-coded bottles: green = 0.5 mg, orange = 1.5 mg
- Kit 1 (lower dose): starts at 0.5 mg; Kit 2 (standard): starts at 1.5 mg
- Each dose in clearly labeled separate container
- Source: Town & Country Compounding Kit Video

Rock Ridge Pharmacy Capsule Kit:

- Capsules in individually labeled packets per dose level (0.5 mg to 4.5 mg)
- Step labeled on each packet (Step 1, Step 2, etc.)
- Source: Rock Ridge Pharmacy LDN Kit

Liquid titration approach:

- Single 1 mg/mL bottle + oral syringe
- Patient measures different volumes for each dose level
- Source: LDN Research Trust Fact Sheet

3.7 Filler/Excipient Considerations (Critical)

Excipient selection has a documented impact on LDN absorption and efficacy:

Excipient	Recommendation	Reason
Avicel (microcrystalline cellulose)	Preferred	Inert, rapid dissolution, no allergens, doesn't form sticky mass with water
Lactose	⚠ Acceptable if no intolerance	Traditional filler; some sensitivity in chronic illness patients
Sucrose	Acceptable	Fast-release; inert
Calcium carbonate	Avoid	Interferes with naltrexone absorption; can form insoluble mass in gut
Methylcellulose / HPMC derivatives	Avoid for standard LDN	Causes slow-release effect; blunts the acute blood spike required for LDN mechanism
DiluCap PSD	Alternative	Overcomes naltrexone's hydrophobic/poor solubility; better dissolution
DiluCap SR	⚠ Slow-release ONLY	Use when slow-release is specifically prescribed (different indication)

Critical note on slow-release: Standard LDN requires a **rapid blood spike** (transient receptor blockade) to work. Slow-release formulations fundamentally alter the mechanism and may reduce or eliminate efficacy. Some patients with severe side effects benefit from slow-release, but it should be explicitly prescribed as an intentional variant, not a default.

Sources: The Compounder Pharmacy — LDN Fillers, LDN Research Trust Fact Sheet, Fagron DiluCap Article

3.8 UI Data Fields — LDN Cascade Dependencies

```
Drug: Naltrexone (Low-Dose / Ultra-Low-Dose)
■■ Dose category
■■ ULDN (< 0.5 mg) → drives liquid/capsule form options
■■ LDN (0.5–4.5 mg) → most common; all forms available
■■ LDN extended (4.5–12 mg) → capsule or liquid

■■ Dosage Form
■■ Capsule (most common)
■ ■■ Strength: [0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 4.5 mg | custom]
■ ■■ Filler/Excipient: [Avicel (recommended) | Lactose | Sucrose | DiluCap]
■ ■■ Release type: [Immediate-release (standard) | Slow-release (explicit)]
■ ■■ Titration kit: [Yes / No]
■ ■■ [If Yes] → Kit format: [Step progression | Monthly escalation]
■ ■■ [If Yes] → Starting strength + target strength + increment size
■
■■ Oral Liquid
■ ■■ Concentration: [1 mg/mL | 2 mg/mL]
■ ■■ Volume: [30 mL | 60 mL | 120 mL]
■ ■■ Starting dose (mg) → auto-calculates volume per dose
■
■■ Sublingual Drops
```

```
■ ■■ Drops per dose (each drop = 0.5 mg standard)
■ ■■ Bottle size
■
■■ Troche/Lozenge
■ ■■ Total strength per troche (e.g., 4 mg)
■ ■■ Prescribed fraction (1/4, 1/2, whole = 1, 2, 4 mg)
■
■■ Transdermal Cream
■■ Concentration + volume (low bioavailability caveat)

Prescriber alert fields:
- Is patient currently on opioids? [Y/N] → [Y] triggers ULDN pathway + contraindication
warning
- Confirm immediate-release intended (not slow-release) unless explicitly marked SR
```

4. BPC-157 (Body Protection Compound)

4.1 Regulatory Status (Critical)

BPC-157 occupies a complex and legally ambiguous position as of April 2026:

- **Late 2023:** FDA added BPC-157 to Category 2 bulk drug substances list — meaning "do not compound" under 503A regulations. Other peptides on same list: TB-500, CJC-1295, Ipamorelin, Thymosin Alpha-1, GHK-Cu (injectable), Melanotan II, KPV, GHRP-2, GHRP-6, Selank, Semax, Kisspeptin-10, Epitalon, DSIP, and others (19 total Category 2 peptides).
- **2024:** Legal challenges from compounders → September 2024 settlement required FDA to submit key peptides for formal Pharmacy Compounding Advisory Committee (PCAC) review before finalizing ban. Source: Safe Harbor Group
- **As of early 2026:** BPC-157 remains Category 2 — under review. **"Likely not legal"** for US compounding pharmacies to provide BPC-157 (per health care attorney David Holt). Some pharmacies continue to supply it despite prohibition.
- **Enforcement reality:** FDA has sent few warning letters to pharmacies supplying BPC-157 despite the prohibition. Gray market thrives via "research chemical" loophole (not for human consumption). RFK Jr. has signaled potential loosening of peptide restrictions.
- **No approved drug exists:** Unlike semaglutide, there is no FDA-approved BPC-157 product. Status is "unapproved drug," not "compounded version of approved drug."

Source: STAT News — BPC-157, Elite NP — Peptide Reclassification 2026

4.2 Formulations and Concentrations

Injectable (Lyophilized Powder — Reconstitutable):

Vial Size	Common Reconstitution	Resulting Concentration	Notes
5 mg	0.5–3.0 mL BAC water	1.67–10 mg/mL	10 mg/mL = 100 mcg per 10 units on U-100 syringe

Vial Size	Common Reconstitution	Resulting Concentration	Notes
10 mg	1.0–6.0 mL BAC water	1.67–10 mg/mL	Most common clinic vial
2 mg	1–2 mL BAC water	1–2 mg/mL	Starter vials

Reconstitution standard (Happy Hormones MD protocol):

- 5 mg vial: inject **0.5 mL** BAC water → 10 mg/mL (10 units = 100 mcg per dose)
- 10 mg vial: inject **1.0 mL** BAC water → 10 mg/mL

Sources: Happy Hormones MD Protocol, PeptideDosages.com BPC-157 5mg

Storage:

- Lyophilized form: Stable up to 3 years frozen (−20°C), 2 years refrigerated; protect from light
- Reconstituted: Refrigerate at 2–8°C; stable ~6 weeks; avoid freeze–thaw cycles

Oral Capsules:

- Typical dose per capsule: 250–500 mcg (0.25–0.5 mg)
- Administered 1–3 times daily
- Best absorbed on empty stomach
- Lower systemic bioavailability than injection; preferred route for GI indications
- Source: Peptides.org BPC-157 Dosage

Nasal Spray:

- Concentration example: 10 mg per 10 mL bottle = 1 mg/mL = 100 mcg per 0.1 mL spray
- Each actuation delivers ~100 mcg
- ~100 sprays per 10 mL bottle
- Sold by wellness clinics (e.g., Gary Brecka nasal spray blend for $375/bottle)
- Source: VPeptide BPC-157 Nasal Spray, STAT News

4.3 Dose Ranges and Protocols

Standard dose range: 200–1,000 mcg/day (body-weight-based: ~2.5–3.75 mcg/kg twice daily)

Body Weight	Conservative	Standard	Higher (supervised)
125 lb (57 kg)	200 mcg	400 mcg	600 mcg
150 lb (68 kg)	250 mcg	500 mcg	750 mcg
180 lb (82 kg)	300 mcg	600 mcg	900 mcg
200 lb (91 kg)	350 mcg	700 mcg	1,000 mcg

Source: Dr. Rogers-Centers, Nulevel Wellness Medspa

Protocol by route and indication:

Route	Typical Daily Range	Frequency	Duration	Best For
SubQ Injection	250–500 mcg	1–2× daily	4–6 weeks	Muscle, tendon, joint recovery
IM Injection	250–500 mcg	1× daily	4–6 weeks	Localized deep injury
Oral/Sublingual	200–500 mcg	1–2× daily	6–8 weeks	GI healing, ulcer, leaky gut
Nasal Spray	~100–300 mcg	1–3 sprays	Variable	Systemic; brain/neuro support

Source: Yunique Medical, Dr. Rogers-Centers

Indications and tailored protocols:

Indication	Dose	Route	Cycle
Tendon/ligament repair	250–500 mcg/day	SubQ near injury	4–6 weeks, then off 2–4 weeks
Post-surgical healing	500–750 mcg/day	SubQ or IM	4–6 weeks minimum
Gut repair (IBD/ulcers)	500–1,000 mcg/day	Oral (empty stomach)	4–6 weeks; extend PRN
Neuroprotection	200–500 mcg/day	SubQ systemic	4–8+ weeks
Joint pain / arthritis	250–750 mcg/day	SubQ near joint	4–6 weeks, reassess
Maintenance	250 mcg	SubQ or oral	Ongoing pulse (2 weeks on, 2 off)

Source: Yunique Medical, Tucson Wellness MD

4.4 Reconstitution Instructions

Standard protocol for lyophilized BPC-157:

1. Allow vial to reach room temperature (reduces condensation)
2. Draw desired volume of bacteriostatic water (BAC water) into syringe
3. Clean vial stopper with alcohol swab
4. Inject BAC water slowly **down the inside wall** of the vial (not directly onto peptide powder)
5. **Do not shake** — gently swirl or roll until fully dissolved
6. Label vial with date of reconstitution
7. Refrigerate at 2–8°C; protected from light

Bacteriostatic water (0.9% benzyl alcohol) is required for multi-dose vials; sterile water can be used for single-dose vials but offers no antimicrobial preservation.

Sources: PeptideDosages.com, Happy Hormones MD, Prime Peptides

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4.5 Common Combinations

BPC-157 + TB-500 (Thymosin Beta-4 fragment):

- Most common BPC-157 combination
- Available as blend vials: 5/5 mg or 10/10 mg
- Also as liposomal oral formulation (BPC-157 + TB-500)
- Some protocols recommend separate injections (different sites), not mixing in same vial, due to pH difference
- TB-500 dose: 2–5 mg weekly (divided into 2–3 doses); BPC-157 dose continues per standard protocol
- Sources: Simon's Compounding, Core Peptides Blend, Freedom Pharmacy Oral, Saving Face Austin

BPC-157 + TB-500 + KPV + GHK-Cu:

- Advanced multi-peptide healing blend (quad formula)
- 5/5/10/5 mg per vial; subcutaneous injection 2–3× weekly
- Source: Beverly Hills Rejuvenation Center

4.6 UI Data Fields — BPC-157 Cascade Dependencies

```
Drug: BPC-157
■■ REGULATORY WARNING field: "BPC-157 is currently Category 2 on FDA's bulk drug
■ substances list (as of 2026). Compounding under 503A is legally restricted.
■ Prescriber must confirm jurisdiction compliance and clinical justification."
■
■■ Formulation
■■ Injectable (lyophilized powder)
■ ■■ Vial size: [2 mg | 5 mg | 10 mg]
■ ■■ Combination: [Single | + TB-500 | + TB-500 + KPV + GHK-Cu]
■ ■ ■■ [If combo] → Specify mg of each component
■ ■■ Reconstitution volume (mL of BAC water) → auto-calculates mg/mL concentration
■ ■■ Dose per injection (mcg) → auto-calculates units (based on mg/mL + U-100 syringe)
■ ■■ Injection route: [SubQ | IM]
■ ■■ Injection site: [Near injury | Abdomen/systemic | Other]
■ ■■ Frequency: [Once daily | Twice daily | 2–3×/week (maintenance)]
■ ■■ Cycle: [Number of weeks on] + [Weeks off / break period]
■
■■ Oral Capsule
■ ■■ Dose per capsule: [250 mcg | 500 mcg | custom]
■ ■■ Frequency: [Once daily | Twice daily | Three times daily]
■ ■■ Timing: [Empty stomach (recommended)]
■ ■■ Indication focus: GI/gut healing vs. systemic
■
■■ Nasal Spray
■■ Concentration: [1 mg/mL standard → 100 mcg/spray]
■■ Dose per use: [# sprays]
■■ Frequency: [1–3× daily]

Body weight field (kg or lbs) → enables weight-based dose calculator
Indication selection → suggests dose range, route, and cycle duration
```

5. UI Data Field & Cascading Dependency Analysis

5.1 Universal Pattern: The Cascade Tree

Every compounded medication prescription requires a **hierarchical cascade** where each selection constrains the valid options for downstream fields:

```
LEVEL 1: Drug selection
↓
LEVEL 2: Dosage form (injectable / cream / capsule / etc.)
↓
LEVEL 3: Salt form / ester / variant (cypionate vs. enanthate; base vs. sodium salt)
↓
LEVEL 4: Concentration (mg/mL)
↓
LEVEL 5: Vial / dispenser size (constrains total mg available per fill)
↓
LEVEL 6: Combination (additive selection → its own concentration field)
↓
LEVEL 7: Vehicle/base (oil type, cream base, excipient)
↓
LEVEL 8: Preservative / special packaging (benzyl alcohol Y/N; color-coded kits)
↓
LEVEL 9: Dose + frequency + route → auto-calculates units/volume per administration
↓
LEVEL 10: Titration protocol → generates week-by-week schedule from above fields
```

5.2 Key Cascading Dependency Pairs

Field Selected	Constrains / Changes
Drug = Semaglutide	Triggers regulatory status warning (post-shortage); must document medical necessity
Salt form = base vs. acetate	Semaglutide base only; block acetate/sodium salt selection with warning
Concentration selected	Changes dose-to-volume/unit calculation; changes compatible vial sizes
Formulation = injectable + lyophilized	Adds BAC water volume field, reconstitution instructions panel
Patient sex = Female (testosterone)	Restricts injectable dose range; restricts cream concentrations to ≤20 mg/mL; removes high-dose pellet options
Combination = +B12/cyanocobalamin	Adds B12 concentration field (typically 0.5 mg/mL fixed); changes label requirements
LDN form = capsule	Adds excipient selector; adds release-type selector (IR vs. SR) with warning
LDN titration kit = Yes	Opens multi-step dose schedule builder
BPC-157 selected	Triggers Category 2 legal status warning; adds reconstitution calculator
BPC-157 vial size + BAC water volume	Auto-calculates mg/mL → drives unit-per-dose calculation
Oil base = sesame	Adds allergy screening flag for sesame sensitivity

Field Selected	Constrains / Changes
Preservative = benzyl alcohol-free	Changes packaging to single-dose vials only (no multi-dose)

5.3 Calculated Fields (Auto-Computed from Inputs)

Calculated Field	Formula	Requires
Volume per dose (mL)	Dose (mg) ÷ Concentration (mg/mL)	Dose + Concentration
Units per dose (U-100 syringe)	Volume (mL) × 100	Volume per dose
Total mg per vial	Concentration (mg/mL) × Vial size (mL)	Concentration + Vial size
Expected days supply	Total mg ÷ Dose per administration ÷ Frequency	Total mg + Dose + Frequency
BPC-157 concentration post-reconstitution	Peptide mg ÷ BAC water volume (mL)	Vial size + Water volume
LDN liquid volume per dose	Dose (mg) ÷ Concentration (mg/mL)	Dose + Liquid concentration
Testosterone cream dose (mg) from pump	Pump volume (mL) × Concentration (mg/mL)	Concentration + Dispenser calibration
Pellet count (testosterone)	Total dose (mg) ÷ 75 mg/pellet	Target total dose

5.4 Regulatory / Safety Alert Fields

Each medication requires specific non-dismissible alert conditions:

Drug	Alert Condition	Alert Content
Semaglutide	Post-shortage compounding	Requires documented medical necessity (different dose, inactive ingredient allergy, or documented clinical need); salt form must be base not sodium/acetate
Semaglutide	Salt form = sodium or acetate	"FDA has stated this salt form is not an approved active ingredient. Do not prescribe."
Testosterone	Patient = pediatric	Benzyl alcohol contraindication warning
Testosterone (women)	Dose > 10 mg/day	Exceeds maximum recommended female dose; confirm clinical justification
LDN	Concurrent opioid use	Opioid antagonism will precipitate withdrawal; use ULDN protocol only

Drug	Alert Condition	Alert Content
LDN	Release type = slow	"Slow-release formulation alters LDN mechanism. Standard LDN requires rapid blood spike for TLR4 modulation. Confirm slow-release is intentional."
LDN	Filler = calcium carbonate	"This filler interferes with naltrexone absorption. Select Avicel, lactose, or sucrose."
BPC-157	Any selection	"BPC-157 is Category 2 on FDA bulk substances list (2023). Compounding under 503A is legally restricted as of 2026. Prescriber assumes full regulatory compliance responsibility."
BPC-157	Patient = cancer diagnosis	"BPC-157 has theoretical angiogenic effects that may promote tumor vascularity. Exercise extreme caution in oncology patients."

Appendix: Source Reference Summary

Topic	Primary Source
Semaglutide FDA shortage resolution	FDA.gov
Semaglutide enforcement dates	Burr & Forman LLP, Alston & Bird
Semaglutide 2026 enforcement status	Heal Nourish Grow
Semaglutide salt form warning	US Pharmacist, Diatribe
Semaglutide concentrations and dosing	Flow Wellness, Rivas Medical
Semaglutide + B12 formulation	Empower Pharmacy, Belmar Pharma
Semaglutide additives guide	Mochi Health
Semaglutide titration schedule	Beauty With Bubbly dosing form, Rock Ridge Pharmacy
Testosterone cypionate concentrations/vials	Empower Pharmacy
Testosterone oil base comparison	ExcelMale Forum, Marek Health
Testosterone cypionate preservative	Pfizer Depo-Testosterone label
Testosterone cream concentrations	Empower Pharmacy Cream
Testosterone troches	Empower Pharmacy Troche
Testosterone pellets	Testopel.com, DailyMed

Topic	Primary Source
Testosterone for women	DUTCH Test, Dr. Oracle
Testosterone + nandrolone combination	Empower Pharmacy Nandrolone/TC/TE
Testosterone + anastrozole pellet	Wells Pharmacy
Compounded BHT dosage forms table	NIH/NCBI National Academies
LDN dose classification	PMC — LDN Review
LDN formulations	McMahan Pharmacy, MayaScript, Bare Compounding
LDN titration protocols	WellPharma, LDN Research Trust, Town & Country
LDN titration kit packaging	Rock Ridge Pharmacy, Town & Country Kit Video
LDN filler/excipient guidance	The Compounder Pharmacy, LDN Research Trust Fact Sheet, Fagron DiluCap
LDN ULDN distinction	Fusion Specialty Pharmacy, Accurate Clinic
BPC-157 regulatory status	STAT News 2026, PeptideLaws.com, Elite NP
BPC-157 legal challenge / settlement	Safe Harbor Group
BPC-157 dose protocols	Yunique Medical, Dr. Rogers-Centers, Tucson Wellness MD
BPC-157 reconstitution	PeptideDosages.com, Happy Hormones MD
BPC-157 + TB-500 combination	Simon's Compounding, Beverly Hills Rejuvenation
BPC-157 nasal spray specs	VPeptide

End of Research Area 7

Section 8: Synthesis & Recommendations for CompoundIQ

This section synthesizes the cross-cutting insights from all seven research areas into actionable recommendations for the CompoundIQ platform.

8.1 Recommended Data Model

Based on the findings from Research Area 3 (Medication Configuration Data Models), the recommended entity-relationship model for CompoundIQ centers on an ingredient-centric architecture that mirrors how compounded medications are actually built — as configurable recipes rather than fixed products.

Entity	Key Attributes	Relationships
ActiveIngredient	name, UNII code, CAS number, salt forms[], therapeutic category, schedule (if controlled)	Has many → IngredientConfig
IngredientConfig	ingredient_id, salt_form, min_concentration, max_concentration, concentration_unit, compatible_dosage_forms[], compatible_routes[]	Belongs to → ActiveIngredient; Referenced by → CompoundTemplate
CompoundTemplate	name, ingredient_configs[], dosage_form, route, default_volume, default_quantity, pharmacy_id, is_standard_catalog	Has many → IngredientConfig; Belongs to → DosageForm, Route; Referenced by → ProtocolTemplate
DosageForm	name (injectable, cream, capsule, troche, etc.), compatible_routes[], requires_beyond_use_date, requires_sterility	Has many → Route
Route	name (SubQ, IM, topical, oral, sublingual, etc.), SNOMED_CT_code	Belongs to → DosageForm
TitrationSchedule	phases[] (each: dose, duration_weeks, frequency), max_dose, escalation_trigger	Referenced by → Prescription, ProtocolTemplate
ProtocolTemplate	name, compound_templates[], titration_schedules[], sex_filter, conditional_rules[], notes	Has many → CompoundTemplate, TitrationSchedule
Prescription	patient_id, prescriber_id, compound_template_id, concentration, quantity, sig_text, sig_structured{}, refills, day_supply, clinical_necessity_note, pharmacy_id	Belongs to → Patient, Prescriber, CompoundTemplate, Pharmacy

This model supports both catalog-based ordering (CompoundTemplate represents a pre-defined pharmacy SKU) and custom ordering (provider assembles ingredients ad hoc). The IngredientConfig layer enforces valid concentration ranges and dosage form compatibility, providing guardrails without eliminating flexibility.

8.2 Recommended UX Pattern: Cascading Progressive Disclosure

Based on Research Area 5 findings, CompoundIQ should implement a cascading progressive disclosure interface that reveals fields sequentially as the prescriber makes selections. This approach is validated by Epic's Order Composer research (Friction per Order benchmarks), academic studies on cognitive load in medical ordering, and the cascading dependency analysis from Research Area 7.

The recommended cascade sequence:

Level	Field	Input Type	What It Constrains
1	Medication Category	Card selector / search	Filters available drugs
2	Drug (Active Ingredient)	Type-ahead search	Determines salt forms, dosage forms, routes
3	Salt / Ester Form	Dropdown (if multiple)	Constrains concentrations, dosage forms
4	Dosage Form	Dropdown	Constrains routes, volume units
5	Route of Administration	Dropdown (auto-set if only one)	Constrains site, frequency defaults
6	Concentration / Strength	Dropdown or numeric input	Constrains dose-per-administration range
7	Volume / Quantity	Dropdown + numeric	Determines day supply, cost
8	Dose per Administration	Numeric with units	Determines units/dose for sig
9	Frequency	Structured selector	Determines day supply calculation
10	Titration Schedule (optional)	Visual timeline builder	Generates multi-phase sig, may affect quantity

Key UX principles: (1) Each level auto-collapses completed selections into a summary chip. (2) Smart defaults pre-select the most common option when there is a clear majority choice. (3) Favorites and recently-used items appear first. (4) The sig preview updates in real-time as each field is completed. (5) Protocol templates bypass the cascade entirely, auto-filling all 10 levels from a saved template.

8.3 Recommended Regulatory Compliance Checklist

Based on Research Area 6, CompoundIQ must satisfy the following regulatory requirements before launch:

- **Prescription Field Validation:** All 12–13 DEA-required fields for controlled substances (21 CFR 1306). Clinical necessity notation for all 503A compounds. Active ingredient(s) + strength must be explicitly specified (not implied by drug name).
- **EPCS Two-Factor Authentication:** 2FA at signing using two distinct factor categories (know/have/are). Identity proofing at NIST SP 800-63-1 Level 3. FIPS 140-2 Security Level 1 for cryptographic module.
- **Pre-Signing Review Screen:** Must display all DEA-required fields plus the mandatory attestation statement (21 CFR 1311.140(a)(3)). No alteration of fields after 'ready to sign' indication.

- **Audit Trail:** Every create/edit/sign/transmit/delete event logged with timestamp, user identity, and outcome. 2-year retention. Daily automated internal audits. Monthly prescription logs to prescribers within 7 days.
- **Third-Party DEA Certification:** Application must be certified by a DEA-approved third-party auditor before transmitting any controlled substance prescriptions electronically. Re-certification every 2 years.
- **NCPDP SCRIPT v2017071 Compliance:** Electronic compound prescriptions must use v2017071 format. Support up to 25 ingredients. 1,000-character sig field.
- **HIPAA / PHI Protections:** BAAs with all vendors. Encryption in transit (TLS 1.2+). No PHI on shipping labels. Minimum necessary standard for all data sharing.
- **State License Verification:** Before routing a prescription, verify the compounding pharmacy is licensed in the patient's state. 503A pharmacies cannot exceed 5% interstate distribution without FDA MOU.
- **Fax Fallback:** Support HIPAA-compliant fax output for pharmacies not on the Surescripts network. All faxed controlled substance prescriptions must include a wet (manual) signature.
- **Time Synchronization:** System clock within 5 minutes of NIST time (21 CFR 1311.120(b)).

8.4 Recommended Protocol Template System Design

Based on Research Area 4 findings, the protocol template system should support the following capabilities:

- One-click instantiation of multi-medication protocol bundles (e.g., 'Male Hormone Optimization' creates Testosterone Cypionate Rx + Anastrozole Rx + HCG Rx simultaneously)
- Sex-specific protocol variants (Male/Female selector loads appropriate medication set)
- Per-medication dose adjustment within the protocol before submission
- Integrated titration timeline for medications requiring dose escalation (e.g., Semaglutide 10-week escalation baked into the protocol)
- Conditional medication suggestions based on lab values (e.g., 'Estradiol is 52 pg/mL — add Anastrozole per protocol?')
- Protocol versioning and practice-wide sharing (all providers in a practice use the same protocol versions)
- Auto-routing to preferred compounding pharmacy with bundled shipment coordination
- Clinical necessity auto-generation from protocol context (e.g., 'Compounded testosterone cypionate required — patient needs concentration not commercially available')

8.5 The Universal Cascade Tree (10-Level Cascade)

Research Area 7's analysis of real-world compounding prescriptions for semaglutide, testosterone, LDN, and BPC-157 reveals a universal 10-level cascading dependency structure that applies across all compound medications. This cascade is both the data model and the UX model:

Level	Semaglutide Example	Testosterone Example	LDN Example
1. Category	Weight Management	Hormone Optimization	Pain / Autoimmune
2. Drug	Semaglutide	Testosterone	Naltrexone
3. Salt/Ester	Semaglutide base	Cypionate	Naltrexone HCl
4. Dosage Form	Injectable solution	Injectable / Cream / Pellet	Capsule / Liquid
5. Route	SubQ	IM or SubQ / Topical	Oral / Sublingual
6. Concentration	1 mg/mL or 5 mg/mL	200 mg/mL	1 mg/mL liquid
7. Volume	2.5 mL vial	10 mL vial	30 mL bottle
8. Dose/Admin	0.25 mg → 2.5 mg	100–200 mg	0.5–4.5 mg
9. Frequency	Once weekly (QW)	QW or BIW	QHS (nightly)
10. Titration	4-phase escalation over 10–16 weeks	Stable dose; adjust at 6-week labs	+0.5 mg q5-7d to 4.5 mg

8.6 Key Competitive Gaps CompoundIQ Can Exploit

Based on the comprehensive portal analysis in Research Area 1, the following features are absent from every existing compounding pharmacy ordering platform:

Gap	Current State	CompoundIQ Opportunity
Titration Schedule Builder	No platform has one. Titration is handled via free-text SIG or sequential single-dose prescriptions.	Visual timeline builder that generates multi-phase sigs and auto-calculates vial quantities per phase. First-to-market advantage.
Protocol Bundles	LifeFile has Multi-Order Management (same med, multiple patients) but no multi-medication protocol templates. Cerbo has Chart Parts but they are documentation-centric, not prescription-centric.	One-click protocol instantiation: select "Male HRT Protocol" and get 3 coordinated Rx orders pre-filled and ready to review.
Progressive Disclosure UI	All existing portals use form-dump interfaces (all fields visible at once) or catalog-browse (SKU selection). No cascading disclosure.	10-level cascading interface where each selection narrows subsequent options. Reduces cognitive load by 50–80% vs. form-dump.
Conditional Lab-Triggered Medications	No functional medicine EHR offers automated lab-triggered medication suggestions in the prescribing workflow.	Soft alerts: "Estradiol is 52 pg/mL — add Anastrozole per your protocol?" Clinician approves; Rx auto-populates.

Gap	Current State	CompoundIQ Opportunity
Multi-Pharmacy Price/Inventory Comparison	Only My Practice Connect offers multi-pharmacy comparison. No pharmacy-native portal does.	Real-time inventory and pricing across connected compounding pharmacies within the prescribing workflow.
Structured Sig Generation	All platforms output free-text sigs. No platform generates NCPDP S&C SIG structured data alongside free text.	Dual output: clean free-text sig for transmission + structured sig data for analytics, CDS, and future interoperability.
Smart Defaults from Prescribing History	LifeFile has MyRX (favorites). WellsPx3 has Clone. No platform uses prescribing analytics to suggest defaults.	ML-powered defaults: pre-select the most common concentration, volume, and frequency based on the provider's historical prescribing patterns for each medication.

CompoundIQ's strategic position is clear: it can be the first compounding pharmacy ordering platform built from the ground up with progressive disclosure UX, native titration scheduling, protocol template bundles, and structured sig generation. The combination of these four capabilities — none of which exists in any current platform — represents a generational leap in compounding pharmacy ordering technology.